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The Pólya–Vinogradov inequality with $\log\log q$

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Abstract

We prove that every primitive Dirichlet character χ modulo $q \geq 3$ has character sums $\sum_{n\leq x}\chi(n)$ of size at most $C\sqrt{q}\log\log q$, with an absolute constant C. Montgomery and Vaughan proved this bound under the generalised Riemann hypothesis, and Paley showed that it is optimal. A complete attributed zero-free argument for $\operatorname{Re} s > 11/12$ is included in a proof supplement. On the major arcs this half-plane gives a power saving for twisted prime sums; on the minor arcs we prove a weak form of the Montgomery–Vaughan bound for exponential sums with multiplicative coefficients, with Selberg's sieve as the only arithmetic input. The same half-plane gives $|L'/L(1,\chi)| \ll \log\log q$ for every nonprincipal χ and, for every prime q, the two-sided bound $-C\log q\log\log q \leq \gamma_q \leq C\log q$ for the Euler–Kronecker constant of the cyclotomic field $\mathbb{Q}(\zeta_q)$.

1. Introduction

Let χ be a primitive Dirichlet character modulo $q \geq 3$. The Pólya–Vinogradov inequality states that $\left|\sum_{n\leq x}\chi(n)\right| \ll \sqrt{q}\log q$ uniformly in x [3, Chapter 23].[1] Paley proved that the factor $\log q$ cannot be replaced by anything smaller than $\log\log q$: for infinitely many primitive quadratic characters the maximum over x exceeds $c\sqrt{q}\log\log q$ [13]. Montgomery and Vaughan [9] proved the upper bound $\sqrt{q}\log\log q$ under the generalised Riemann hypothesis. Without hypothesis, Granville and Soundararajan lowered the exponent of

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[1] The proof expands χ in additive characters through the Gauss sum $\tau(\bar{\chi})$, whose modulus is $\sqrt{q}$ for primitive χ, and bounds the resulting geometric series trivially; Lemma 5.1 below is the same argument with the harmonic sums kept.

$\log q$ for characters of fixed odd order [5].[2] We prove the bound of Montgomery and Vaughan for every primitive character.

THEOREM 1.1. *There is an absolute constant C such that for every primitive character χ modulo $q \geq 3$ and every $x \geq 1$,*

$$\Big|\sum_{n\leq x} \chi(n)\Big| \leq C\sqrt{q}\log\log q.$$

By Paley's theorem, Theorem 1.1 is optimal up to the value of C. The input that replaces the Riemann hypothesis is the following theorem.

THEOREM 1.2 (OpenAI; proof included). *If χ is a nonprincipal Dirichlet character, then $L(s,\chi) \neq 0$ for $\operatorname{Re} s > 11/12$.*

Proof. Appendix A includes the complete argument of OpenAI [11]. Its Proposition 3.1 is proved in its Section 5.6; its Section 3 derives nonvanishing for finite-order Hecke characters over $\mathbb{Q}(\sqrt{-3})$ and transfers it to Dirichlet characters. No assertion from the stronger 7/8 preprint is used in that derivation. For an imprimitive character, the removed factors $1-\chi^*(p)p^{-s}$ cannot vanish when $\operatorname{Re} s > 0$, because their second term has modulus less than one. Thus the same half-plane applies to every nonprincipal character. □

The included argument is OpenAI's work. The attribution is retained; its reproduction supplies a proof within this document and is not a new independent proof by the present authors. Appendix B checks every numerical change needed to use 11/12 in place of 7/8. The separate attempt to replace the zero-free argument independently has not produced a replacement theorem; Appendix C records two precise barriers encountered in that attempt. The existing Lean development based on the stronger theorem remains separately attributed to OpenAI [12]; it is not represented as a formalisation of the newly changed parameters or of every sentence in the proof supplement.

Montgomery and Vaughan use the Riemann hypothesis in one place: on the major arcs, for prime sums of the characters $\chi\psi$ twisted by additive characters, where ψ runs over the characters of a small modulus r. A power saving in these sums is enough there, and Theorem 1.2 provides one (Corollary 3.8). On the minor arcs they use their unconditional bound for exponential sums with multiplicative coefficients. A weaker form of it suffices, and we prove that form in Section 4 with Selberg's sieve as the only arithmetic input. Figure 1 shows how the parts fit together.

[2] For a primitive character of odd order g they obtain $\sqrt{q}(\log q)^{1-\delta_g/2+o(1)}$ with $\delta_g = 1-\frac{g}{\pi}\sin\frac{\pi}{g}$, and under the generalised Riemann hypothesis the same exponent of $\log\log q$.

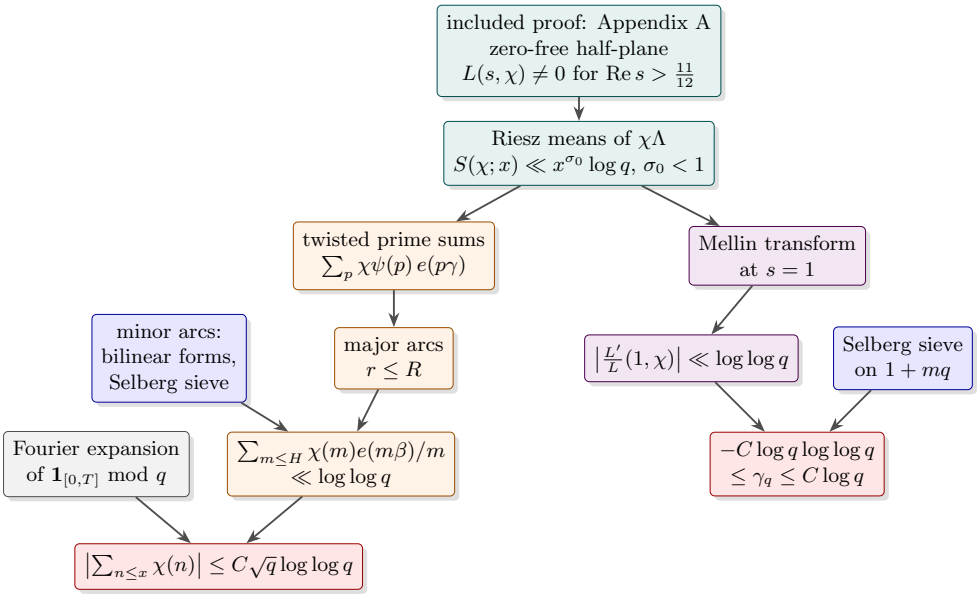


Figure 1. The structure of the proofs. The left branch proves Theorem 1.1: the Fourier expansion of a character sum reduces it to the sums $\sum_{m\le H}\chi(m)e(m\beta)/m$, which are bounded on the major arcs by the half-plane and on the minor arcs by bilinear forms and the sieve. The right branch proves Theorem 1.3.

The same half-plane controls the logarithmic derivative of $L(s,\chi)$ at $s=1$. For a number field K write $\zeta_K(s) = c_{-1}(s-1)^{-1} + c_0 + O(s-1)$ near $s=1$; the Euler–Kronecker constant of K is $\gamma_K = c_0/c_{-1}$. Let γ_q be the Euler–Kronecker constant of $\mathbb{Q}(\zeta_q)$. For a prime q the factorisation $\zeta_{\mathbb{Q}(\zeta_q)}(s) = \zeta(s)\prod_{\chi\neq\chi_0} L(s,\chi)$, the product over the nonprincipal characters modulo q [14, Theorem 4.3],[3] gives

$$\gamma_q = \gamma + \sum_{\chi\neq\chi_0} \frac{L'}{L}(1,\chi), \tag{1.1}$$

where γ is Euler's constant.

THEOREM 1.3. *There is an absolute constant C with the following properties.*

[3]All nonprincipal characters modulo a prime are primitive, so no Euler factors have to be removed: the conductor of each nonprincipal χ is q itself, and $L(s,\chi)$ is the L-function of the corresponding primitive character.

(i) *For every nonprincipal character χ modulo q,*

$$\Big|\frac{L'}{L}(1,\chi)\Big| \le C\log\log q.$$

(ii) *For every prime $q \ge 3$, $-C\log q\log\log q \le \gamma_q \le C\log q$.*

Under the generalised Riemann hypothesis the bound (i) is classical, Ihara proved $\gamma_q \le (2+o(1))\log q$ and Badzyan proved a lower bound of the order of (ii) [7, 1]. Ford, Luca and Moree proved the two bounds of (ii) without hypothesis for a set of primes q of density one [4, 10].[4] For all primes the best unconditional bound we know is $|\gamma_q| \ll_\varepsilon q^\varepsilon$, due to Ihara, Murty and Shimura [8]. The upper bound in (ii) needs only Chebyshev's bound. The lower bound needs an upper bound for the prime powers $n \le q^{25}$ with $n \equiv 1 \pmod q$, which we obtain from Selberg's sieve applied in dyadic blocks (Lemma 6.3).

Theorems 1.1 and 1.3 are not among the consequences of Theorem 1.2 drawn in [12]. The arguments work, with other constants, whenever $L(s,\chi) \ne 0$ on $\operatorname{Re} s > \theta$ for all nonprincipal χ and some fixed $\theta < 1$. The paper is organised as follows. Section 2 collects Chebyshev's bounds and Selberg's sieve. Section 3 bounds Riesz means of $\chi\Lambda$ by a contour shift. Section 4 proves the bound for exponential sums with multiplicative coefficients, Section 5 proves Theorem 1.1 and Section 6 proves Theorem 1.3.

2. **Preliminaries**

Throughout, p and ℓ denote primes, Λ is the von Mangoldt function, $\theta(x) = \sum_{p\le x}\log p$, $\psi(x) = \sum_{n\le x}\Lambda(n)$, $e(x) = e^{2\pi i x}$, and $\|x\|$ is the distance from x to the nearest integer. We write $\omega(n)$ for the number of prime factors of n, $\tau_3(n)$ for the number of ordered factorisations $n = abc$, and $P(n)$ for the largest prime factor of $n \ge 2$, with $P(1) = 1$.

LEMMA 2.1. *For $x \ge 1$ and integers $N \ge 2$,*

(i) $\theta(x) \le x\log 4$;
(ii) $\psi(x) - \theta(x) \le 2\sqrt{x}\log x$ *and* $\psi(x) \le 4x$;
(iii) $\sum_{p\le N}\frac{\log p}{p} \le \log N + \log 4$;
(iv) $\sum_{p\le N}\frac{1}{p} \le \log\log N + 10$.

Proof. (i) It suffices to treat integers $m \ge 1$, and we use induction on m. The cases $m = 1, 2$ are clear, and for even $m > 2$ we have $\theta(m) = \theta(m-1)$. Let $m = 2n+1$. The primes in $(n+1, 2n+1]$ divide $\binom{2n+1}{n}$, and $2\binom{2n+1}{n} \le$

[4]The difficulty for an individual prime q lies in the prime powers $n \equiv 1 \pmod q$, which enter γ_q with the weight $(q-1)\Lambda(n)/n$. Below q^{25} we bound them for each prime q separately by Lemma 6.3; beyond q^{25} the zero-free half-plane makes their contribution bounded.

2^{2n+1} because $\binom{2n+1}{n}$ and $\binom{2n+1}{n+1}$ are equal terms of the binomial expansion of $(1+1)^{2n+1}$. Hence $\theta(2n+1) \le \theta(n+1) + n\log 4 \le (2n+1)\log 4$.

(ii) We have $\psi(x) - \theta(x) = \sum_{k\ge 2} \theta(x^{1/k})$, where only $k \le \log x/\log 2$ contribute. Each term is at most $\sqrt{x}\log 4$ by (i), so the sum is at most $(\log x/\log 2)\sqrt{x}\log 4 = 2\sqrt{x}\log x$. Since $\log x \le \sqrt{x}$, $\psi(x) \le (\log 4 + 2)x \le 4x$.

(iii) By Legendre's formula and (i),

$$N\log N \ge \log N! \ge \sum_{p\le N} \Big\lfloor \frac{N}{p} \Big\rfloor \log p \ge N \sum_{p\le N} \frac{\log p}{p} - N\log 4.$$

(iv) Let $A(u) = \sum_{p\le u} \log p/p$. Partial summation and (iii) give

$$\sum_{p\le N} \frac{1}{p} = \frac{A(N)}{\log N} + \int_2^N \frac{A(u)}{u\log^2 u}\,du \le 1 + \frac{\log 4}{\log 2} + \int_2^N \Big(\frac{1}{u\log u} + \frac{\log 4}{u\log^2 u}\Big)du,$$

and the integral is at most $\log\log N - \log\log 2 + 2$. □

We use Selberg's upper bound sieve in the following form; for the method see [6, Chapter 3], and we give the proof of the form we need.[5]

LEMMA 2.2. *Let $a_1, \dots, a_X$ be integers, P a squarefree integer, ν a multiplicative function with $0 < \nu(p) < 1$ for $p \mid P$, and $y \ge 1$. For $d \mid P$ put $R_d = \#\{n \le X : d \mid a_n\} - \nu(d)X$, and for $l \mid P$ put $g(l) = \prod_{p\mid l} \nu(p)/(1-\nu(p))$. Then*

$$\#\{n \le X : (a_n, P) = 1\} \le \frac{X}{S} + \sum_{\substack{d\mid P\\ d\le y}} 3^{\omega(d)}|R_d|, \qquad S = \sum_{\substack{l\mid P\\ l\le\sqrt{y}}} g(l).$$

Proof. Let D be the set of divisors d of P with $d \le \sqrt{y}$, and let $(\lambda_d)_{d\in D}$ be real numbers with $\lambda_1 = 1$. If $(a_n, P) = 1$ then $\sum_{d\in D,\, d\mid a_n} \lambda_d = 1$, and otherwise this sum is real, so

$$\begin{aligned}\#\{n : (a_n, P) = 1\} &\le \sum_{n\le X} \Big(\sum_{\substack{d\in D\\ d\mid a_n}} \lambda_d\Big)^2 \\ &= X \sum_{d_1,d_2\in D} \lambda_{d_1}\lambda_{d_2}\nu([d_1,d_2]) + \sum_{d_1,d_2\in D} \lambda_{d_1}\lambda_{d_2} R_{[d_1,d_2]}.\end{aligned}$$

Write Q for the double sum in the first term. Since ν is multiplicative and d_1, d_2 are squarefree, $\nu([d_1,d_2]) = \nu(d_1)\nu(d_2)/\nu((d_1,d_2))$. Let h be the multiplicative function on the divisors of P with $h(p) = 1/\nu(p) - 1 = 1/g(p)$; then $1/\nu(m) =$

[5] The weights λ_d are those of Selberg's quadratic form, and the only feature used later is the bound $|\lambda_d| \le 1$, which makes the remainder term at most a sum of $3^{\omega(d)}|R_d|$ over $d \le y$.

$\sum_{l\mid m} h(l)$ for $m \mid P$, and

$$Q = \sum_{l\mid P} h(l)\Big(\sum_{\substack{d\in D\\ l\mid d}} \lambda_d \nu(d)\Big)^2 = \sum_{l\in D} \frac{y_l^2}{g(l)}, \qquad y_l = \sum_{\substack{d\in D\\ l\mid d}} \lambda_d \nu(d).$$

Möbius inversion gives $\lambda_d \nu(d) = \sum_{l\in D,\, d\mid l} \mu(l/d) y_l$, so the condition $\lambda_1 = 1$ reads $\sum_{l\in D} \mu(l) y_l = 1$. By Cauchy's inequality

$$1 \le \Big(\sum_{l\in D} \frac{y_l^2}{g(l)}\Big)\Big(\sum_{l\in D} g(l)\Big) = QS,$$

with equality for $y_l = \mu(l) g(l)/S$. We make this choice, so that $Q = 1/S$. Then, writing $l = de$,

$$\lambda_d = \frac{1}{\nu(d) S} \sum_{\substack{l\in D\\ d\mid l}} \mu(l/d)\mu(l) g(l) = \frac{\mu(d) g(d)}{\nu(d) S} \sum_{\substack{e\in D,\ (e,d)=1\\ de\le \sqrt{y}}} g(e).$$

Since $g(d)/\nu(d) = \prod_{p\mid d}(1-\nu(p))^{-1} = \prod_{p\mid d}(1+g(p)) = \sum_{f\mid d} g(f)$, we get $|\lambda_d| \le S^{-1} \sum_{f\mid d} \sum_e g(fe)$, and the products fe are distinct elements of D; hence $|\lambda_d| \le 1$. Finally $[d_1, d_2] \le d_1 d_2 \le y$, and every $d \mid P$ is the least common multiple of at most $3^{\omega(d)}$ pairs (d_1, d_2), so the second term is at most $\sum_{d\mid P,\, d\le y} 3^{\omega(d)} |R_d|$. □

3. **Riesz means of prime sums**

For an arithmetic function a and $x \ge 1$ put

$$S(a; x) = \sum_{n\le x} a(n)\Big(1 - \frac{n}{x}\Big), \qquad S(\chi; x) = S(\chi\Lambda; x) = \sum_{n\le x} \chi(n)\Lambda(n)\Big(1 - \frac{n}{x}\Big).$$

The weight $1 - n/x$ makes the Mellin kernel absolutely integrable, which keeps the contour shift below free of truncation errors. The aim of this section is Proposition 3.7.

LEMMA 3.1. *Let $a\colon \mathbb{N} \to \mathbb{C}$ satisfy $\sum_n |a(n)| n^{-\sigma} < \infty$ for $\sigma > 1$, and let $F(s) = \sum_n a(n) n^{-s}$ for $\operatorname{Re} s > 1$.*

(i) *For $\operatorname{Re} s > 1$, $\displaystyle\int_1^\infty S(a; x)\, x^{-s-1}\, dx = \frac{F(s)}{s(s+1)}$.*

(ii) *For $x > 0$, $\displaystyle S(a; x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(2+it)\, \frac{x^{2+it}}{(2+it)(3+it)}\, dt$.*

In other words the following square commutes, and (ii) inverts its right column on the line $\operatorname{Re} s = 2$*:*

$$\begin{array}{ccc} a & \xrightarrow{\text{Riesz mean}} & S(a;\cdot) \\ \Big\downarrow{\scriptstyle \text{Dirichlet series}} & & \Big\downarrow{\scriptstyle \text{Mellin transform}} \\ F & \xrightarrow[\times\,\frac{1}{s(s+1)}]{} & \dfrac{F(s)}{s(s+1)} \end{array}$$

Proof. (i) For $n \ge 1$, $\int_n^\infty (1-n/x)x^{-s-1}dx = n^{-s}/s - n^{-s}/(s+1) = n^{-s}/(s(s+1))$. Since $\sum_n |a(n)| \int_n^\infty x^{-\sigma-1}dx = \sigma^{-1}\sum_n |a(n)|n^{-\sigma} < \infty$, the sum and the integral may be exchanged.

(ii) For $y > 0$ let $k(y) = \frac{1}{2\pi i}\int_{(2)} y^s \frac{ds}{s(s+1)}$. If $y \ge 1$, moving the line of integration to $\operatorname{Re} s = -U$ and letting $U \to \infty$ picks up the residues 1 at $s = 0$ and $-1/y$ at $s = -1$, so $k(y) = 1 - 1/y$; if $y < 1$, moving it to the right gives $k(y) = 0$. As $|s(s+1)|^{-1}$ is integrable on $\operatorname{Re} s = 2$ and $\sum_n |a(n)|n^{-2} < \infty$, summation and integration may be exchanged, and the right side equals $\sum_n a(n)k(x/n) = S(a;x)$. □

Lemma 3.2. *Let* χ *be nonprincipal modulo* q*. For* $\operatorname{Re} s = \sigma > 0$, $|L(s,\chi)| \le q|s|/\sigma$.

Proof. Put $A(t) = \sum_{n \le t} \chi(n)$; then $|A(t)| \le q$, and partial summation gives $L(s,\chi) = s\int_1^\infty A(t)t^{-s-1}\,dt$ for $\sigma > 0$. Hence $|L(s,\chi)| \le q|s| \int_1^\infty t^{-\sigma-1}dt = q|s|/\sigma$. □

Lemma 3.3. *Let* $A_0 = \frac{3}{2}\sum_\ell \ell^{-3/2}$*. For every Dirichlet character* χ *the series* $E(s) = -\sum_\ell \log(1-\chi(\ell)\ell^{-s})$ *is holomorphic on* $\operatorname{Re} s > 1$*,* $e^{E(s)} = L(s,\chi)$ *there, and* $|E(s)| \le A_0$ *for* $\operatorname{Re} s \ge 3/2$*.*

Proof. For $\operatorname{Re} s \ge 1$ we have $|\chi(\ell)\ell^{-s}| \le 1/2$, and $|\log(1-w)| \le \sum_k |w|^k/k \le \frac{3}{2}|w|$ for $|w| \le 1/2$. The series therefore converges locally uniformly on $\operatorname{Re} s > 1$ and is bounded by A_0 on $\operatorname{Re} s \ge 3/2$, and the Euler product gives $e^E = L$. □

Lemma 3.4 (Borel–Carathéodory). *Let* h *be holomorphic on a neighbourhood of the disc* $|z-c| \le R$ *and let* $0 < r < R$*. Then*

$$\max_{|z-c|\le r} |h(z)-h(c)| \le \frac{2r}{R-r} \max_{|w-c|=R} \operatorname{Re}\big(h(w)-h(c)\big).$$

Proof. Put $g(z) = h(c+z) - h(c)$ and $A = \max_{|w|=R} \operatorname{Re} g(w)$. The mean of $\operatorname{Re} g$ over the circle $|w| = R$ is $\operatorname{Re} g(0) = 0$, so $A \ge 0$. If $A = 0$, then $\operatorname{Re} g$ vanishes on the circle, hence on the disc, and g is an imaginary constant equal to $g(0) = 0$. Let $A > 0$. By the maximum principle $\operatorname{Re} g \le A$ on $|z| \le R$, so

$|2A-g| \geq A$ there and $\phi = g/(2A-g)$ is holomorphic. Since $|g|^2 - |2A-g|^2 = 4A(\operatorname{Re} g - A) \leq 0$, we have $|\phi| \leq 1$, and $\phi(0) = 0$. Schwarz's lemma gives $|\phi(z)| \leq |z|/R$, and from $g = 2A\phi/(1+\phi)$ we get $|g(z)| \leq 2A|z|/(R-|z|)$. □

LEMMA 3.5. *Let $0 < \delta < 1/1000$, $1/2 \leq a \leq 1$, and let χ be nonprincipal modulo q with $L(s,\chi) \neq 0$ for $\operatorname{Re} s > a + 2\delta$. Put $K_1 = (1+2A_0)/(2\delta^2)$. For $a + 8\delta \leq \sigma \leq 2$ and $t \in \mathbb{R}$,*

$$\Big|\frac{L'}{L}(\sigma + it, \chi)\Big| \leq K_1 \log\big(2q(3+|t|)^2\big).$$

Proof. Let $c = 2 + it$, $R = 2 - a - 3\delta$, $r = 2 - a - 6\delta$ and $\Lambda_0 = \log(2q(3+|t|)^2) \geq \log 18 > 1$. On the disc $|z-c| < 2 - a - 2\delta$ we have $\operatorname{Re} z > a + 2\delta$, so $L(z,\chi)$ has no zero there and has a holomorphic logarithm h. On $|z-c| < 1/2$ both h and the function E of Lemma 3.3 are logarithms of L, so they differ by a constant in $2\pi i\mathbb{Z}$; we choose $h = E$ there. Then $|h(c)| \leq A_0$.

If $|z-c| \leq R$, then $\operatorname{Re} z \geq a + 3\delta \geq 1/2$ and $|z| \leq |c| + 3/2 \leq 4 + |t|$, so Lemma 3.2 gives $\operatorname{Re} h(z) = \log|L(z,\chi)| \leq \log(2q(4+|t|)) \leq \Lambda_0$. Lemma 3.4 gives, for $|z-c| \leq r$,

$$|h(z) - h(c)| \leq \frac{2r}{R-r}\big(\Lambda_0 + A_0\big) \leq \frac{\Lambda_0 + A_0}{\delta},$$

as $r \leq 3/2$ and $R - r = 3\delta$. Hence $|h(z)| \leq A_0 + (\Lambda_0 + A_0)/\delta \leq (1+2A_0)\Lambda_0/\delta$ on $|z-c| \leq r$. For $w = \sigma + it$ with $a + 8\delta \leq \sigma \leq 2$ we have $|w - c| = 2 - \sigma \leq r - 2\delta$, so the circle $|z - w| = 2\delta$ lies in $|z - c| \leq r$, and Cauchy's estimate gives $|L'/L(w,\chi)| = |h'(w)| \leq (1+2A_0)\Lambda_0/(2\delta^2)$. □

LEMMA 3.6. *For $q \geq 1$, $t \in \mathbb{R}$ and $s = \sigma + it$ with $\sigma \geq 1/2$,*

$$\log\big(2q(3+|t|)^2\big) \leq 10\log(q+2)(1+|t|)^{1/2}, \qquad |s(s+1)| \geq \tfrac{1}{8}(1+|t|)^2.$$

Proof. We have $\log(2q) \leq 2\log(q+2)$, and $\log y \leq 2y^{1/2}$ with $3 + |t| \leq 4(1+|t|)$ gives $\log(3+|t|) \leq 4(1+|t|)^{1/2}$; the first bound follows from $\log(q+2) \geq 1$. For the second, $|s| \geq \max(\sigma, |t|) \geq (1+|t|)/4$ and $|s+1| \geq \max(\sigma+1, |t|) \geq (1+|t|)/2$. □

PROPOSITION 3.7. *Let $0 < \delta < 1/1000$. There is $K = K(\delta)$ such that the following holds. Let $1/2 \leq a \leq 1$ and let χ be nonprincipal modulo q with $L(s,\chi) \neq 0$ for $\operatorname{Re} s > a + 2\delta$. Then for $x \geq 1$,*

$$|S(\chi; x)| \leq K\, x^{a+8\delta} \log(q+2).$$

Proof. Put $\sigma_0 = a + 8\delta$ and

$$\Phi(s) = -\frac{L'}{L}(s,\chi)\,\frac{x^s}{s(s+1)}.$$

For $\operatorname{Re} s > 1$, $-L'/L(s,\chi) = \sum_n \chi(n)\Lambda(n)n^{-s}$, so Lemma 3.1(ii) gives $S(\chi;x) = \frac{1}{2\pi}\int \Phi(2+it)\,dt$. The function Φ is holomorphic on $\sigma_0 \le \operatorname{Re} s \le 2$, and by Lemmas 3.5 and 3.6,

$$|\Phi(\sigma+it)| \le 80K_1 \log(q+2)\, x^{\sigma}(1+|t|)^{-3/2} \qquad (\sigma_0 \le \sigma \le 2).$$

This tends to 0 as $|t| \to \infty$ uniformly in σ, so Cauchy's theorem on the rectangles $[\sigma_0, 2] \times [-T, T]$, $T \to \infty$, moves the integral to $\operatorname{Re} s = \sigma_0$ (Figure 2). Since $\int_{\mathbb{R}}(1+|t|)^{-3/2}dt = 4$, we get $|S(\chi;x)| \le \frac{160K_1}{\pi} x^{\sigma_0}\log(q+2)$. □

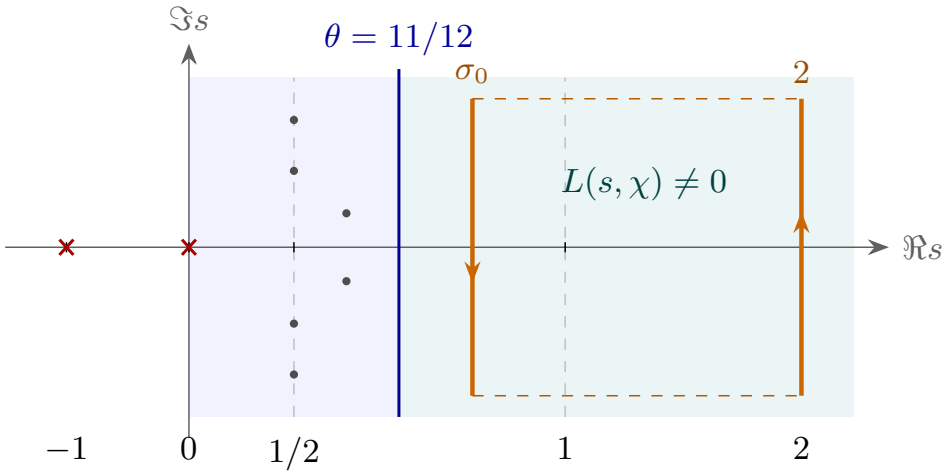


Figure 2. The contour shift in the proof of Proposition 3.7. The zeros of $L(s,\chi)$ lie in the strip $0 \le \operatorname{Re} s \le 11/12$ (dots), the kernel $x^s/(s(s+1))$ has poles at 0 and -1 (crosses), and the line of integration moves from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = \sigma_0$ inside the zero-free half-plane. Horizontal spacing is schematic.

From now on fix $\delta = 1/2000$ and $a = 11/12 - 1/1000$, so that $a + 2\delta = 11/12$ and

$$\sigma_0 = a + 8\delta = \frac{11}{12} + \frac{3}{1000},$$

and let K be the constant of Proposition 3.7 for this δ. By Theorem 1.2, Proposition 3.7 applies to every nonprincipal character.

Corollary 3.8. *There is an absolute constant C_3 such that for every nonprincipal character ξ modulo k:*

(i) $|S(\xi;x)| \le Kx^{\sigma_0}\log(k+2)$ *for* $x \ge 1$;
(ii) $\left|\sum_{p\le x}\xi(p)\log p\right| \le C_3 x^{49/50}\log(k+2)$ *for* $x \ge 1$;

(iii) *for integers* $2 \le A \le B$ *and* $\gamma \in \mathbb{R}$,

$$\Big|\sum_{A<p\le B} \xi(p)e(p\gamma)\Big| \le 2C_3\,\frac{B^{49/50}\log(k+2)(2+2\pi|\gamma|B)}{\log A}. \tag{3.1}$$

Proof. (i) is Proposition 3.7. For (ii) put $T(y) = yS(\xi;y)$, that is, $T(y) = \sum_{n\le y}\xi(n)\Lambda(n)(y-n)$, and let $0 < h \le x$. Then

$$T(x+h) - T(x) = h\sum_{n\le x}\xi(n)\Lambda(n) + \sum_{x<n\le x+h}\xi(n)\Lambda(n)(x+h-n),$$

and the last sum has at most $h+1$ terms, each of modulus at most $h\log(2x)$. With $|T(y)| \le Ky^{1+\sigma_0}\log(k+2)$ this gives

$$\Big|\sum_{n\le x}\xi(n)\Lambda(n)\Big| \le \frac{2K(2x)^{1+\sigma_0}\log(k+2)}{h} + (h+1)\log(2x).$$

Take $h = x^{47/50}$. Since $1+\sigma_0-47/50 < 49/50$ and $\log(2x) \le 1+100x^{1/100}$, the right side is at most $(8K+202)x^{49/50}\log(k+2)$. The prime powers contribute at most $\psi(x)-\theta(x) \le 2\sqrt{x}\log x \le 200x^{49/50}$ by Lemma 2.1(ii), and (ii) follows with $C_3 = 8K+402$.

(iii) Let $\Pi(u) = \sum_{p\le u}\xi(p)\log p$ and $f(u) = e(u\gamma)/\log u$. Partial summation gives

$$\sum_{A<p\le B}\xi(p)e(p\gamma) = f(B)\Pi(B) - f(A)\Pi(A) - \int_A^B \Pi(u)f'(u)\,du,$$

and $|f'(u)| \le 2\pi|\gamma|/\log A + 1/(u\log^2 u)$ for $u \ge A$. By (ii) the right side is at most $C_3B^{49/50}\log(k+2)\big(3+2\pi|\gamma|B\big)/\log A$, which gives (3.1). □

4. **Exponential sums with multiplicative coefficients**

For $M > 0$ put $w_M(\theta) = M$ if $\|\theta\| < 1/M$ and $w_M(\theta) = 1/\|\theta\|$ otherwise; then $0 \le w_M(\theta) \le M$ and $w_M(-\theta) = w_M(\theta)$.

THEOREM 4.1. *There is an absolute constant* C_1 *with the following property. Let* f *be completely multiplicative with* $|f(n)| \le 1$*, let* $\alpha \in \mathbb{R}$*, and let* a, r *be coprime integers with* $r \ge 1$*,* $|\alpha - a/r| \le r^{-2}$ *and* $2 \le R \le r \le N/R$*. Then*

$$\Big|\sum_{n\le N} f(n)e(n\alpha)\Big| \le C_1\Big(\frac{N}{\log N} + \frac{N(\log N)^2}{R^{1/4}}\Big).$$

Montgomery and Vaughan [9, Corollary 1] proved this for multiplicative f with $N(\log R)^{3/2}R^{-1/2}$ in place of $N(\log N)^2R^{-1/4}$. The proof below follows their approach, a bilinear decomposition together with the spacing of the multiples of α and an upper bound for prime pairs, with cruder estimates, which suffice for Theorem 1.1.

LEMMA 4.2. *Let a, r be coprime with $r \geq 2$ and $|\alpha - a/r| \leq r^{-2}$, and let $H \geq 0$ be an integer.*

(i) $\|k\alpha\| \geq 1/(2r)$ *for every integer k with $0 < |k| \leq r/2$.*

(ii) $\#\{1 \leq h \leq H : \|h\alpha\| < t\} \leq (4H/r + 1)(4rt + 1)$ *for every $t \geq 0$.*

(iii) *For $M > 0$ and every integer $K \geq 0$ with $M \leq 2^K$,*

$$\sum_{h \leq H} w_M(h\alpha) \leq \Big(\frac{4H}{r} + 1\Big)(3M + 4r + 8rK),$$

$$\sum_{h \leq H} w_M(h\alpha)^2 \leq \Big(\frac{4H}{r} + 1\Big)(20rM + 3M^2).$$

Proof. (i) Let z be the integer nearest to $k\alpha$. Since $0 < |k| < r$ and $(a, r) = 1$, $r \nmid ka$, so $|ka/r - z| \geq 1/r$ and $\|k\alpha\| \geq 1/r - |k|\,|\alpha - a/r| \geq 1/(2r)$.

(ii) Let $L = \lfloor r/2 \rfloor$, so $L \geq r/4$. For k in a block of L consecutive integers let $y_k = k\alpha - z_k$ with z_k the integer nearest to $k\alpha$. For $k \neq k'$ in the block, $0 < |k - k'| < L \leq r/2$ and $y_k - y_{k'} \equiv (k - k')\alpha \pmod 1$, so $|y_k - y_{k'}| \geq 1/(2r)$ by (i). Points of $(-t, t)$ at mutual distances at least $1/(2r)$ number at most $4rt + 1$, and $[1, H]$ is covered by at most $H/L + 1 \leq 4H/r + 1$ blocks.

(iii) If $\|\theta\| \geq 1/M$, let $k \geq 1$ be least with $\|\theta\| < 2^k/M$; then $k \leq K$ because $\|\theta\| \leq 1/2 < 2^K/M$, and $\|\theta\| \geq 2^{k-1}/M$, so $w_M(\theta) \leq 2^{1-k}M$. Hence

$$w_M(\theta)^j \leq M^j\,[\|\theta\| < 1/M] + \sum_{k=1}^{K} (2^{1-k}M)^j\,[\|\theta\| < 2^k/M] \qquad (j = 1, 2).$$

Summing over $h \leq H$ with (ii), the first sum is at most $(4H/r + 1)\big(4r + M + \sum_{k \leq K}(8r + 2^{1-k}M)\big)$ and the second at most $(4H/r + 1)\big(4rM + M^2 + \sum_{k \geq 1}(16rM2^{-k} + 4M^2 4^{-k})\big)$. □

LEMMA 4.3. *Let $\theta, \alpha \in \mathbb{R}$, $U > 0$, and let A, B, N be nonnegative integers with $B \leq A + U$.*

(i) $\big|\sum_{A < x \leq B} e(x\theta)\big| \leq w_U(\theta)$.

(ii) *For a finite set V of positive integers and $b : V \to \mathbb{C}$,*

$$\sum_{A < u \leq B} \Big| \sum_{\substack{v \in V \\ uv \leq N}} b(v) e(uv\alpha) \Big|^2 \leq \sum_{v, v' \in V} |b(v)|\,|b(v')|\,w_U\big((v - v')\alpha\big).$$

(iii) *For $M > 0$ and an integer $H \geq 0$, $\sum_{1 \leq m, m' \leq H} w_M((m - m')\alpha) \leq H\big(M + 2\sum_{h \leq H} w_M(h\alpha)\big)$.*

Proof. (i) If $\|\theta\| < 1/U$, the sum has at most U terms of modulus 1. Otherwise $\theta \notin \mathbb{Z}$, the sum has modulus at most $|e((B - A)\theta) - 1|/|e(\theta) - 1|$, and $|e(\theta) - 1| = 2|\sin \pi\theta| \geq 4\|\theta\|$, so the modulus is at most $1/(2\|\theta\|)$. (ii) Expanding the square gives $\sum_{v, v'} b(v)\overline{b(v')} \sum_u e(u(v - v')\alpha)$, where u runs over the integers of $(A, \min(B, N/v, N/v')]$, an interval of length at most U;

apply (i). (iii) For fixed m, the term $m' = m$ is $w_M(0) = M$, and each $h = |m - m'| \in [1, H]$ arises from at most two m'. □

LEMMA 4.4. *Let $C_5 = 1 + 4\zeta(3/2)$. For every completely multiplicative f with $|f(n)| \le 1$, every $\alpha \in \mathbb{R}$ and every integer $N \ge 1$, the sum $S = \sum_{n\le N} f(n)e(n\alpha)$ satisfies*

$$\Big|S\log N - \sum_{p\le N} \log p\, f(p) \sum_{m\le N/p} f(m)e(pm\alpha)\Big| \le C_5 N.$$

Proof. Write $S\log N = S_1 + S_2$ with $S_1 = \sum_{n\le N} f(n)e(n\alpha)\log n$ and $S_2 = \sum_{n\le N} f(n)e(n\alpha)\log(N/n)$. Then $|S_2| \le N\log N - \log N! \le N$, since $N^N/N! \le e^N$. In S_1 write $\log n = \sum_{d|n}\Lambda(d)$ and $n = dm$; complete multiplicativity gives $\sum_{d\le N}\Lambda(d)f(d)\sum_{m\le N/d} f(m)e(dm\alpha)$. The terms with d prime form the sum in the statement, and the others have modulus at most $N\sum_p\sum_{k\ge 2}\log p/p^k \le 2N\sum_p \log p/p^2 \le 4\zeta(3/2)N$, since $\log p \le 2\sqrt{p}$. □

Let $G(h) = \prod_{p|h} p/(p-1)$.

LEMMA 4.5. (i) *For every integer $X \ge 2$ and every even $h \ge 2$,*

$$\#\{0 \le n < X : n \text{ and } n+h \text{ prime}\} \le \frac{400\,G(h)^2 X}{(\log X)^2} + 3X^{3/5}.$$

(ii) $\sum_{h\le X} G(h)^4 \le e^{30}X$ *for every $X \ge 1$.*

Proof. (i) Apply Lemma 2.2 to $a_n = n(n+h)$, $0 \le n < X$, with $y = X^{1/5}$, $z = \lfloor y\rfloor$, P the product of the primes $p \le z$ with $p \nmid h$, and $\nu(d) = 2^{\omega(d)}/d$. Every $p \mid P$ is odd, so $\nu(p) = 2/p < 1$. For $d \mid P$ the residues n with $d \mid n(n+h)$ form $2^{\omega(d)}$ classes modulo d (namely $n \equiv 0$ or $-h \pmod p$ for each $p \mid d$, distinct as $p \nmid h$), and each class contains between $X/d-1$ and $X/d+1$ of the integers $0 \le n < X$; hence $|R_d| \le 2^{\omega(d)}$. Since the prime factors of d are at least 3, $3^{\omega(d)}|R_d| \le 6^{\omega(d)} \le d^2 \le y^2$, and the error term is at most $y^3 = X^{3/5}$.

Let $\mathcal{M}$ be the set of $m \le \sqrt{y}$ whose prime factors divide P. Since $\nu(p)/(1-\nu(p)) = \sum_{k\ge 1}\nu(p)^k$ and $\nu(m) = 2^{\Omega(m)}/m$ for these m, $g(l)$ is the sum of $2^{\Omega(m)}/m$ over the m whose set of prime factors is that of l. An $m \in \mathcal{M}$ is counted in $g(l)$ for l the product of its prime factors, and $l \le m \le \sqrt{y}$. Hence, with $\tau(m) \le 2^{\Omega(m)}$ the number of divisors of m,

$$S \ge \sum_{m\in\mathcal{M}} \frac{2^{\Omega(m)}}{m} \ge \sum_{m\in\mathcal{M}} \frac{\tau(m)}{m}.$$

Let $T = \lfloor y^{1/4}\rfloor$ and let B be the set of $b \le T$ that are prime to h; these b are odd. For $b, b' \in B$ the product bb' lies in $\mathcal{M}$, and a given m is a product bb' in at most $\tau(m)$ ways, so $S \ge \big(\sum_{b\in B} 1/b\big)^2$. If E is a set of primes and $p \notin E$, every $b \le T$ prime to E is uniquely $p^k b'$ with b' prime to $E \cup \{p\}$, so the sum

of $1/b$ over the $b \le T$ prime to E is at most $p/(p-1)$ times the same sum for $E \cup \{p\}$. Adding the prime factors of h one at a time gives

$$\sum_{b\in B} \frac{1}{b} \ge \frac{1}{G(h)} \sum_{b\le T} \frac{1}{b} \ge \frac{\log(T+1)}{G(h)}.$$

As $T+1 > y^{1/4} = X^{1/20}$, we get $S \ge (\log X)^2/(400G(h)^2)$, that is, $X/S \le 400G(h)^2 X/(\log X)^2$. Finally, if $n < X$ and both n and $n+h$ are prime, then $n \le z$ or $(n(n+h), P) = 1$, so the count is at most $z + 1 + X/S + y^3 \le 400G(h)^2 X/(\log X)^2 + 3X^{3/5}$.

(ii) For $t = p-1 \ge 1$, $(t+1)^5 \le (t+31)t^4$, that is, $(p/(p-1))^4 \le 1 + 30/p$. Hence $G(h)^4 \le \sum_d 30^{\omega(d)}/d$ over the squarefree $d \mid h$, and

$$\sum_{h\le X} G(h)^4 \le \sum_{d\le X} \mu(d)^2 \frac{30^{\omega(d)}}{d} \cdot \frac{X}{d} \le X \prod_p \Big(1 + \frac{30}{p^2}\Big)$$
$$\le X \exp\Big(30 \sum_{n\ge 2} \frac{1}{n^2}\Big) \le e^{30} X. \qquad \square$$

Proof of Theorem 4.1. Let S be the sum in the theorem, $L = \log N$ and $\ell = \lfloor \log_2 N \rfloor$. Since $N \ge rR \ge 4$, we have $L \ge 1$. If $\ell < 64$, then $L < 64 \log 2 < 45$ and $|S| \le N \le 45N/L$. Let now $\ell \ge 64$, and put $I = \lfloor \log_2 \ell \rfloor + 1$ and $J = \ell + 1 - 4I$. Then $16^I > \ell^4$, $1 \le J \le \ell \le 2L$, $I \le \ell \le 2L$, $2^I \le 2\ell$, $N < 2^{\ell+1} = 2^J 16^I$ and $J^2 2^J \le (\ell+1)^2 2^{\ell+1} \ell^{-4} \le 2N$. By Lemma 4.4, $|S|L \le C_5 N + |T_1| + |T_2|$, where T_1 and T_2 are the parts of $\sum_p \log p\, f(p) \sum_{m\le N/p} f(m) e(pm\alpha)$ with $p \le 2^J$ and with $p > 2^J$.

The small primes. Let $0 \le j < J$, $U = 2^j$, $H = \lfloor N/U \rfloor$, and for $U < x \le 2U$ let $B(x) = \sum_{m\le H,\, xm\le N} f(m) e(xm\alpha)$; the inner sum for a prime $p \in (U, 2U]$ is $B(p)$. By Cauchy's inequality, Lemma 4.3(ii), (iii) and Lemma 4.2(iii) with $M = U$, $K = j$,

$$\sum_{U<p\le 2U} \log p\, |B(p)| \le \big(U \log^2(2U)\big)^{1/2} Q_j^{1/2},$$

where $Q_j = H\big(U + 2(4H/r + 1)(3U + 4r + 8rj)\big)$. Since $HU \le N$, $UQ_j \le 7NU + (8+16j)Nr + 24N^2/r + (32+64j)N^2/U$, and $\log(2U) \le j+1$. Summing over $j < J$ with $j+1 \le J$, $\sum_{j<J} 2^{j/2} \le 3 \cdot 2^{J/2}$, $\sqrt{(32+64j)2^{-j}} \le 8(j+1)(3/4)^j$ and $\sum_{j\ge 0} (j+1)^2 (3/4)^j = 112$,

$$|T_1| \le 3\sqrt{7} J \sqrt{N 2^J} + J^2 N \sqrt{24/r} + 896N + J^2 \sqrt{24JNr}.$$

Now $J^2 2^J \le 2N$ gives $3\sqrt{7} J \sqrt{N 2^J} \le 12N$; $r \ge R \ge \sqrt{R}$ gives $\sqrt{24/r} \le 5R^{-1/4}$; and $r\sqrt{R} \le rR \le N$ gives $\sqrt{24JNr} \le 5JNR^{-1/4}$. With $J \le 2L$ and $L \ge 1$, $|T_1| \le 908N + 60L^3 NR^{-1/4}$.

The large primes. For $p > 2^J$ we have $N/p < 16^I$, so the m in T_2 lie in $(0, 16^I - 1]$, the union of the blocks $(16^i - 1, 16^{i+1} - 1]$ with $0 \le i < I$. Fix i, let

$A = 16^i - 1$, $U = 15 \cdot 16^i$ and $X = \lfloor N/16^i \rfloor$, and assume $X \geq 1$, the block being empty otherwise. Only the primes $p \in (2^J, X]$ contribute, since $p \geq X+1$ and $m \geq 16^i$ give $pm > N$. Let V be the set of these primes, which are odd, and $B(m) = \sum_{p \in V,\, pm \leq N} \log p \, f(p) e(pm\alpha)$. The block is $\sum_{A < m \leq A+U} f(m) B(m)$, and by Cauchy's inequality, Lemma 4.3(ii) and $\log p \leq \lambda = \log(X+1)$, its square is at most $U\lambda^2 W$, where

$$W = \sum_{p,p' \in V} w_U((p-p')\alpha) = |V| U + 2 \sum_{h=1}^{X} w_U(h\alpha) \, \#\{p \in V : p + h \in V\}.$$

The count vanishes for odd h, and for even h Lemma 4.5(i), applied to $X+1$, bounds it by $400 G(h)^2 (X+1)\lambda^{-2} + 3(X+1)^{3/5}$. By Cauchy's inequality, Lemma 4.5(ii) and Lemma 4.2(iii), $\sum_{h \leq X} G(h)^2 w_U(h\alpha) \leq e^{15} \sqrt{X} Q^{1/2}$ with $Q = (4X/r + 1)(20rU + 3U^2)$, and $\sum_{h \leq X} w_U(h\alpha) \leq XU$. Hence $U\lambda^2 W \leq A_1 + 800 e^{15} Z + A_3$, where

$$A_1 = U^2 \lambda^2 X, \qquad Z = U(X+1)\sqrt{X} Q^{1/2}, \qquad A_3 = 6U^2 \lambda^2 (X+1)^{3/5} X.$$

Let $\kappa = \log(N+1) \geq \lambda$ and $T_i = 4^i \kappa \, (N(2N)^{3/5})^{1/2}$. Since $U^2 X = 225 \cdot 16^i (16^i X) \leq 225 \cdot 16^i N$ and $X + 1 \leq 2N$, we get $A_1 \leq 225 T_i^2$ and $A_3 \leq 1350 T_i^2 \leq 37^2 T_i^2$. With $w = UX \leq 15N$, $X \leq N/16^i$ and $R \leq r \leq N/R$,

$$\begin{aligned} Z^2 \leq 4U^2 X^3 Q &= 320 w^3 X + \frac{48 w^4}{r} + 80 r w^3 + 12 U w^3 \\ &\leq \frac{320 \cdot 15^3 N^4}{16^i} + \frac{48 \cdot 15^4 N^4}{R} + \frac{80 \cdot 15^3 N^4}{R} \\ &\quad + 180 \cdot 15^3 \cdot 16^i N^3, \end{aligned}$$

and every coefficient is at most 40^4, so $Z^{1/2} \leq 40(N 2^{-i} + 2NR^{-1/4} + 2^i N^{3/4})$. The block is therefore at most

$$\begin{aligned} &15 T_i + 40\sqrt{800 e^{15}} \big(N 2^{-i} + 2NR^{-1/4} + 2^i N^{3/4}\big) + 37 T_i \\ &\quad \leq K_b \big(N 2^{-i} + NR^{-1/4} + 2^i N^{3/4} + T_i\big), \end{aligned}$$

with $K_b = 80\sqrt{800 e^{15}} + 52$. Summing over $i < I$,

$$|T_2| \leq K_b \big(2N + INR^{-1/4} + 2^I N^{3/4} + 4^I \kappa (N(2N)^{3/5})^{1/2}\big).$$

Here $I \leq 2L^3$; $2^I \leq 4L \leq 16 N^{1/4}$; $4^I \leq 16 L^2$ and $\kappa \leq 2L$; and $\log x \leq 15 x^{1/15}$ gives $L^3 (N(2N)^{3/5})^{1/2} \leq 2 \cdot 15^3 N$. Hence $|T_2| \leq K_b (216018 N + 2L^3 N R^{-1/4})$.

Altogether $|S| L \leq (C_5 + 908 + 216018 K_b) N + (60 + 2K_b) L^3 N R^{-1/4}$, and dividing by L gives the theorem with $C_1 = 45 + C_5 + 908 + 216018 K_b + 60 + 2K_b$. □

5. Proof of Theorem 1.1

In this section χ is a primitive character modulo $q \geq 3$.

LEMMA 5.1. *Let $B \geq 0$ be such that $\left|\sum_{m\leq H} \bar{\chi}(m)e(m\beta)/m\right| \leq B$ for every integer $H \leq q/2$ and every $\beta \in \mathbb{R}$. Then $\left|\sum_{n\leq x} \chi(n)\right| \leq \sqrt{q}\,(B+6)$ for every $x \geq 0$.*

Proof. We may take $x = T$ to be an integer. Let $\tau = \sum_{a \bmod q} \bar{\chi}(a)e(a/q)$. Since $\bar{\chi}$ is primitive, $\sum_a \bar{\chi}(a)e(an/q) = \chi(n)\tau$ for every n, and summing the squares of the moduli of these identities over n mod q gives $q\varphi(q) = |\tau|^2\varphi(q)$, so $|\tau| = \sqrt{q}$. Summing over $1 \leq n \leq T$,

$$\tau \sum_{n\leq T} \chi(n) = \sum_{0<|a|\leq q/2} \bar{\chi}(a)\,\frac{e(aT/q)-1}{1-e(-a/q)},$$

where a runs over the integers with $-q/2 < a \leq q/2$, $a \neq 0$. For $0 < |\theta| \leq 1/2$ put $r(\theta) = (1-e(-\theta))^{-1} - (2\pi i\theta)^{-1}$; then $|1-e(-\theta)| = 2|\sin\pi\theta| \geq 4|\theta|$. If $2\pi|\theta| \leq 1$, then $|e(-\theta)-1+2\pi i\theta| \leq (2\pi\theta)^2$ and $|r(\theta)| \leq (2\pi\theta)^2/(4|\theta|\cdot 2\pi|\theta|) < 3$; otherwise $|r(\theta)| \leq 1/(4|\theta|) + 1/(2\pi|\theta|) < 3$. The terms with $r(a/q)$ contribute at most $6q$. Writing $a = \pm m$ with $1 \leq m \leq q/2$, the remaining terms are $q/(2\pi i)$ times a combination, with coefficients of modulus at most 1, of four sums $\sum_{m\leq H} \bar{\chi}(m)e(m\beta)/m$ with $H \leq q/2$, so they contribute at most $2qB/\pi$. Dividing by $|\tau|$ gives the claim. □

LEMMA 5.2. *For $Y \geq 2$ and $H \geq 1$,*

$$\sum_{\substack{n\leq H\\ P(n)\leq Y}} \frac{1}{n} \leq e^{11}\log Y.$$

Proof. The sum is at most $\prod_{p\leq Y}(1-1/p)^{-1} \leq \exp\bigl(\sum_{p\leq Y}\frac{1}{p-1}\bigr)$, and $\sum_p\bigl(\frac{1}{p-1}-\frac{1}{p}\bigr) \leq 1$; apply Lemma 2.1(iv). □

LEMMA 5.3. *There is an absolute constant C_2 such that for $0 < r < q$, $b \in \mathbb{Z}$, $\beta \in \mathbb{R}$, integers $Y \geq \max(2,r)$ and $t \geq 1$,*

$$\Bigl|\sum_{\substack{n\leq t\\ P(n)>Y}} \chi(n)e\Bigl(n\Bigl(\frac{b}{r}+\beta\Bigr)\Bigr)\Bigr| \leq C_2\,\frac{r\log(qr+2)\,(2+2\pi|\beta|t)\,t}{Y^{1/50}\log Y}.$$

Proof. Every $n \leq t$ with $P(n) > Y$ is uniquely $n = pm$ with $p = P(n)$ prime, $Y < p \leq t/m$ and $P(m) \leq p$, and then $m \leq t/Y$. For such p we have $p > r$, so $p \nmid r$, and on the units modulo r the function $u \mapsto e(umb/r)$ equals $\sum_\psi c_\psi\psi(u)$, the sum over the characters ψ modulo r, with $|c_\psi| \leq 1$.

The character $\chi\psi$ modulo qr is nonprincipal. Otherwise $\chi(n) = \bar{\psi}(n)$ for every n prime to qr. Let $d = (q,r)$ and let n be prime to q with $n \equiv 1 \pmod d$.

By the Chinese remainder theorem there is n' prime to qr with $n' \equiv n \pmod q$ and $n' \equiv 1 \pmod r$, and then $\chi(n) = \chi(n') = \bar{\psi}(n') = 1$. So χ is trivial on the units congruent to 1 modulo d, and is induced by a character modulo $d \le r < q$, which contradicts primitivity.

Applying (3.1) to $\xi = \chi\psi$ with $\gamma = m\beta$, $A = \max(Y, P(m) - 1)$ and $B = \lfloor t/m \rfloor$ (the inner sum is empty if $A \ge B$) gives

$$\Big| \sum_{\substack{n \le t \\ P(n) > Y}} \chi(n) e\Big(n\Big(\frac{b}{r} + \beta\Big)\Big) \Big| \le 2C_3 r \, \frac{\log(qr+2)(2+2\pi|\beta|t)}{\log Y} \sum_{m \le t/Y} \Big(\frac{t}{m}\Big)^{49/50},$$

and $\sum_{m \le M} m^{-49/50} \le 50 M^{1/50}$. Hence $C_2 = 100 C_3$ will do. □

Proposition 5.4. *There is an absolute constant C_4 such that if $\log q \ge 200$, then $\big|\sum_{m \le H} \chi(m) e(m\beta)/m\big| \le C_4 \log\log q$ for every integer $H \le q$ and every $\beta \in \mathbb{R}$.*

Proof. Let $L = \log q$, $M = \lceil L \rceil$, $R = M^{16}$ and $Y = M^{1800}$. Since $e^{L/16} \ge (L/16)^4/24 > L + 1 \ge M$, we have $2 \le R < q$ and $R \le Y$. Split the sum according to $P(m) \le Y$ or $P(m) > Y$. By Lemma 5.2 the first part is at most $e^{11} \log Y$. For the second put $A(t) = \sum_{n \le t,\, P(n) > Y} \chi(n) e(n\beta)$ and $w(t) = 1/t - 1/(t+1)$. Partial summation and $|A(H)| \le H$ give

$$\Big| \sum_{\substack{m \le H \\ P(m) > Y}} \frac{\chi(m) e(m\beta)}{m} \Big| \le 1 + \sum_{1 \le t < H} |A(t)| \, w(t).$$

If $t \le Y$, then $A(t) = 0$. Let $t > Y$ and $Q = \lfloor t/R \rfloor \ge 1$. By Dirichlet's theorem there are coprime a, r with $1 \le r \le Q$ and $|\beta - a/r| \le 1/(r(Q+1))$; then $|\beta - a/r|\, t \le R$ (Figure 3). If $r \le R$, then Lemma 5.3 with $b = a$ gives $|A(t)| \le C_2 k t$, where

$$k = \frac{R \log(qR+2)(2+2\pi R)}{Y^{1/50} \log Y}.$$

If $r > R$, then $R \le r \le t/R$ and $|\beta - a/r| \le r^{-2}$, so Theorem 4.1 for $f = \chi$ bounds the full sum $\sum_{n \le t} \chi(n) e(n\beta)$, and since $\log t \le L$, removing the $\Psi(t) = \#\{n \le t : P(n) \le Y\}$ smooth terms gives

$$|A(t)| \le C_1\Big(\frac{t}{\log t} + t\rho\Big) + \Psi(t), \qquad \rho = \frac{L^2}{R^{1/4}} = \frac{L^2}{M^4}.$$

Since $t w(t) \le 1/t$ and $w(t) \le 1/t^2$,

$$\sum_{1 \le t < H} |A(t)| \, w(t) \le C_1 \sum_{2 \le t \le H} \frac{1}{t \log t} + (C_1\rho + C_2 k) \sum_{t \le H} \frac{1}{t} + \sum_{1 \le t < H} \Psi(t) w(t).$$

The first sum is at most $3 + \log L$, the second at most $1 + L$, and by partial summation the third is at most $\sum_{n \le H,\, P(n) \le Y} 1/n \le e^{11} \log Y$. Finally $L \le M \le$

$2L$ gives $\rho(1+L) \le 2/M \le 1$; $\log(qR+2) \le 18M$, $2+2\pi R \le 8M^{16}$, $Y^{1/50} = M^{36}$ and $\log Y = 1800\log M \ge 1800$ give $k(1+L) \le 288M^{34}/(1800M^{36}) \le 1$; and $\log Y = 1800\log M \le 3600\log L$. Collecting terms, the sum is at most $1 + 4C_1 + C_2 + (7200e^{11} + C_1)\log L \le (7200e^{11} + 1 + 5C_1 + C_2)\log L$. □

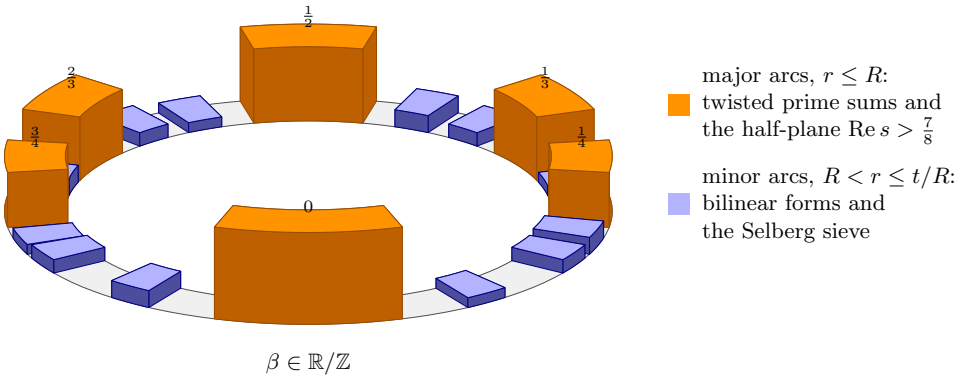


Figure 3. The dissection of $\mathbb{R}/\mathbb{Z}$ in the proof of Proposition 5.4, drawn for $R = 4$. Near a fraction a/r with $r \le R$ (major arcs) the sum $A(t)$ is controlled by the twisted prime sums of Corollary 3.8; elsewhere (minor arcs) by Theorem 4.1. The heights only indicate that the major arcs carry the large values.

Proof of Theorem 1.1. The character $\bar\chi$ is primitive modulo q. If $\log q \ge 200$, then Proposition 5.4 for $\bar\chi$ and Lemma 5.1 give

$$\Big|\sum_{n\le x} \chi(n)\Big| \le \sqrt{q}\,(C_4 \log\log q + 6) \le (C_4+6)\sqrt{q}\log\log q.$$

If $\log q < 200$, then every sum $\sum_{m\le H} \bar\chi(m)e(m\beta)/m$ with $H \le q$ has modulus at most $1 + \log q < 201$, and Lemma 5.1 gives

$$\Big|\sum_{n\le x} \chi(n)\Big| \le 207\sqrt{q} \le \frac{207}{\log\log 3}\sqrt{q}\log\log q.$$ □

6. The logarithmic derivative at $s = 1$

Recall that $\sigma_0 = 11/12 + 3/1000$ and that K is the constant of Corollary 3.8.

Lemma 6.1. *For every nonprincipal character χ modulo q,*

$$-\frac{L'}{L}(1,\chi) = 2\int_1^\infty \frac{S(\chi;t)}{t^2}\,dt,$$

and the integral converges absolutely.

Proof. By Lemma 3.1(i) with $a = \chi\Lambda$, for real $s > 1$,

$$\int_1^\infty S(\chi;t)\, t^{-s-1}\, dt = -\frac{L'}{L}(s,\chi)\,\frac{1}{s(s+1)}.$$

As $s \to 1^+$ the left side tends to $\int_1^\infty S(\chi;t)t^{-2}dt$ by dominated convergence, since $|S(\chi;t)| \le Kt^{\sigma_0}\log(q+2)$ and $\sigma_0 < 1$. The function $L(s,\chi)$ is entire and does not vanish at $s = 1$ by Theorem 1.2, so the right side tends to $-\frac{1}{2}\frac{L'}{L}(1,\chi)$. □

Proof of Theorem 1.3(i). Let χ be nonprincipal modulo q, so $q \ge 3$, and put $Y = \big(\log(q+2)\big)^{1/(1-\sigma_0)} \ge 1$. Since $|S(\chi;t)| \le \psi(t) \le 4t$ by Lemma 2.1(ii), Lemma 6.1 gives

$$\Big|\frac{L'}{L}(1,\chi)\Big| \le 2\int_1^Y \frac{4\,dt}{t} + 2K\log(q+2)\int_Y^\infty t^{\sigma_0-2}\,dt = \frac{8\log\log(q+2)+2K}{1-\sigma_0}.$$

As $q+2 \le q^2$, $\log\log(q+2) \le \log 2 + \log\log q$, and since $\log\log q \ge \log\log 3 > 0$ the right side is at most $C\log\log q$. □

For the lower bound in Theorem 1.3(ii) we need to control the prime powers $n \le q^{25}$ with $n \equiv 1 \pmod q$. We count them in dyadic blocks with Selberg's sieve.

LEMMA 6.2. *Let q be prime, let $M \ge 1$ be an integer and let $1 < z < q$. The number of m with $M \le m < 2M$ for which $1 + mq$ is a prime power is at most*

$$\frac{2M}{\log z} + z(1+\log z)^2 + z.$$

Proof. If $1 + mq = p^k$ with $p \le z$, then $p^k \in (Mq, 2Mq)$, and for each prime p at most one power of p lies in this interval, since $p^{k+1} \ge 2p^k$. These m number at most z. If $p > z$, then $1 + mq$ has no prime factor up to z, and we apply Lemma 2.2 to the M integers $a_m = 1 + mq$, $M \le m < 2M$, with P the product of the primes up to z, $\nu(d) = 1/d$ and $y = z$. Every prime $p \le z$ differs from q, so for $d \mid P$ the condition $d \mid 1 + mq$ fixes m in one residue class modulo d, and $|R_d| \le 1$. Here $g(l) = \prod_{p\mid l}(p-1)^{-1}$, which is the sum of $1/n$ over the n whose set of prime factors is that of l. Every $n \le \sqrt{z}$ has all its prime factors at most z and the product of its distinct prime factors is at most $\sqrt{z}$, so

$$S = \sum_{\substack{l\mid P\\ l\le\sqrt{z}}} g(l) \ge \sum_{n\le\sqrt{z}} \frac{1}{n} \ge \frac{\log z}{2}.$$

For squarefree d, $3^{\omega(d)} = \tau_3(d)$, and $\sum_{d\le z}\tau_3(d) = \#\{(a,b,c) : abc \le z\} \le z\big(\sum_{a\le z} 1/a\big)^2 \le z(1+\log z)^2$. Lemma 2.2 therefore bounds the number of these m by $2M/\log z + z(1+\log z)^2$. □

LEMMA 6.3. *There is an absolute constant C_0 such that for every prime $q \geq 3$,*

$$\sum_{\substack{1<n\leq q^{25}\\ n\equiv 1 \pmod q}} \frac{\Lambda(n)}{n} \leq C_0 \frac{\log q \log\log q}{q}.$$

Proof. Write $n = 1 + mq$ with $1 \leq m < q^{24}$, and let $J = \lfloor 24 \log_2 q \rfloor + 1$, so that $2^J > q^{24}$ and $J \leq 36 \log q$. We split the range of m into the blocks $2^j \leq m < 2^{j+1}$, $0 \leq j < J$ (Figure 4), and let N_j be the number of m in the j-th block with $1 + mq$ a prime power. We claim $N_j \leq 600 \cdot 2^j/(j+1)$. For $j < 48$ this holds trivially. For $j \geq 48$ apply Lemma 6.2 with $M = 2^j$ and $z = 2^u$, $u = j/48 \geq 1$. Since $2^j < 2^J \leq 2q^{24}$, we have $z < 2^{1/48} q^{1/2} < q$. Then $2M/\log z = 96 \cdot 2^j/(j \log 2) \leq 150 \cdot 2^j/(j+1)$, and since $\log z \leq u$,

$$z(1+\log z)^2 + z \leq 2^u (2+u)^2 \leq \frac{441 \cdot 2^{4u}}{48u+1} \leq \frac{441 \cdot 2^j}{j+1},$$

because $(2+u)^2(48u+1) \leq 9u^2 \cdot 49u = 441u^3$ and $u^3 \leq 2^{3u}$. This proves the claim.

A term of the j-th block has $\Lambda(n) \leq \log(1 + 2^{j+1}q) \leq (j+2)\log 2 + \log q$ and $n > 2^j q$, so the block contributes at most

$$\frac{600}{(j+1)q}\big((j+2)\log 2 + \log q\big) \leq \frac{600}{q}\Big(2\log 2 + \frac{\log q}{j+1}\Big).$$

Summing over $j < J$ and using $\sum_{j<J} 1/(j+1) \leq 1 + \log J$, $J \leq 36 \log q$ and $\log J \leq 4 + \log\log q$, the sum in the lemma is at most $\frac{600}{q}\big(51 \log q + 5 \log q + \log q \log\log q\big)$. Since $\log\log q \geq \log\log 3 > 1/11$, this is at most $C_0 \log q \log\log q/q$ with $C_0 = 600 \cdot 617$. □

Proof of Theorem 1.3(ii). Let $q \geq 3$ be prime and put $c(n) = \sum_{\chi \neq \chi_0} \chi(n)$, the sum over the nonprincipal characters modulo q. By orthogonality $c(n) = (q-1)\mathbf{1}[n \equiv 1 \pmod q] - \mathbf{1}[q \nmid n]$. Put

$$T(t) = \sum_{\chi \neq \chi_0} S(\chi; t) = \sum_{n \leq t} c(n)\Lambda(n)\Big(1 - \frac{n}{t}\Big), \qquad I = \int_1^\infty \frac{T(t)}{t^2}\, dt.$$

By Lemma 6.1, $\sum_{\chi \neq \chi_0} \frac{L'}{L}(1, \chi) = -2I$, so by (1.1) it suffices to show $-C \log q \leq I \leq C \log q \log\log q$.

By Corollary 3.8(i), $|T(t)| \leq qK \log(q+2)\, t^{\sigma_0}$. Let $Y = q^{25}$. Since $25(\sigma_0 - 1) < -2$ and $\log(q+2) \leq q$,

$$\Big|\int_Y^\infty \frac{T(t)}{t^2}\, dt\Big| \leq \frac{qK \log(q+2)\, Y^{\sigma_0 - 1}}{1 - \sigma_0} \leq \frac{K \log(q+2)}{q(1-\sigma_0)} \leq \frac{K}{1-\sigma_0}.$$

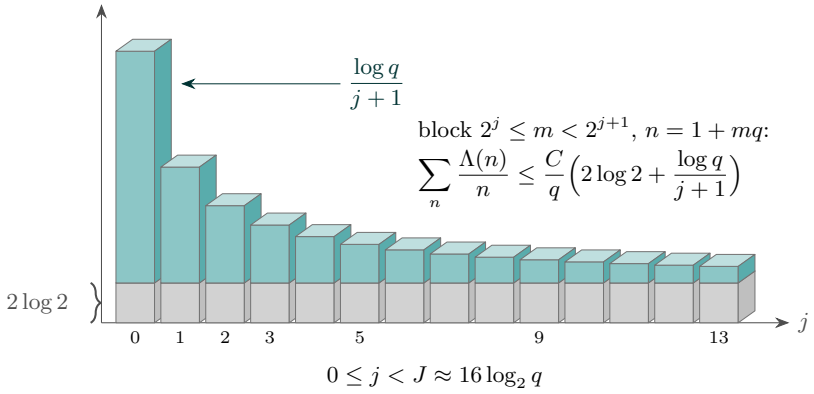


Figure 4. The dyadic blocks in the proof of Lemma 6.3. The block $2^j \le m < 2^{j+1}$ contributes a constant part, of total size $\ll J \ll \log q$, and a part of size $\log q/(j+1)$, whose total is $\ll \log q \log J$. The second part produces the factor $\log\log q$.

For the lower bound, $c(n) \ge -1$ gives $-T(t) \le \psi(t) \le 4t$, so

$$-I \le \int_1^Y \frac{4\,dt}{t} + \frac{K}{1-\sigma_0} = 100\log q + \frac{K}{1-\sigma_0}.$$

For the upper bound, $c(n) \le (q-1)\mathbf{1}[n \equiv 1 \pmod q]$ gives

$$\int_1^Y \frac{T(t)}{t^2}\,dt \le (q-1) \sum_{\substack{n \le Y \\ n \equiv 1 \pmod q}} \Lambda(n) \int_n^\infty \Big(1 - \frac{n}{t}\Big)\frac{dt}{t^2}$$

$$= \frac{q-1}{2} \sum_{\substack{1 < n \le q^{25} \\ n \equiv 1 \pmod q}} \frac{\Lambda(n)}{n},$$

which is at most $\frac{C_0}{2}\log q \log\log q$ by Lemma 6.3. Hence

$$I \le \frac{C_0}{2}\log q \log\log q + \frac{K}{1-\sigma_0},$$

and both bounds follow since $\log q > 1$ and $\log\log q > 1/11$. □

Appendix A. The included zero-free argument: source, statement and attribution

This appendix reproduces the complete 49-page repository document *The Quasi-Riemann Hypothesis*, dated 5 October 2026, by OpenAI. The document proves nonvanishing on $\operatorname{Re} s > 11/12$ for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and transfers this to Dirichlet L-functions. Its original page labels

are retained, so references below to its sections, propositions and pages can be checked directly.

The copy is pinned to the official repository commit `1581d4cbdd69a0faf34b2089614fe2a814316a6a`; its SHA-256 is `f919b57829b178c8e60e7c17b018cf773e7907cf642ef5a3347d8a826e8dbf18`. It is included under the repository's Apache License 2.0; the source, license and provenance notice are retained in the accompanying files. No wording or formula in the reproduced document has been changed. OpenAI's attribution and original reference list remain visible. Its incidental applications are not presented as new results of this article.

The logical chain used here is specific. The reproduced Proposition 3.1 bounds the mean square of a family of twisted Möbius sums. Section 4 reduces that statement to an auxiliary norm estimate. Section 5 proves the latter by cube inversion, a transfer estimate and iteration; Sections 6 and 7 and Appendices A and B supply the estimates and identities used in that iteration. Section 3 obtains cancellation in a fixed Möbius sum and then continues its Mellin transform to exclude zeros. The Dirichlet statement follows by quadratic base change and restoration of the finitely omitted Euler factors. These parts of the argument, rather than the stronger 7/8 statement mentioned in its introduction, are the input to the present article.

Including the complete argument removes an omitted *external proof* from this review document. It does not turn the OpenAI argument into an independent discovery by the present authors. Nor does reproduction alone certify every proof step: its analytic estimates remain part of the mathematics that a referee must assess. The separately recorded formal audit and the exact parameter audit have the scopes stated below.

The Quasi-Riemann Hypothesis

OpenAI

October 5, 2026

Abstract

We establish the quasi-Riemann hypothesis by proving that every Dirichlet L-function, including Riemann's zeta function, has no zeros in the half-plane $\operatorname{Re} s > 11/12$. More generally, we prove the same zero-free half-plane for every finite-order Hecke L-function over $K = \mathbb{Q}(\sqrt{-3})$. In particular, this rules out the existence of Landau–Siegel zeros.

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1 Introduction

For a primitive Dirichlet character χ of conductor q, the associated *Dirichlet L-function* is defined for $\operatorname{Re} s > 1$ by the formula

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s}, \qquad \operatorname{Re} s > 1,$$

and admits a meromorphic continuation to $s \in \mathbb{C}$ [22, §4.6]. When χ is the trivial character, $L(s,\chi)$ recovers the Riemann zeta function $\zeta(s)$ [41].

The zeros of Dirichlet L-functions are of significant interest, such as for their role in governing the distribution of primes in arithmetic progressions. The *Generalized Riemann Hypothesis* predicts that all zeros of $L(s,\chi)$ in the critical strip $0 < \operatorname{Re} s < 1$ satisfy $\operatorname{Re} s = 1/2$. For $\zeta(s)$, the weaker assertion that there exists $\varepsilon > 0$ such that $\zeta(s)$ has no zeros in $\operatorname{Re} s > 1 - \varepsilon$ is called a *quasi-Riemann Hypothesis* (e.g., in [2, §1], [34, p. 274], and [4, §2]). For Dirichlet L-functions, we consider the analogous assertion with a single $\varepsilon > 0$ valid for every primitive character χ, independently of both conductor and height.

Let $K = \mathbb{Q}(\sqrt{-3})$. The main result of this paper is the following.

Theorem 1.1. *Every finite-order Hecke L-function over K has no zeros in the half-plane* $\operatorname{Re} s > 11/12$. *Thus every Dirichlet L-function, including* $\zeta(s)$, *has no zeros for* $\operatorname{Re} s > 11/12$.

Section 3 proves the theorem assuming Proposition 3.1, whose proof is completed in Section 5.6.

Theorem 1.1 is a natural intermediate step toward the stronger zero-free region $\operatorname{Re} s > 7/8$ established in [36, Theorem 1.1]. For simplicity, we have isolated Theorem 1.1 and its proof here. By a standard explicit-formula argument (see [6, Chapters 19–20]), Theorem 1.1 gives the following quantitative form of the prime number theorem in arithmetic progressions.

Corollary 1.2. *Let* $\pi(x; q, a)$ *count the primes* $p \le x$ *with* $p \equiv a \pmod q$. *Write* φ *for Euler's totient function. For* $x \ge 2$,

$$\sup_{\substack{1 \le q \le x \\ (a,q)=1}} \left| \pi(x; q, a) - \frac{1}{\varphi(q)} \int_2^x \frac{dt}{\log t} \right| \ll x^{11/12} \log x,$$

where the implied constant is absolute and effective.

Theorem 1.1 has a number of additional arithmetic consequences. By the zero-free-region method originating with Rodosskiĭ [42], in the form recorded in [33, Theorem 13.12], the least quadratic nonresidue modulo an odd prime p is bounded by a fixed power of $\log p$, which in particular proves Vinogradov's conjecture [49] that this nonresidue is $\ll_\varepsilon p^\varepsilon$ for every $\varepsilon > 0$. See also Bhargava, Ivanyos, Mittal, and Saxena [3, Theorem 6.7] for this consequence of a fixed zero-free half-plane. From a computational number theory perspective, this bound allows square roots modulo p to be extracted deterministically in time polynomial in $\log p$ using the Tonelli–Shanks algorithm (see [11, §2.9]). Theorem 1.1 also yields a deterministic polynomial-time implementation of Miller's primality test [32]; note that polynomial-time primality testing was previously known unconditionally via the Agrawal–Kayal–Saxena algorithm [1]. For negative fundamental discriminants D, Littlewood's short Euler-product argument [29], applied to the fixed zero-free half-plane in Theorem 1.1, together with Dirichlet's class-number formula (see [6, Chapter 6]), yields the effective class-number bound $h(D) \gg \sqrt{|D|}/\log\log|D|$, with an absolute computable implied constant. The class-group computations of Elsenhans, Klüners, and Nicolae [8, Theorem 2], building on Weinberger [50], give a complete list of imaginary quadratic fields with class-group exponent dividing two when there are no Landau–Siegel zeros. Together with Grube's characterization of idoneal numbers and his reduction to fundamental discriminants [15] (see [23, Theorem 6 and §2.3]), this shows that Theorem 1.1 confirms the completeness of Euler's list of 65 idoneal numbers.

1.1 Prior work

There is a long line of work establishing zero-free regions for Dirichlet L-functions. Hadamard and de la Vallée Poussin proved the Prime Number Theorem in 1896 by establishing that $\zeta(s)$ has no zeros on the line $\operatorname{Re} s = 1$ [16, 46]. De la Vallée Poussin subsequently obtained a quantitative zero-free region to the left of this line [47]. For nonprincipal primitive Dirichlet characters, the classical zero-free-region theorem of Grönwall and Titchmarsh [14, 45], in the modern form given in [22, Theorem 5.26] (see also [6, Chapter 14]), gives an absolute, effective constant $c_0 > 0$ such that

$$L(\sigma + it, \chi) \neq 0 \quad \text{if} \quad \sigma > 1 - \frac{c_0}{\log(q(|t| + 3))}, \tag{1.1}$$

with at most one exception for each primitive character. Any exception is a simple real zero, can occur only if χ is quadratic, and is called a *Landau–Siegel zero*. Theorem 1.1 rules out the existence of such a zero.

There are two parameters in (1.1): the conductor q and the height $|t|$. Even when q is fixed, its width tends to zero with the height. Vinogradov and Korobov developed methods [48, 24] to prove that

$$\zeta(\sigma+\mathrm{i}t)\neq 0 \quad \text{if} \quad \sigma > 1-\frac{c_1}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \qquad |t|\geq 3, \tag{1.2}$$

for an absolute $c_1>0$ (Ford [9] gives an explicit value of c_1). The assertion of Theorem 1.1 is a half-plane of fixed width, independent of both conductor and height.

As we will see in the outline, the proof of Theorem 1.1 draws on modern developments in character large sieves and metaplectic theta series. In particular, the connection between cubic Gauss sums and cubic theta coefficients [26, 38] links the argument to work on Patterson's conjecture by Heath-Brown and Patterson [20], Heath-Brown [19], and Dunn and Radziwiłł [7]. The recursive large-sieve arguments also build on Heath-Brown's proof of the quadratic large sieve inequality [18].

1.2 Organization

Section 2 gives a high-level outline of the argument. Section 3 deduces the zero-free region from a mean-square estimate for twisted Möbius sums. Section 4 reduces this estimate to a dual mean square with cubic Gauss-sum coefficients. Section 5 states the completed mean-square and transfer estimates, combines them into a recursive inequality, and proves the required mean-square bound. Sections 6 and 7 prove the completed estimate and the bounds for the remaining cube-divisor sums, respectively. Appendix A supplies the arithmetic identities and the detailed theta calculation. Appendix B records the smooth separation lemma used throughout.

1.3 Notation

Write $\mathcal{O}=\mathcal{O}_K=\mathbb{Z}[\omega]$, where $\omega=e^{2\pi\mathrm{i}/3}$. For $a\in K$, write $\mathrm{N}_{K/\mathbb{Q}}(a)=a\overline{a}=|a|^2$ for its norm. For a nonzero integral ideal $\mathfrak{a}\subseteq\mathcal{O}$, write $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{a})=\#(\mathcal{O}/\mathfrak{a})$. We use $\mathbf{1}_{\mathcal{C}}$ for the indicator of a condition $\mathcal{C}$. For a prime ideal represented by p and a nonzero element or ideal a, write $v_p(a)$ for its exponent in the prime factorization of a.

We use standard asymptotic notation, writing $f=O(g)$ or $f\ll g$ when $|f|\leq Cg$, where $g\geq 0$ and $C>0$ is an absolute constant unless otherwise specified. Subscripts indicate possible dependence of the implied constant: for example, $f\ll_{\nu,W,\varepsilon} g$ means $|f|\leq C_{\nu,W,\varepsilon}g$, where the constant may depend on ν, W, ε but is uniform in all other varying parameters. For nonnegative f, g, we write $f\asymp g$ when $f\ll g$ and $g\ll f$. A dyadic norm range is an interval $R\leq \mathrm{N}_{K/\mathbb{Q}}(a)<2R$; a sum over dyadic scales uses $R=2^j$.

We use $D\geq 2$ as an ambient size parameter; in the main argument, it is the original norm scale introduced in the outline below. Auxiliary scales may vary within ranges bounded by fixed powers of D. We write

$$A\preccurlyeq B \quad \text{if} \quad A\ll_\varepsilon D^\varepsilon B \qquad \text{for every } \varepsilon>0,$$

uniformly in the varying parameters over their stated ranges. Implied constants may also depend on the fixed data specified in each statement.

2 Outline of the argument

In this section, we sketch the proof of Theorem 1.1, suppressing various coprimality conditions, local factors, and details of smoothing. The precise statements appear in Sections 3–7.

2.1 Step 1: Reduction to power-saving estimates for twisted Möbius sums

Fix a finite-order Hecke character ν of K. Let $L_K(s,\nu)$ be the corresponding Hecke L-function. We extend ν by zero to ideals not coprime to its conductor. We will deduce Theorem 1.1 from a power-saving estimate for ν-twisted Möbius sums.

Let μ denote the ideal Möbius function on K. All original ideal sums run over nonzero integral ideals. As in Section 3, the ideals in these sums are understood to be prime to 2, 3, and the conductor of ν. For a norm scale $D > 0$, consider

$$A_1(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\nu(\mathfrak{n})W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D),$$

where $W \in C_c^\infty((0,\infty);\mathbb{C})$ is a smooth cutoff function. Thus $A_1(D)$ is a smoothed ν-twisted Möbius sum over ideals of norm comparable to D, with roughly D terms. We seek a power saving over this size: for a fixed $\delta > 0$ and every $\varepsilon > 0$,

$$A_1(D) \ll_{\nu,W,\varepsilon} D^{1-\delta+\varepsilon} \text{ for every } W \in C_c^\infty((0,\infty);\mathbb{C}). \tag{2.1}$$

The implication from (2.1) to a zero-free half-plane is the smoothed Hecke version of the classical relation between Möbius sums and zero-free regions. In the zeta-function case, the equivalence between the Riemann Hypothesis and the bound $\sum_{n\le x}\mu(n) \ll_\varepsilon x^{1/2+\varepsilon}$ is due to Littlewood [28]; see also [31, §1]. Section 3 gives the complete argument needed here.

2.2 Step 2: Embedding the sum in a family

In order to estimate $A_1(D)$, we embed it into a family of such sums, parametrized by $u \in \mathcal{O}_K$, by introducing a sextic twist. We call an element of $\mathcal{O}$ *primary* if it is congruent to 1 modulo 3. The ring $\mathcal{O} = \mathbb{Z}[\omega]$ is Euclidean, so it has unique factorization and every ideal is principal; for an ideal coprime to 3, multiplying any generator by a unique unit gives $n \equiv 1 \pmod 3$, since the six units represent the six invertible residue classes modulo 3. For a primary element n, we write $\mu(n) = \mu((n))$ and $\nu(n) = \nu((n))$; divisor sums count each ideal divisor once, using its primary generator. For an ideal $\mathfrak{n}$ prime to 6, use its unique primary generator n and write

$$\chi_{\mathfrak{n}}(u) = \chi_n(u) = (u/n)_6.$$

Here $(u/n)_6$ is the sextic residue symbol, extended by zero when $(u,n) \neq 1$.[1] We also write $(u/n)_2$ and $(u/n)_3$ for the quadratic and cubic residue symbols, defined by the same convention with 6 replaced by 2 and 3, respectively. When $(n,6) = 1$, they equal $\chi_n(u)^3$ and $\chi_n(u)^2$. Restrictions that keep these symbols defined are understood when omitted in the outline. The family is

$$A_u(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\nu(\mathfrak{n})\chi_{\mathfrak{n}}(u)W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D). \tag{2.2}$$

Fix $0 < \vartheta \le 1/10$. The crucial estimate, Proposition 3.1, is that

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(u)\le H} |A_u(D)|^2 \ll_{\nu,W,\vartheta,\varepsilon} D^{1+\varepsilon}H, \qquad H = D^{1+\vartheta}. \tag{2.3}$$

In a mean square, we call the variable in the outer sum the *row* and the variable in the inner sum the *column*. In (2.3) these are u and $\mathfrak{n}$, respectively. Up to the factor of D^ε, (2.3) can be thought of as saying that the family $\{A_u(D)\}$ exhibits "square-root cancellation" on average (in the L^2 sense) over u.

[1] For a primary prime $p \nmid 6$ and $p \nmid u$, $(u/p)_6$ is the unique sixth root of unity congruent to $u^{(\mathrm{N}_{K/\mathbb{Q}}(p)-1)/6}$ modulo p; set $(u/p)_6 = 0$ when $p \mid u$. Extend multiplicatively in the denominator to define $\chi_n(u) = (u/n)_6$ for primary n coprime to 6. For the unit ideal, $\chi_1(u) = 1$ for every u, including $u = 0$.

The mean-square estimate (2.3) gives the desired power saving for $A_1(D)$ because $A_{p^6}(D) \approx A_1(D)$ for many primes p. For a primary prime p with $Y/2 < \mathrm{N}_{K/\mathbb{Q}}(p) \le Y$, the identity $\chi_n(p^6) = \mathbf{1}_{p\nmid n}$ leaves only $O_W(D/Y)$ differing terms and hence gives

$$A_{p^6}(D) = A_1(D) + O_W(D/Y). \tag{2.4}$$

By Landau's prime ideal theorem [27] (see [22, Theorem 5.33]), there are $\asymp Y/\log Y$ choices of such p. Taking $Y = H^{1/6}$, considering the contribution of such terms to (2.3) and using (2.4) gives

$$|A_1(D)|^2 \ll D^{1+\varepsilon}H^{5/6} + D^2H^{-1/3}. \tag{2.5}$$

With $H = D^{1+\vartheta}$, this gives $A_1(D) \ll D^{11/12+5\vartheta/12+\varepsilon}$ for every fixed $0 < \vartheta \le 1/10$. Choosing ϑ sufficiently small for each requested exponent loss yields $A_1(D) \ll_{\nu,W,\varepsilon} D^{11/12+\varepsilon}$. Therefore it remains to establish (2.3).

2.3 Step 3: Poisson summation

We turn to the task of proving the mean-square estimate (2.3). Now that we have introduced the row variable u, we can apply Poisson summation in this variable. This introduces sextic Gauss sums, arising from the Fourier transforms of the sextic characters. Upon application of the Gauss–Jacobi identities, these sextic Gauss sums absorb the factor of μ and turn into cubic Gauss sums. We interpret the resulting cubic Gauss sums as coefficients of Kubota's cubic theta function. In the next step, we use the automorphy of this theta function. For now, we explain the appearance of these Gauss sums in more detail.

Define the additive character

$$e(z) = \exp(4\pi\mathrm{i}\ \mathrm{Im}\, z/\sqrt{3}) \quad (z \in \mathbb{C}).$$

For a squarefree Eisenstein integer $n \equiv 1 \pmod 3$ prime to 6 and an integer j, define the normalized Gauss sums

$$\gamma_j(n) = \frac{1}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)}} \sum_{x \bmod n} \chi_n(x)^j e(x/n). \tag{2.6}$$

For $j = -1$, we interpret χ_n^{-1} as the conjugate character $\overline{\chi_n}$.

Consider the classical identity

$$\left(\frac{-1}{m}\right) = \left(\frac{1}{\sqrt{m}} \sum_{x \bmod m} \left(\frac{x}{m}\right) e^{2\pi i x/m}\right)^2,$$

where m is an odd squarefree positive integer and the symbols are Jacobi symbols. We would like a similar decomposition for μ instead of $\left(\frac{-1}{m}\right)$. Following the work of Hasse [17, pp. 443–445] and Heath-Brown [19, (2)], we derive the following identity in Appendix A.1, valid for squarefree primary n away from a fixed set of excluded primes:

$$\mu(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n), \qquad \alpha(n) = \frac{n}{|n|}, \quad G(n) = \overline{\chi_n(4)}\gamma_3(n).$$

Here $G(n)$ is a fixed ray-class factor. The factor $\gamma_{-1}(n)$ comes from Poisson summation; as promised, it combines with $\mu(n)$ to produce the cubic coefficient $\overline{\alpha(n)}\gamma_2(n)$ (up to ray-class factors). Further details are given in Section 4.

We call the sums obtained after applying Poisson summation *dual sums*.[2] Rearranging these dual sums reduces the problem of estimating $\sum_u |A_u(D)|^2$ to estimating the following family of

[2]For this terminology, see [40, p. 440], [25, §2.3], and [10, arXiv abstract].

column sums (for clarity, we have suppressed auxiliary twists and coprimality conditions):

$$B_h(X) = \sum_{\substack{n\in\mathcal{O}\\ n\equiv 1\ (3)\\ n\ \text{squarefree}}} \overline{\alpha(n)}\gamma_2(n)\xi(n)\chi_n(h)U(\mathrm{N}_{K/\mathbb{Q}}(n)/X). \tag{2.7}$$

Here h is the new row variable, Fourier dual to u, while U is a smooth weight restricting the column variable n to norm comparable to X. We choose ξ to be either $\nu\eta$ or $\overline{\nu}\eta$, where η ranges over a finite set of ray class characters of fixed modulus.[3] For each choice of ξ, this defines a family of sums $B_h(X)$.

After treating the diagonal and separating the smooth weights, we roughly get that

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(u)\le H} |A_u(D)|^2 \preccurlyeq DH + \frac{H}{D}\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(D)|^2. \tag{2.8}$$

Here $\mathcal{H} \asymp D^2/H$ is the dual norm scale. This schematic comparison suppresses the common factors arising when the square is expanded. Thus the desired estimate is

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(D)|^2 \preccurlyeq D^2. \tag{2.9}$$

2.4 Step 4: Relation to cubic Gauss sums and Kubota's cubic theta function

To estimate (2.9), we first identify the cubic Gauss coefficients with Fourier coefficients of a theta function. Its transformation law applies to a sum with extra cube factors, which we will remove in Step 5.

Let $\theta(z,v)$ be Kubota's cubic theta function on hyperbolic three-space ($z\in\mathbb{C}$, $v\in\mathbb{R}_{>0}$), obtained as a residue of a cubic metaplectic Eisenstein series [26, 38]. We use the normalization of [7, §5.1, (5.6)–(5.8)], recalled below in (6.1).

Put $\lambda = 1+2\omega$. For primary elements $n, b\in\mathcal{O}=\mathbb{Z}[\omega]$, with n squarefree and $(nb,6)=1$, let $c_\theta(nb^3)$ denote the Fourier coefficient of $\overline{\theta}$ indexed by $\lambda^{-3}nb^3$ in its expansion in z. Patterson's formula [38, Theorem 8.1], recorded in [7, (5.7)], gives

$$c_\theta(nb^3) = 3^{5/2}|b|\,\overline{\chi_n(\lambda)^2}\gamma_2(n). \tag{2.10}$$

Taking $b=1$ in (2.10) and substituting into (2.7) yields

$$B_h(X) = 3^{-5/2}\sum_{\substack{n\in\mathcal{O}\\ n\equiv 1\ (3)\\ n\ \text{squarefree}}} c_\theta(n)\overline{\alpha(n)}\chi_n(\lambda)^2\xi(n)\chi_n(h)U(\mathrm{N}_{K/\mathbb{Q}}(n)/X). \tag{2.11}$$

Thus (2.11) expresses $B_h(X)$ as a smoothed, twisted sum of the squarefree-index coefficients of $\overline{\theta}$. This connection was used by Heath-Brown and Patterson to study Kummer sums [20].

To use the summation formula for theta coefficients, we embed the sum (2.11) in a completed sum, by including terms indexed by nb^3. This use of cube completion follows Dunn and Radziwiłł [7, Lemma 5.4 and Proposition 5.3], with the underlying theta coefficients given by Patterson [38, Theorem 8.1]. For fixed h, ξ, and U, define

$$T_h(X) := \frac{1}{3^{5/2}\sqrt{X}}\sum_{\substack{n,b\in\mathcal{O}\\ n,b\equiv 1\ (3)\\ n\ \text{squarefree}}} c_\theta(nb^3)\overline{\alpha(nb^3)}\chi_{nb^3}(\lambda)^2\xi(nb^3)\chi_{nb^3}(h)U(\mathrm{N}_{K/\mathbb{Q}}(nb^3)/X). \tag{2.12}$$

[3] The ray class group modulo an ideal $\mathfrak{m}$ is the group of fractional ideals prime to $\mathfrak{m}$, modulo principal ideals generated by elements congruent to 1 modulo $\mathfrak{m}$. A ray class character is a character of this finite group. For an element prime to $\mathfrak{m}$, its ray class means the class of its principal ideal.

The $b = 1$ terms are exactly $X^{-1/2}B_h(X)$; the other terms supply the cube indices in the theta expansion. The first goal is to bound the mean square of $T_h(X)$ over h.

The automorphy of θ gives a summation formula that transforms the completed column sum $T_h(X)$, with the row h held fixed, into dual sums of theta coefficients with new smooth weights and character twists. A cusp is represented by a boundary point in $K \cup \{\infty\}$. A *cusp expansion* is the Fourier expansion in the horizontal variable after a change of coordinates taking infinity to that point. We call its Fourier coefficients *cusp coefficients*. Our starting point is a variant (established in Appendix A.2) of the theta transformation of Dunn and Radziwiłł [7, §5], extending work of Patterson [38] and Yoshimoto [51]. The key point is how the character twists change. Suppose for illustration that h is squarefree and primary, with $(h, 6) = 1$. Then, by Proposition 6.2, the transformed expression is a finite linear combination of sums of the form

$$\sum_{0\neq m\in\mathcal{O}} \frac{d_\theta(m)\alpha(m)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(m)}}\,\chi_h(m)^3 V_*^\sharp\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(m)X}{\mathrm{N}_{K/\mathbb{Q}}(h)^2}\Big). \tag{2.13}$$

The dual column index is m, while h remains the row index. Here $d_\theta(m)$ denotes the coefficient at the Fourier index $\lambda^{-4}m$ in a cusp expansion of $\overline{\theta}$, and $V_*^\sharp$ is the transform in (6.6), applied to $V_*(y) = y^{1/2}U(y)$. We have suppressed a finite sum over these cusp expansions, fixed periodic twists, bounded prefactors, and fixed scale constants. Within each fixed ray class of h, the coefficient sequences and transformed weights are independent of h by Lemma 6.3. Proposition 6.2 gives the precise formula, writing $d(\ell)$ for the cusp coefficient at $\ell = \lambda^{-4}m$.

Since $V_*^\sharp$ decays rapidly at infinity, the effective norm range in (2.13) is $\mathrm{N}_{K/\mathbb{Q}}(m) \ll \mathrm{N}_{K/\mathbb{Q}}(h)^2/X$, in place of the original range $\mathrm{N}_{K/\mathbb{Q}}(nb^3) \asymp X$. These dual sums arise from the theta transformation and involve theta coefficients on the transformed scale, now twisted by the quadratic character χ_h^3. This character arises because, at each prime $p \mid h$, the Fourier and theta factors combine as

$$\chi_p^{-1}\chi_p^{-2} = \chi_p^{-3} = \chi_p^3. \tag{2.14}$$

Now a key point is that since χ_p^3 is quadratic, Goldmakher and Louvel's quadratic large sieve [13, Theorem 1.1 and Corollary 1.2] (a generalization of Heath-Brown's quadratic large sieve [18] to number fields) bounds the mean square of the sums in (2.13) as h varies. For squarefree quadratic families with row and column norm ranges M, L, the quadratic large sieve gives the factor $M + L$, up to $(ML)^\varepsilon$. Crucially, this avoids the additional term $(ML)^{2/3}$ in Blomer, Goldmakher, and Louvel's general higher-order large sieve [5, Theorem 1.3].

We then apply Cauchy–Schwarz in the cube variable and the quadratic large sieve in the squarefree column variable to estimate the mean square of $T_h(X)$. The details are given in the proof of Proposition 5.2, which gives, for $\mathcal{H}, X \geq 1$,

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |T_h(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^\varepsilon\Big(\mathcal{H} + \frac{\mathcal{H}^2}{X}\Big). \tag{2.15}$$

At $X = D$ and $\mathcal{H} \leq D$, the heuristic comparison $B_h(D) \approx \sqrt{D}\,T_h(D)$ would therefore give the desired D^2 bound. The remaining task is to justify the corresponding mean-square bound by removing the cube factors.

2.5 Step 5: Removing the cube factors

We now pass from a mean-square bound for $T_h(X)$ to one for $X^{-1/2}B_h(X)$, with both averages taken over h. Note that one cannot simply discard the terms with $b \neq 1$, because the contributions from different b can cancel. We begin by undoing the addition of cube factors using Möbius inversion. Related completions appear in Patterson [38, Theorem 6.1] and Heath-Brown [19, §3], with explicit removal of the cube factors by Möbius inversion in Dunn and Radziwiłł [7, Proposition 5.3 and (8.2)].

Fix h and ξ, and write $P_h(X) = X^{-1/2}B_h(X)$ for the $b = 1$ part of $T_h(X)$. Substituting the coefficient formula (2.10) into (2.12) gives

$$T_h(X) = \sum_{\substack{b\in\mathcal{O}\\ b\equiv 1 \ (3)}} \frac{w_h(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)}\, P_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}\Big), \tag{2.16}$$

where

$$w_h(b) = \overline{\alpha(b)}^{\,3}\xi(b)^3\chi_b(h)^3. \tag{2.17}$$

Each factor in (2.17) is completely multiplicative in b. Thus $w_h(bc) = w_h(b)w_h(c)$ even when b, c share prime factors, and $|w_h(b)| \le 1$. Möbius inversion (cf. [22, §1.3]) gives

$$P_h(X) = \sum_{\substack{d\in\mathcal{O}\\ d\equiv 1 \ (3)}} \frac{\mu(d)w_h(d)}{\mathrm{N}_{K/\mathbb{Q}}(d)}\, T_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(d)^3}\Big). \tag{2.18}$$

Indeed, substituting (2.16) into (2.18) and grouping by the total cube index b gives the factor $w_h(b)\sum_{d|b}\mu(d)$, which is 1 for $b = 1$ and 0 otherwise.

The objective is now an estimate for P_h, rather than for the completed sum T_h. For the simplified family and the parameter ranges arising from Step 3, the required bound is

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^{\varepsilon}X. \tag{2.19}$$

Since $B_h(X) = \sqrt{X}\,P_h(X)$, this is equivalent to a bound of size X^2, up to the same small power, for the mean square of $B_h(X)$.

For $1 \le H_c \le X^{1/3}$, let $P_{h,\le H_c}(X)$ be the part of (2.18) with $\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c$. Applying weighted Cauchy–Schwarz for each fixed h, then summing over h and using (2.15), gives

$$\begin{aligned}
&\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_{h,\le H_c}(X)|^2\\
&\le \Big(\sum_{\mathrm{N}_{K/\mathbb{Q}}(e)\le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(e)}\Big)\Big(\sum_{\mathrm{N}_{K/\mathbb{Q}}(d)\le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(d)} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |T_h\big(X/\mathrm{N}_{K/\mathbb{Q}}(d)^3\big)|^2\Big)\\
&\preccurlyeq \sum_{\mathrm{N}_{K/\mathbb{Q}}(d)\le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(d)}\Big(\mathcal{H} + \frac{\mathcal{H}^2\mathrm{N}_{K/\mathbb{Q}}(d)^3}{X}\Big) \preccurlyeq \mathcal{H} + \frac{\mathcal{H}^2H_c^3}{X}.
\end{aligned}$$

Thus

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_{h,\le H_c}(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^{\varepsilon}\Big(\mathcal{H} + \frac{\mathcal{H}^2H_c^3}{X}\Big). \tag{2.20}$$

For $\mathcal{H} \le X$, the estimate (2.20) is within the target (2.19) provided $\mathcal{H}^2H_c^3/X \le X$. Thus we apply the completed mean-square bound directly only up to the cutoff

$$H_c = \min\Big\{X^{1/3}, \Big(\frac{X}{\mathcal{H}}\Big)^{2/3}\Big\}. \tag{2.21}$$

At the basic scales from Step 3,

$$X \asymp D, \qquad \mathcal{H} \asymp D^{1-\vartheta}, \qquad H_c \asymp D^{2\vartheta/3}.$$

The inverse sum can extend to $\mathrm{N}_{K/\mathbb{Q}}(d) \asymp D^{1/3}$, so we must still control the larger divisors.

Write $\tau_{\mathrm{div}}(b)$ for the number of nonzero integral ideal divisors of (b). Let $P_{h,>H_c}(X)$ denote the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) > H_c$ in (2.18), and put $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3$. Substituting (2.16) into the truncated inversion formula for $P_{h,>H_c}(X)$ obtained from (2.18), and grouping by $b = dc$, gives

$$\begin{aligned} P_{h,>H_c}(X) &= \sum_{\mathrm{N}_{K/\mathbb{Q}}(d)>H_c} \sum_{c} \frac{\mu(d) w_h(d) w_h(c)}{\mathrm{N}_{K/\mathbb{Q}}(d)\mathrm{N}_{K/\mathbb{Q}}(c)} P_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(dc)^3}\Big) \\ &= \sum_{\mathrm{N}_{K/\mathbb{Q}}(b)>H_c} \frac{\beta_0(b)\chi_b(h)^3}{\mathrm{N}_{K/\mathbb{Q}}(b)} P_h(L_b), \end{aligned}$$

where all indices are primary and

$$\beta_0(b) = \overline{\alpha(b)}^{\,3}\xi(b)^3 \sum_{\substack{d\mid b \\ \mathrm{N}_{K/\mathbb{Q}}(d)>H_c}} \mu(d), \qquad |\beta_0(b)| \le \tau_{\mathrm{div}}(b).$$

The support of U restricts $\mathrm{N}_{K/\mathbb{Q}}(b) \ll_U X^{1/3}$. Weighted Cauchy–Schwarz for each h, followed by summation, gives

$$\begin{aligned} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_{h,>H_c}(X)|^2 &\le \Big(\sum_b \frac{\tau_{\mathrm{div}}(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)}\Big) \sum_b \frac{\tau_{\mathrm{div}}(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(L_b)|^2 \\ &\preccurlyeq \sup_b \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(L_b)|^2. \end{aligned} \tag{2.22}$$

Put $a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n)$. Write

$$E(\mathcal{H}, X) := \frac{1}{X} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} \Big|\sideset{}{^*}\sum_n a_\xi(n)\chi_n(h) U(\mathrm{N}_{K/\mathbb{Q}}(n)/X)\Big|^2.$$

A star restricts an index to squarefree primary elements. By (2.7) and the definition of P_h, we have

$$E(\mathcal{H}, X) = \frac{1}{X} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(X)|^2 = \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(X)|^2.$$

At the scales from Step 3, our goal is to prove

$$E(\mathcal{H}, X) \preccurlyeq X. \tag{2.23}$$

At $X = D$, this is precisely the bound (2.9) for the mean square of $B_h(D)$. Substitution into the Poisson comparison (2.8) then gives the required original mean-square estimate (2.3). Combining (2.20) for $P_{h,\le H_c}$ (the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c$) and (2.22) for $P_{h,>H_c}$ (the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) > H_c$), with the cutoff (2.21), gives

$$E(\mathcal{H}, X) \preccurlyeq X + \sup_{b:\, 1\le L_b\le X/H_c^3} E(\mathcal{H}, L_b).$$

It therefore remains to prove $E(\mathcal{H}, L_b) \preccurlyeq X$ for $1 < L_b \le X/H_c^3$. Here the row range $\mathcal{H}$ stays fixed, and the required bound is still of size X even though the column scale has decreased to L_b.

We now expand the square and apply Poisson summation in h as before; we now record the proof in terms of $E(\mathcal{H}, L_b)$. The Gauss-sum coefficients become Möbius coefficients, and

the new row variable y has norm at most about $L_b^2/\mathcal{H}$. After separating the weights, this gives schematically

$$E(\mathcal{H}, L_b) \preccurlyeq X + \frac{\mathcal{H}}{L_b^2} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(y)\ll L_b^2/\mathcal{H}} |M(y)|^2,$$

$$M(y) = \sum_{\mathrm{N}_{K/\mathbb{Q}}(n)\asymp L_b}^{*} \mu(n)\xi_1(n)\overline{\chi_n(y)}U_1(\mathrm{N}_{K/\mathbb{Q}}(n)/L_b).$$

Here ξ_1 is another fixed ray class character and U_1 is a smooth compactly supported weight produced by separating the variables. The term X includes the diagonal and zero-frequency contributions, using $\mathcal{H} \le X$.

Since $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3 \le X$, we may enlarge the nonnegative sum over y to

$$\mathrm{N}_{K/\mathbb{Q}}(y) \ll Y, \qquad Y = \frac{XL_b}{\mathcal{H}} \ge \frac{L_b^2}{\mathcal{H}}.$$

The purpose of this enlargement is that a second application of Poisson summation gives a shorter row range:

$$\mathcal{H}' = \frac{L_b^2}{Y} = \frac{\mathcal{H}L_b}{X}.$$

The coefficients return to cubic Gauss-sum coefficients, and

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(y)\ll Y} |M(y)|^2 \preccurlyeq YL_b + YE'(\mathcal{H}', L_b).$$

Here E' has the same form as E, with possibly different smooth weights and fixed characters; YL_b accounts for the zero-frequency contribution. Consequently,

$$E(\mathcal{H}, L_b) \preccurlyeq X + \frac{X}{L_b}E'\Big(\frac{\mathcal{H}L_b}{X}, L_b\Big), \qquad \frac{E(\mathcal{H}, L_b)}{X} \preccurlyeq 1 + \frac{E'(\mathcal{H}L_b/X, L_b)}{L_b}.$$

Thus it suffices to prove

$$E'\Big(\frac{\mathcal{H}L_b}{X}, L_b\Big) \preccurlyeq L_b.$$

This is the original type of estimate at smaller parameters:

$$(\mathcal{H}', X') = \Big(\frac{\mathcal{H}}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}, \frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}\Big), \qquad \frac{\mathcal{H}'}{X'} = \frac{\mathcal{H}}{X}.$$

Both scales decrease while their ratio stays fixed. The factor X/L_b in the preceding inequality is exactly what converts the new target bound L_b into the required bound X. These two Poisson summations give the *transfer estimate*: a bound for the remaining mean square in terms of new mean squares of the same type. For related uses of an enlarged summation range, see Goldmakher–Louvel [13, Lemma 4.4 and the proof of Theorem 4.1], following Heath-Brown [18, Lemma 9].

Combining the bounds (2.20) and (2.22) from the first part of Step 5 with the transfer estimate above gives the recursive bound

$$\frac{E(\mathcal{H}, X)}{X} \preccurlyeq 1 + \sup_{\substack{\mathrm{N}_{K/\mathbb{Q}}(b)>H_c \\ L_b>1}} \frac{E'(\mathcal{H}', X')}{X'}, \qquad X' = L_b, \quad \mathcal{H}' = \frac{\mathcal{H}L_b}{X}.$$

For each b in this supremum, the cutoff (2.21) gives $\mathcal{H}' < \mathcal{H}(\mathcal{H}/X)^2 \ll D^{-2\vartheta}\mathcal{H}$. Since $\mathcal{H}'/X' = \mathcal{H}/X$, the same contraction applies at every step. After $O_\vartheta(1)$ steps, every resulting mean square either has an empty remainder or has row parameter at most 1, where counting gives the desired

bound at its reduced scales. Applying the recursive inequality back through these steps proves (2.23) for the original $E(\mathcal{H}, X)$, completing the sketch of the proof.

The preceding sketch suppresses auxiliary twists and common factors for the purpose of illustration. To carry out this argument with the auxiliary twists included, we use the family

$$\mathcal{E}(\mathcal{H}, X, F) = \frac{1}{XF} \sideset{}{^*}\sum_{\mathrm{N}_{K/\mathbb{Q}}(f) \asymp F} \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(k) \ll \mathcal{H}} \Big| \sideset{}{^*}\sum_{n} a_\xi(n) \chi_n(k) \chi_n(f)^4 U(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \Big|^2$$

and prove $\mathcal{E}(\mathcal{H}, X, F) \preccurlyeq XF$ in the parameter ranges of Proposition 5.1. Unlike the sketch above, the full argument must also handle the common factors and the resulting dyadic ranges. The precise family is defined in (4.9), and Proposition 5.4 states the transfer estimate. Combining it with the bounds for the two parts of the inverse sum in Step 5 reduces the row range at each step. Section 5 proves the desired bound by a finite iteration, following the admissible-exponent method of Heath-Brown [18, Lemma 8 and §8], [19, Lemma 9]; see also [13, Theorem 4.1] and [5, §3.2].

3 From the mean-square estimate to the zero-free region

We first carry out Steps 1 and 2 of the outline: extract cancellation in $A_1(D)$ from a mean-square estimate for the family, then use a Mellin transform to deduce nonvanishing. Fix a finite-order Hecke character ν of K and a finite set S of prime ideals that contains all prime ideals above 2 or 3, as well as all prime ideals dividing the conductor of ν. For an ideal or element a, write $(a, S) = 1$ if no prime ideal in S divides a. An ideal is supported on S if all its prime factors belong to S. For $W \in C_c^\infty((0,\infty); \mathbb{C})$, recall the family

$$A_u(D) := \sum_{(n,S)=1} \mu(n) \nu(n) \chi_n(u) W(\mathrm{N}_{K/\mathbb{Q}}(n)/D),$$

introduced in (2.2). Here and throughout the original family, n runs over ideals prime to S, represented by their primary generators. The key estimate is the following.

Proposition 3.1. *For every fixed $0 < \vartheta \le 1/10$ and $\varepsilon > 0$, there exists an integer $k = k(\vartheta, \varepsilon) \ge 1$ such that, for every compact interval $I \subset (0,\infty)$, all smooth W supported in I, and $D \ge 2$,*

$$\sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(u) \le D^{1+\vartheta}} |A_u(D)|^2 \ll_{\nu,S,I,\vartheta,\varepsilon} \Big(\max_{0 \le j \le k} \|W^{(j)}\|_\infty \Big)^2 D^{2+\vartheta+\varepsilon}. \tag{3.1}$$

We first deduce Theorem 1.1 assuming Proposition 3.1. The proof of the proposition is completed in Section 5.6.

Proof of Theorem 1.1 from Proposition 3.1. Fix $0 < \vartheta \le 1/10$, and put $H = D^{1+\vartheta}$ and $Y = H^{1/6} = D^{(1+\vartheta)/6}$. For prime ideals $Y/2 < \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p}) \le Y$, $\mathfrak{p} \notin S$, let p be their primary generators, chosen with $p \equiv 1 \pmod 3$. Then $\chi_n(p^6) = \mathbf{1}_{\mathfrak{p} \nmid n}$ and therefore

$$|A_1(D) - A_{p^6}(D)| \le \|W\|_\infty \#\{n : (n,S) = 1,\ \mathfrak{p} \mid n,\ \mathrm{N}_{K/\mathbb{Q}}(n) \asymp D\} \ll_W D/Y.$$

For this fixed field, Landau's prime ideal theorem [27] (see [22, Theorem 5.33]) gives $J \asymp Y/\log Y$ such primes. Their sixth powers are distinct rows of norm at most H. Apply (3.1) with loss $\varepsilon/2$ and use $\log Y \ll_\varepsilon D^{\varepsilon/2}$ to obtain

$$\begin{aligned} |A_1(D)|^2 &\le \frac{2}{J} \sum_{\substack{Y/2 < \mathrm{N}_{K/\mathbb{Q}}(p) \le Y \\ (p,S)=1}} |A_{p^6}(D)|^2 + O_W(D^2/Y^2) \\ &\ll_{\nu,S,W,\vartheta,\varepsilon} D^{2+\vartheta+\varepsilon}/Y + D^2/Y^2 = D^{11/6+5\vartheta/6+\varepsilon} + D^{5/3-\vartheta/3}. \end{aligned} \tag{3.2}$$

Thus $A_1(D) \ll_{\nu,S,W,\vartheta,\varepsilon} D^{11/12+5\vartheta/12+\varepsilon}$. Given any requested exponent loss, choose $\vartheta > 0$ and then the loss in Proposition 3.1 sufficiently small. Renaming the resulting loss ε, we obtain

$$A_1(D) \ll_{\nu,S,W,\varepsilon} D^{11/12+\varepsilon}. \tag{3.3}$$

We now use (3.3) to rule out a zero of $L_K(s,\nu)$ in $\operatorname{Re} s > 11/12$. Suppose such a zero ϱ exists. Choose $0 \neq \phi \in C_c^\infty((1,2))$, $\phi \geq 0$, and $W(y) = y^{-\varrho}\phi(y)$. With the Mellin convention $\widehat{W}(s) = \int_0^\infty W(y) y^s \, dy/y$, we have $\widehat{W}(\varrho) = \int \phi(y)\, dy/y > 0$. The function

$$\mathcal{M}_W(s) = \int_0^\infty A_1(D) D^{-s} \frac{dD}{D}$$

is holomorphic on $\operatorname{Re} s > 11/12$: the estimate (3.3) controls the integral at infinity, and the compact support of W makes $A_1(D)$ vanish for sufficiently small D. Termwise integration for $\operatorname{Re} s > 1$ gives

$$\mathcal{M}_W(s) = \widehat{W}(s) \sum_{(n,S)=1} \frac{\mu(n)\nu(n)}{\mathrm{N}_{K/\mathbb{Q}}(n)^s} = \frac{\widehat{W}(s)}{L_K^S(s,\nu)},$$

where L_K^S is the Euler product outside S:

$$L_K^S(s,\nu) := L_K(s,\nu) \prod_{\mathfrak{p}\in S} (1 - \nu(\mathfrak{p})\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s}).$$

By Hecke's meromorphic continuation theorem [21] (see [22, §5.10]) and the identity theorem, the identity $L_K^S(s,\nu)\mathcal{M}_W(s) = \widehat{W}(s)$ holds on $\operatorname{Re} s > 11/12$. The omitted Euler factors are nonzero here, so evaluation at $s = \varrho$ gives $0 = \widehat{W}(\varrho) > 0$, a contradiction.

To deduce the assertion for Dirichlet L-functions in Theorem 1.1, let χ_{-3} be the nontrivial character modulo 3. For any Dirichlet character χ, quadratic base change gives, up to Euler factors nonzero in $\operatorname{Re} s > 0$,

$$L_K(s, \chi \circ \mathrm{N}_{K/\mathbb{Q}}) = L(s,\chi)L(s,\chi\chi_{-3}).$$

The Hecke conclusion excludes zeros of either factor away from $s = 1$. At $s = 1$, the only possible pole–zero cancellation is ruled out by Dirichlet's nonvanishing theorem, which gives $L(1,\chi_{-3}) > 0$ (see [6, Chapters 4 and 6]). □

4 Poisson summation and the dual mean square

We now turn to the mean-square estimate in Proposition 3.1. Following Step 3 of the outline, we expand the square and apply Poisson summation in the row variable u. The finite Fourier transforms of the characters supply Gauss sums. Arithmetic identities then combine these Gauss sums with the original Möbius coefficients to give the normalized cubic Gauss-sum coefficients that appear in the theta function.

The column indices produced by expanding the square need not be coprime. We first extract their common factor, then use Möbius inversion to separate the remaining coprimality condition. These operations introduce an auxiliary twisting index and an exclusion ideal. The comparison below removes the exclusion without changing the row range or the product of the column and auxiliary scales.

We record the arithmetic and Poisson identities first, then define this family. Proposition 4.5 states the estimate for it that suffices to prove (3.1); the rest of the section proves that implication.

4.1 The arithmetic identities

We first record the identities that convert between Möbius coefficients and normalized cubic Gauss sums. Recall that, for a squarefree primary element n with $(n, S) = 1$,

$$\alpha(n) = \frac{n}{|n|}, \qquad \gamma_j(n) = \frac{1}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)}} \sum_{x \bmod n} \chi_n(x)^j e(x/n) \quad (j \in \mathbb{Z}),$$

where $e(z) = \exp(4\pi\mathrm{i}\,\mathrm{Im}\,z/\sqrt{3})$, as in (2.6). Put

$$a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n) \qquad (n \text{ squarefree}),$$

where ξ is a fixed ray class character. The character ξ accounts for the original twist ν and for the residue-class factors in the identities below.

By character orthogonality [22, §3.1], we can absorb functions on a fixed ray class group into a finite sum of twists ξ. Choose a fixed modulus, supported on S, divisible by the conductor of ν and sufficiently large that all reciprocity factors below depend only on the corresponding ray classes. We then expand these factors in characters of this finite group.

The following lemma collects standard consequences of the Gauss–Jacobi identities, quadratic Gauss-sum evaluations, and reciprocity.

Lemma 4.1. *On a fixed ray class group, there are a function G with values in $\{z \in \mathbb{C} : |z| = 1\}$ and a symmetric $\{\pm 1\}$-valued bicharacter $\mathcal{R}$. This means that $\mathcal{R}(a, b) = \mathcal{R}(b, a)$ and $\mathcal{R}$ is multiplicative in each argument separately:*

$$\mathcal{R}(aa', b) = \mathcal{R}(a, b)\mathcal{R}(a', b),$$
$$\mathcal{R}(a, bb') = \mathcal{R}(a, b)\mathcal{R}(a, b'),$$

for all classes a, a', b, b' in the ray class group. These functions have the following properties. Let a, b, n be primary elements prime to S, with $(a, b) = 1$ and n squarefree. Then

$$\chi_b(a) = \mathcal{R}(a, b)\chi_a(b), \qquad G(ab) = G(a)G(b)\mathcal{R}(a, b), \tag{4.1}$$
$$\gamma_2(n)^3 = \mu(n)\alpha(n), \qquad \gamma_1(n)\gamma_2(n) = \mu(n)\alpha(n)G(n), \tag{4.2}$$
$$G(n) = \overline{\chi_n(4)}\gamma_3(n), \qquad \gamma_1(n)\gamma_{-1}(n) = \chi_n(-1). \tag{4.3}$$

Consequently

$$\overline{\alpha(n)}\gamma_2(n)\gamma_1(n) = \mu(n)G(n), \tag{4.4}$$
$$\mu(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n), \tag{4.5}$$
$$a_\xi(ab) = a_\xi(a)a_\xi(b)\chi_b(a)^4, \tag{4.6}$$
$$\chi_a(-1)\overline{G(a)}G(b)\mathcal{R}(a, b) = G(ba^{-1}). \tag{4.7}$$

In (4.6), a, b are also squarefree. The identity for $G(ab)$ in (4.1) extends to all classes of the fixed ray class group; the quotient in (4.7) is taken in that group.

The proof, including the dependence of G and $\mathcal{R}$ on fixed ray classes, is given in Appendix A.1. Under Poisson summation, (4.5) converts the Möbius coefficients to $a_\xi(n)$, up to fixed ray class factors, while (4.4) converts them back.

4.2 Poisson summation with excluded primes

The row sum to which we apply Poisson summation will have an additional coprimality restriction. We record the formula with that restriction included, so that its effect on the Fourier frequencies and the normalization is explicit.

For a nonzero ideal $\mathfrak{r}$, write $\operatorname{rad}\mathfrak{r}$ for the product of its distinct prime divisors. We use the additive character e introduced in Step 3. In quotients and Gauss sums, use a fixed generator for each ideal, chosen primary when the ideal is prime to 3.

Lemma 4.2. *Let $\mathcal{H} > 0$, let $\mathfrak{r}$ be a nonzero ideal, and let $\Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H})$ be a smooth radial Schwartz weight on the row lattice. Let χ be a primitive multiplicative character of $(\mathcal{O}/\mathfrak{m})^\times$, viewed as a function on $\mathcal{O}$ by reduction modulo $\mathfrak{m}$ and extension by zero on nonunits. Write*

$$\gamma(\chi) = \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})^{-1/2} \sum_{x \bmod \mathfrak{m}} \chi(x) e(x/\mathfrak{m})$$

for its normalized Gauss sum. Since χ is primitive, its modulus $\mathfrak{m}$ is determined by χ and is suppressed in the notation. Then

$$\begin{aligned} &\sum_k \chi(k) \mathbf{1}_{(k,\mathfrak{r})=1} \Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H}) \\ &\quad = \frac{\mathcal{H}\gamma(\chi)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})}} \sum_{d \mid \mathrm{rad}\,\mathfrak{r}} \frac{\mu(d)\chi(d)}{\mathrm{N}_{K/\mathbb{Q}}(d)} \sum_h \overline{\chi(h)} \widehat{\Phi}\Big(\frac{\mathcal{H}\mathrm{N}_{K/\mathbb{Q}}(h)}{\mathrm{N}_{K/\mathbb{Q}}(d)\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})} \Big). \end{aligned} \tag{4.8}$$

Here $\widehat{\Phi}$ is defined by

$$\widehat{\Phi}(|w|^2) = \frac{2}{\sqrt{3}} \int_{\mathbb{C}} \Phi(|z|^2) e(-zw)\, dx\, dy, \qquad z = x + \mathrm{i}y.$$

The measure is normalized so that $\mathcal{O}$ has covolume one. Here $k, h \in \mathcal{O}$, and h is the Fourier frequency. We have

$$\chi \text{ nonprincipal} \quad \Longrightarrow \quad \overline{\chi(0)} = 0.$$

For the principal primitive character ($\chi = \mathbf{1}$, $\mathfrak{m} = 1$), the zero-frequency contribution is

$$\mathcal{H}\widehat{\Phi}(0) \prod_{\mathfrak{p} \mid \mathfrak{r}} \Big(1 - \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})}\Big).$$

Proof. Inclusion–exclusion followed by $k = d\ell$ gives

$$\sum_k \chi(k) \mathbf{1}_{(k,\mathfrak{r})=1} \Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H}) = \sum_{d \mid \mathrm{rad}\,\mathfrak{r}} \mu(d)\chi(d) \sum_\ell \chi(\ell) \Phi\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(\ell)}{\mathcal{H}/\mathrm{N}_{K/\mathbb{Q}}(d)} \Big).$$

Apply lattice Poisson summation [22, Theorem 4.5] in ℓ on residue classes modulo $\mathfrak{m}$, at scale $\mathcal{H}/\mathrm{N}_{K/\mathbb{Q}}(d)$. The primitive Gauss-sum identity [22, (3.12)] evaluates the finite Fourier transform as $\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})}\gamma(\chi)\overline{\chi(h)}$ for every h, giving (4.8). For a nonprincipal primitive character, $\overline{\chi(0)} = 0$, so the zero-frequency term vanishes. For the principal character, sum $\mu(d)/\mathrm{N}_{K/\mathbb{Q}}(d)$ over $d \mid \mathrm{rad}\,\mathfrak{r}$ to obtain the displayed zero-frequency contribution. □

4.3 The dual mean squares

We use the following mean square of the column sums with coefficients $a_\xi(n)$.

Definition 4.3. The parameters $\mathcal{H}$, X, and F are the norm scales of k, n, and f, respectively. Here k ranges over elements of $\mathcal{O}$, and a star restricts a sum to squarefree ideals prime to S, represented by their primary generators. For a smooth compactly supported weight W on $(0, \infty)$, define the *dual mean square* by

$$\begin{aligned} &\mathcal{E}(\mathcal{H}, X, F; \xi, W) \\ &\quad = \frac{1}{XF} \sideset{}{^*}\sum_{\substack{F \le \mathrm{N}_{K/\mathbb{Q}}(f) < 2F \\ (f,S)=1}} \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(k) \le \mathcal{H}} \Big| \sideset{}{^*}\sum_{(n,S)=1} a_\xi(n) \chi_n(k) \chi_n(f)^4 W(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \Big|^2. \end{aligned} \tag{4.9}$$

Below, ξ ranges over all characters of the fixed ray class group chosen above, through which ν, G, and $\mathcal{R}$ factor.

Write $\mathcal{E}_{\mathfrak{r}}$ for the same expression with the additional restriction $(n,\mathfrak{r})=1$. This notation is only needed in the Poisson reductions; the following comparison returns to $\mathcal{E}$.

Lemma 4.4. *Let r_0 be the product of the primes dividing $\mathfrak{r}$ outside S. For $\mathcal{H}, X > 0$, $F \geq 1$, and smooth compactly supported W,*

$$\mathcal{E}_{\mathfrak{r}}(\mathcal{H},X,F;\xi,W) \leq \tau_{\mathrm{div}}(r_0)\sum_{d\mid r_0}\mathcal{E}\Big(\mathcal{H},\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(d)},F\mathrm{N}_{K/\mathbb{Q}}(d);\xi,W\Big). \tag{4.10}$$

Proof. Let $C_{\mathfrak{r}}(X;k,f)$ denote the inner column sum defining $\mathcal{E}_{\mathfrak{r}}$, with ξ, W fixed. Inclusion–exclusion, $n=dm$, and (4.6) give

$$\begin{aligned} C_{\mathfrak{r}}(X;k,f) &= \sum_{d\mid r_0}\mu(d)\sideset{}{^*}\sum_{\substack{(n,S)=1\\ d\mid n}} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \\ &= \sum_{\substack{d\mid r_0\\(d,f)=1}}\mu(d)a_\xi(d)\chi_d(k)\chi_d(f)^4 C_1(X/\mathrm{N}_{K/\mathbb{Q}}(d);k,df). \end{aligned}$$

Indeed, $\chi_m(d)^4$ enforces $(m,d)=1$ and combines with $\chi_m(f)^4$; terms with $(d,f)\neq 1$ vanish. Each exterior coefficient has modulus at most one. Apply Cauchy–Schwarz in d, then enlarge the injective image $f\mapsto df$ to the squarefree range $F\mathrm{N}_{K/\mathbb{Q}}(d) \leq \mathrm{N}_{K/\mathbb{Q}}(df) < 2F\mathrm{N}_{K/\mathbb{Q}}(d)$. The normalizing product is unchanged: $(X/\mathrm{N}_{K/\mathbb{Q}}(d))(F\mathrm{N}_{K/\mathbb{Q}}(d)) = XF$. □

4.4 Reduction to the dual mean square

The following proposition gives the dual estimate sufficient for (3.1). Its proof applies Poisson summation directly to the original Möbius sums.

Proposition 4.5. *Fix $0<\vartheta\leq 1/10$ and put $H=D^{1+\vartheta}$. For each fixed $C\geq 1$ and all real $B,F\geq 1$, consider the ranges*

$$X=\frac{D}{BF},\qquad 0<\mathcal{H}\leq\frac{CD^2}{HB^2},\qquad XF=\frac{D}{B}. \tag{4.11}$$

Suppose that for every $\varepsilon>0$ there exists an integer $J=J(\vartheta,\varepsilon)\geq 1$ such that, for every compact interval $I\subset(0,\infty)$ and every smooth W supported in I,

$$\mathcal{E}(\mathcal{H},X,F;\xi,W) \ll_{\nu,S,I,C,\vartheta,\varepsilon} \|W\|^2_{C^J(I)} D^{\varepsilon} XF, \tag{4.12}$$

where

$$\|W\|_{C^J(I)} = \max_{0\leq j\leq J}\sup_{x\in I}|W^{(j)}(x)|.$$

For functions of several variables, the norm uses all partial derivatives of total order at most J. Then the original mean-square estimate in Proposition 3.1 holds.

Proof. Write $I=[a_0,b_0]$. Choose a nonnegative radial Schwartz weight Φ such that $\Phi(x)\geq 1$ on $[0,1]$ and $\operatorname{supp}\widehat{\Phi}\subset[0,C_\Phi]$ for some fixed $C_\Phi>0$. It suffices to prove $\mathcal{M}_D \ll_{\nu,S,I,\vartheta,\varepsilon} \|W\|^2_{C^{J'}(I)} HD^{\varepsilon}$ for some $J'=J'(\vartheta,\varepsilon)$, where

$$\mathcal{M}_D := \frac{1}{D}\sum_{u\in\mathcal{O}}\Phi(\mathrm{N}_{K/\mathbb{Q}}(u)/H)\Big|\sum_{(n,S)=1}\mu(n)\nu(n)\chi_n(u)W(\mathrm{N}_{K/\mathbb{Q}}(n)/D)\Big|^2. \tag{4.13}$$

Conjugate the inner sum, expand the square, and put $g=(n_1,n_2)$ and $n_j=gz_j$. The squarefree indices satisfy $(z_1,z_2)=(z_1z_2,g)=1$, and

$$\overline{\chi_{n_1}(u)}\chi_{n_2}(u)=\mathbf{1}_{(u,g)=1}\overline{\chi_{z_1}(u)}\chi_{z_2}(u).$$

Apply Lemma 4.2 with $\chi=\overline{\chi_{z_1}}\chi_{z_2}$ and exclusion g:

$$\begin{aligned}&\sum_u \Phi(\mathrm{N}_{K/\mathbb{Q}}(u)/H)\mathbf{1}_{(u,g)=1}\chi(u)\\&\quad=\sum_{e\mid g}\frac{H\mu(e)\chi(e)\gamma(\chi)}{\mathrm{N}_{K/\mathbb{Q}}(e)\sqrt{\mathrm{N}_{K/\mathbb{Q}}(z_1)\mathrm{N}_{K/\mathbb{Q}}(z_2)}}\sum_h\overline{\chi(h)}\widehat{\Phi}\Big(\frac{H\mathrm{N}_{K/\mathbb{Q}}(h)}{\mathrm{N}_{K/\mathbb{Q}}(e)\mathrm{N}_{K/\mathbb{Q}}(z_1)\mathrm{N}_{K/\mathbb{Q}}(z_2)}\Big).\end{aligned}\tag{4.14}$$

The zero frequency occurs only when $z_1=z_2=1$ and contributes

$$Z=\frac{H}{D}\widehat{\Phi}(0)\sideset{}{^*}\sum_{(g,S)=1}|W(\mathrm{N}_{K/\mathbb{Q}}(g)/D)|^2\prod_{p\mid g}\Big(1-\frac{1}{\mathrm{N}_{K/\mathbb{Q}}(p)}\Big)\ll_{\Phi,I}H\|W\|_\infty^2.$$

Retain all $h\neq0$, including those with $z_1=z_2=1$.

Expand the fixed ray class function $\overline{\nu(t)}G(t^{-1})=\sum_\xi c_\xi\xi(t)$. The Chinese remainder theorem and (4.5)–(4.7) give

$$\mu(z_1)\mu(z_2)\overline{\nu(z_1)}\nu(z_2)\gamma(\overline{\chi_{z_1}}\chi_{z_2})=\sum_\xi c_\xi a_\xi(z_1)\overline{a_\xi(z_2)}.\tag{4.15}$$

Write $N=\mathrm{N}_{K/\mathbb{Q}}$ for the remainder of this proof, and put $W_0(x)=x^{-1/2}\overline{W(x)}$. Substituting (4.15) into (4.14), including the normalization $1/D$ in (4.13), gives

$$\mathcal{M}_D-Z=\sum_\xi c_\xi\mathcal{S}_\xi,\qquad|\mathcal{M}_D-Z|\ll\max_\xi|\mathcal{S}_\xi|,$$

since the character sum is fixed and finite. Fix ξ. Using $\overline{\chi_z(e)}=\chi_z(e^5)$, its contribution is

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_g\sum_{e\mid g}\frac{\mu(e)N(g)}{N(e)}\sum_{h\neq0}\sideset{}{^*}\sum_{\substack{z_1,z_2\\(z_1,z_2)=1\\(z_1z_2,g)=1}}&a_\xi(z_1)\overline{a_\xi(z_2)}\chi_{z_1}(he^5)\overline{\chi_{z_2}(he^5)}\\&\times W_0\Big(\frac{N(gz_1)}{D}\Big)\overline{W_0\Big(\frac{N(gz_2)}{D}\Big)}\widehat{\Phi}\Big(\frac{HN(h)}{N(e)N(z_1)N(z_2)}\Big).\end{aligned}$$

Here and below every starred variable is squarefree, primary, and prime to S, while h,k range over nonzero elements of $\mathcal{O}$. We must show $|\mathcal{S}_\xi|\ll_{\nu,S,I,\vartheta,\varepsilon}\|W\|^2_{C^{J'}(I)}HD^\varepsilon$.

Insert the coprimality identity and change variables:

$$\mathbf{1}_{(z_1,z_2)=1}=\sum_{v\mid z_1,\ v\mid z_2}\mu(v),\qquad z_j=vm_j.$$

By (4.6), the common factor from the two columns satisfies

$$a_\xi(vm)=a_\xi(v)a_\xi(m)\chi_m(v)^4,\qquad|a_\xi(v)\chi_v(he^5)|^2=\mathbf{1}_{(v,h)=1},$$

where $(v,e)=1$. Thus the same sum becomes

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_{\substack{g,v\\(g,v)=1}}\sum_{e\mid g}\frac{\mu(e)\mu(v)N(g)}{N(e)}\sum_{\substack{h\neq0\\(h,v)=1}}\sideset{}{^*}\sum_{\substack{m_1,m_2\\(m_1m_2,gv)=1}}&a_\xi(m_1)\overline{a_\xi(m_2)}\\&\times\chi_{m_1}(he^5v^4)\overline{\chi_{m_2}(he^5v^4)}W_0\Big(\frac{N(gvm_1)}{D}\Big)\overline{W_0\Big(\frac{N(gvm_2)}{D}\Big)}\\&\times\widehat{\Phi}\Big(\frac{HN(h)}{N(e)N(v)^2N(m_1)N(m_2)}\Big).\end{aligned}$$

There is now no restriction $(m_1,m_2)=1$. Make the bijective change of variables

$$b=g/e,\qquad f=ev,\qquad k=eh,$$
$$e=(f,k),\qquad v=f/e,\qquad g=be,\qquad h=k/e.$$

The resulting b,f are coprime and squarefree, $k\neq 0$ is arbitrary, and

$$gv=bf,\qquad he^5v^4=kf^4,\qquad \mu(e)\mu(v)=\mu(f),\qquad \frac{N(g)}{N(e)}=N(b).$$

Consequently,

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_{\substack{b,f\\(b,f)=1}}\mu(f)N(b)\sum_{k\neq 0}\sideset{}{^*}\sum_{\substack{m_1,m_2\\(m_1m_2,b)=1}}a_\xi(m_1)\overline{a_\xi(m_2)}\chi_{m_1}(kf^4)\overline{\chi_{m_2}(kf^4)}\\ \times W_0\Big(\frac{N(bfm_1)}{D}\Big)\overline{W_0\Big(\frac{N(bfm_2)}{D}\Big)}\widehat{\Phi}\Big(\frac{HN(k)}{N(f)^2N(m_1)N(m_2)}\Big).\end{aligned}\tag{4.16}$$

The characters enforce $(m_j,f)=1$, leaving only the displayed exclusion $(m_j,b)=1$. The supports of W_0 and $\widehat{\Phi}$ give

$$N(b)N(f)\le b_0D,\qquad 0<N(k)\le\frac{C_\Phi b_0^2D^2}{HN(b)^2}.$$

Partition $B\le N(b)<2B$ and $F\le N(f)<2F$ into dyadic ranges. Denote the corresponding contribution to (4.16) by $\mathcal{S}_{\xi;B,F}$, and put

$$X=\frac{D}{BF},\qquad \mathcal{H}=\frac{C_{I,\Phi}D^2}{HB^2},\qquad C_{I,\Phi}=\max(2,C_\Phi b_0^2).$$

These satisfy (4.11). In the variables $x_j=N(m_j)/X$, the coupled smooth weight is

$$\begin{aligned}\mathcal{K}_{b,f,k}(x_1,x_2)&=W_0(rx_1)\overline{W_0(rx_2)}\widehat{\Phi}\Big(\frac{a}{x_1x_2}\Big),\\ r=\frac{N(b)N(f)}{BF}\in[1,4),&\qquad a=\frac{HN(k)}{N(f)^2X^2}\le C_{I,\Phi}.\end{aligned}$$

For every integer $q\ge 0$, these kernels satisfy

$$\sup_{b,f,k}\|\mathcal{K}_{b,f,k}\|_{C^q([a_0/4,b_0]^2)}\ll_{I,\Phi,q}\|W\|^2_{C^q(I)}.$$

For $U\in C_c^\infty(I_*)$, where $I_*=[a_0/8,2b_0]$, put

$$S_U(b,f,k)=\sideset{}{^*}\sum_{\substack{(n,S)=1\\(n,b)=1}}a_\xi(n)\chi_n(k)\chi_n(f)^4U(N(n)/X).$$

The shifts $(X,F)\mapsto(X/N(d),FN(d))$ preserve (4.11). Thus Lemma 4.4 and (4.12), with $J=J(\vartheta,\varepsilon/4)$, give

$$\begin{aligned}\sum_{f,k}|S_U(b,f,k)|^2&=XF\,\mathcal{E}_{(b)}(\mathcal{H},X,F;\xi,U)\\ &\le XF\,\tau_{\mathrm{div}}(b)\sum_{d\mid b}\mathcal{E}(\mathcal{H},X/N(d),FN(d);\xi,U)\\ &\ll_{\nu,S,I,\vartheta,\varepsilon}D^{\varepsilon/2}(XF)^2\|U\|^2_{C^J(I_*)}.\end{aligned}$$

Here $N(b) \ll_I D$, so the divisor factors are absorbed in $D^{\varepsilon/4}$. We use the following smooth-weight principle, stated and proved in Lemma B.2: a mean-square bound valid for every common test function, with a C^J norm, also bounds the corresponding quadratic sum with a kernel depending on the row, at a cost given by the kernel's C^{2J+4} norm. For each fixed b, apply Lemma B.2 with row index (f,k), coefficient $\mu(f)\mathbf{1}_{(f,b)=1}$, and kernel $\mathcal{K}_{b,f,k}$. The larger row range $0 < N(k) \le \mathcal{H}$ adds only terms whose original kernel vanishes. With $q = 2J+4$, this gives

$$\begin{aligned} |\mathcal{S}_{\xi;B,F}| &\ll_{\nu,S,I,\vartheta,\varepsilon} D^{\varepsilon/2}\|W\|^2_{C^q(I)} \sideset{}{^*}\sum_{B\le N(b)<2B} \frac{HN(b)}{D^2}(XF)^2 \\ &\ll D^{\varepsilon/2}\|W\|^2_{C^q(I)} \frac{HB}{D^2}\, B\,(XF)^2 = D^{\varepsilon/2}\|W\|^2_{C^q(I)} H, \end{aligned} \tag{4.17}$$

where $XF = D/B$ and ideal counting gives $O(B)$ choices of b. Summing the $O((\log D)^2)$ nonempty dyadic ranges bounds $\mathcal{S}_\xi$ as required. Sum over the fixed finite set of ξ, include Z, and restore the factor D to obtain (3.1). The required derivative order depends only on ϑ, ε. □

5 Iteration of the dual mean-square estimate

Put $\Sigma = XF$. We prove the following estimate for the family (4.9), with exclusions removed by Lemma 4.4. The two inputs are proved in Sections 6 and 7.

Proposition 5.1. *Fix $\kappa > 0$ and $C_0 \ge 1$. Suppose that*

$$\mathcal{H}, X, F \ge 1, \qquad \Sigma = XF \le D^{C_0}, \qquad \mathcal{H} \le \Sigma D^{-\kappa}. \tag{5.1}$$

For every $\varepsilon > 0$ there is an integer $J = J(\kappa, C_0, \varepsilon) \ge 1$ such that

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I,\nu,S,\kappa,C_0,\varepsilon} \|W\|^2_{C^J(I)} D^\varepsilon \Sigma \tag{5.2}$$

for every compact interval $I \subset (0,\infty)$ and smooth W supported in I.

The proof is given in Section 5.5, using Lemma 5.3 and Proposition 5.4.

5.1 The completed sums

Let Ψ be a completely multiplicative $\mathbb{C}$-valued function on the nonzero integral ideals of $\mathcal{O}_K = \mathbb{Z}[\omega]$ coprime to 3, vanishing on ideals divisible by a prime in S. For a primary element n, write $\Psi(n) = \Psi((n))$. Define

$$T(X;\Psi) = \sideset{}{^*}\sum_{(n,S)=1} \sum_{\substack{b\in\mathcal{O}\\ b\equiv 1 \ (3)\\ (b,S)=1}} \frac{\overline{\alpha(n)}\gamma_2(n)\Psi(n)\overline{\alpha(b)}^3\Psi(b)^3}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)\,\mathrm{N}_{K/\mathbb{Q}}(b)}}\, V_*(\mathrm{N}_{K/\mathbb{Q}}(n)\mathrm{N}_{K/\mathbb{Q}}(b)^3/X), \tag{5.3}$$

where $V_*(y) = \sqrt{y}\,W(y)$. The extra index b supplies the cubes in the Fourier expansion of the Kubota theta function. Here n, b are primary elements of $\mathcal{O}$; only n is required to be squarefree.

The $b = 1$ part is exactly $X^{-1/2}\sum^*_{(n,S)=1}\overline{\alpha(n)}\gamma_2(n)\Psi(n)W(\mathrm{N}_{K/\mathbb{Q}}(n)/X)$. To identify this with the normalized column sum in (4.9), fix a ray class character ξ. For a nonzero row $k \in \mathcal{O}$ and a squarefree primary f prime to S, set

$$\Psi_k(n) := \xi(n)\chi_n(k)\chi_n(f)^4, \qquad T(X;k,f) := T(X;\Psi_k). \tag{5.4}$$

The displayed product defines $\Psi_k(n)$ for $(n,S) = 1$; set $\Psi_k(n) = 0$ otherwise. Thus the zero extension required in (5.3) is part of this definition.

5.2 The completed mean-square estimate

For the twist (5.4), the theta transformation converts the relevant sextic twists into quadratic characters. Combining it with the quadratic large sieve gives the following estimate, proved in Section 6.

Proposition 5.2. *Fix $\varepsilon > 0$, $C_0 \geq 1$, and a character ξ of the fixed ray class group. There is $J = J(\varepsilon, C_0) \geq 1$ such that*

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(k)\ll\mathcal{H}} |T(X;k,f)|^2 \ll_{I,\xi,S,\varepsilon,C_0} D^{\varepsilon}\|W\|^2_{C^J(I)}\Big(\mathcal{H} + \frac{\mathcal{H}^2\mathrm{N}_{K/\mathbb{Q}}(f)}{X}\Big) \tag{5.5}$$

whenever

$$1 \leq \mathcal{H}, X, \mathrm{N}_{K/\mathbb{Q}}(f) \leq D^{C_0}.$$

Here f is squarefree and primary with $(f,S) = 1$, W is smooth and supported in a compact interval $I \subset (0,\infty)$, and T is defined by (5.4).

5.3 Cube inversion and the remaining sums

Set

$$H_c^3 = \min\Big(X, \frac{X^2}{\mathcal{H}^2}\Big), \qquad L_b = \frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}. \tag{5.6}$$

Lemma 5.3. *Fix $C_0 \geq 1$ and $\varepsilon > 0$. Suppose $1 \leq \mathcal{H}, X, F \leq D^{C_0}$ and $\mathcal{H} \leq \Sigma = XF$. There is an integer $J = J(\varepsilon, C_0) \geq 1$ such that*

$$\mathcal{E}(\mathcal{H},X,F;\xi,W) \ll_{I,\nu,S,C_0,\varepsilon} D^{\varepsilon}\Big(\Sigma\|W\|^2_{C^J(I)} + \sup_{\substack{b\equiv 1\ (3),\ (b,S)=1\\ \mathrm{N}_{K/\mathbb{Q}}(b)>H_c,\ L_b>1}} \mathcal{E}(\mathcal{H},L_b,F;\xi,W)\Big) \tag{5.7}$$

for every compact interval $I \subset (0,\infty)$ and $W \in C_c^\infty(I)$. An empty supremum is zero; in particular it is empty when $\mathcal{H}^2 \leq X$.

Proof. Write $N = \mathrm{N}_{K/\mathbb{Q}}$ and $I = [u,v]$. Complete multiplicativity in (5.3), including at the zeros of Ψ_k, gives

$$\begin{aligned} X^{-1/2} \sideset{}{^*}\sum_{(n,S)=1} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(N(n)/X) \\ = \sum_{(h,S)=1} \frac{\mu(h)\overline{\alpha(h)}^3\Psi_k(h)^3}{N(h)} T(X/N(h)^3;k,f). \end{aligned} \tag{5.8}$$

Indeed, after substitution of (5.3), the coefficient of the total cube index b contains $\sum_{h|b}\mu(h) = \mathbf{1}_{b=1}$. This is Möbius inversion; compare [22, (1.18)] and [7, (8.2)].

Split the right-hand side of (5.8) into $P_{\mathrm{short}}(k,f) + P_{\mathrm{long}}(k,f)$, where

$$P_{\mathrm{short}}(k,f) = \sum_{\substack{(h,S)=1\\ N(h)\leq H_c}} \frac{\mu(h)\overline{\alpha(h)}^3\Psi_k(h)^3}{N(h)} T(X/N(h)^3;k,f),$$

$$\mathcal{E}_{\mathrm{short}} = \frac{1}{F}\sideset{}{^*}\sum_{\substack{F\leq N(f)<2F\\ (f,S)=1}} \sum_{0<N(k)\leq\mathcal{H}} |P_{\mathrm{short}}(k,f)|^2.$$

Define $\mathcal{E}_{\mathrm{long}}$ in the same way using P_{long}. Weighted Cauchy–Schwarz for each k, f, followed by Proposition 5.2, gives

$$\begin{aligned}\mathcal{E}_{\mathrm{short}} &\le \Big(\sum_{N(h)\le H_c}\frac{1}{N(h)}\Big)\sum_{N(h)\le H_c}\frac{1}{N(h)}\frac{1}{F}\sideset{}{^*}\sum_{\substack{F\le N(f)<2F\\(f,S)=1}}\sum_{0<N(k)\le\mathcal{H}}|T(X/N(h)^3;k,f)|^2\\ &\ll_{I,\nu,S,C_0,\varepsilon} D^{\varepsilon/2}\|W\|^2_{C^J(I)}\Big(\mathcal{H}+\frac{\mathcal{H}^2FH_c^3}{X}\Big)\ll D^{\varepsilon/2}\Sigma\|W\|^2_{C^J(I)}.\end{aligned}\tag{5.9}$$

Here $X/N(h)^3\ge 1$, and the reciprocal-norm sum contributes only a logarithm. We use Proposition 5.2 with exponent $\varepsilon/4$ and range C_0+1, since $N(f)<2D^{C_0}$. If $H_c<1$, the sum is empty.

Expand T using (5.3) in the remaining terms, with cube index c:

$$\begin{aligned}P_{\mathrm{long}}(k,f)=\frac{1}{\sqrt{X}}\sum_{\substack{(h,S)=1\\N(h)>H_c}}\sum_{(c,S)=1}\mu(h)\sqrt{N(hc)}\,\overline{\alpha(hc)}^{\,3}\Psi_k(hc)^3\\ \times\sideset{}{^*}\sum_{(n,S)=1}a_\xi(n)\chi_n(k)\chi_n(f)^4W\Big(\frac{N(n)N(hc)^3}{X}\Big).\end{aligned}$$

Put $b=hc$; the pairs giving b are exactly $(h,b/h)$ with $h\mid b$ and $N(h)>H_c$. Since $\Psi_k(b)^3=\xi(b)^3\chi_b(k)^3\mathbf{1}_{(b,f)=1}$, this gives

$$\begin{aligned}P_{\mathrm{long}}(k,f)&=\sum_{\substack{(b,S)=1\\N(b)>H_c}}\frac{\beta_0(b,f)\chi_b(k)^3}{N(b)}\frac{1}{\sqrt{L_b}}\sideset{}{^*}\sum_{(n,S)=1}a_\xi(n)\chi_n(k)\chi_n(f)^4W(N(n)/L_b),\\ \beta_0(b,f)&:=\overline{\alpha(b)}^{\,3}\xi(b)^3\mathbf{1}_{(b,f)=1}\sum_{\substack{h\mid b\\N(h)>H_c}}\mu(h),\qquad |\beta_0(b,f)|\le\tau_{\mathrm{div}}(b).\end{aligned}$$

The divisor function $\tau_{\mathrm{div}}(b)$ counts ideal divisors. All these indices are primary and prime to S; b need not be squarefree. The support of W restricts the sum to $N(b)^3\le vX$. Weighted Cauchy–Schwarz now gives

$$\mathcal{E}_{\mathrm{long}}\le\Big(\sum_{\substack{H_c<N(b)\le(vX)^{1/3}\\(b,S)=1}}\frac{\tau_{\mathrm{div}}(b)}{N(b)}\Big)\sum_{\substack{H_c<N(b)\le(vX)^{1/3}\\(b,S)=1}}\frac{\tau_{\mathrm{div}}(b)}{N(b)}\mathcal{E}(\mathcal{H},L_b,F;\xi,W).$$

The reciprocal divisor sum is $\ll_I\log^2(2+X)$, since

$$\sum_{N(b)\le R}\frac{\tau_{\mathrm{div}}(b)}{N(b)}=\sum_{N(cd)\le R}\frac{1}{N(c)N(d)}\le\Big(\sum_{N(c)\le R}\frac{1}{N(c)}\Big)^2\ll\log^2(2R)\qquad(R\ge1).$$

If $L_b\le 1$, the column sum is empty unless $L_b\ge 1/v$; otherwise it has $O_I(1)$ terms, so

$$\mathcal{E}(\mathcal{H},L_b,F;\xi,W)\ll_I\frac{\mathcal{H}}{L_b}\|W\|^2_\infty\ll_I\Sigma\|W\|^2_\infty.$$

Absorb the logarithms in D^ε and combine the two parts using $\mathcal{E}\le 2\mathcal{E}_{\mathrm{short}}+2\mathcal{E}_{\mathrm{long}}$. This proves (5.7). If $\mathcal{H}^2\le X$, then $H_c=X^{1/3}$, so $N(b)>H_c$ implies $L_b<1$. □

5.4 The transfer estimate

Fix a nonnegative radial Schwartz weight Φ with $\Phi(t) \geq 1$ for $0 \leq t \leq 1$ and $\operatorname{supp} \widehat{\Phi} \subset [0, C_\Phi]$. Such a weight is obtained by squaring and rescaling a real radial Schwartz function with compactly supported Fourier transform. Write $N = \mathrm{N}_{K/\mathbb{Q}}$. For fixed $\mathcal{H}, L, F, \xi$, put

$$\mathcal{A}(W) := \frac{1}{LF} \sideset{}{^*}\sum_{\substack{F \leq N(f) < 2F \\ (f,S)=1}} \sum_{k \in \mathcal{O}} \Phi(N(k)/\mathcal{H}) \Big| \sideset{}{^*}\sum_{(n,S)=1} a_\xi(n) \chi_n(k) \chi_n(f)^4 W(N(n)/L) \Big|^2. \tag{5.10}$$

Thus $\mathcal{E}(\mathcal{H}, L, F; \xi, W) \leq \mathcal{A}(W)$. In the following proposition Σ is an independent target scale; we will apply it at column scale L_b with $\Sigma = XF$.

Proposition 5.4. *Assume $1 \leq \mathcal{H}, L, F, \Sigma \leq D^{C_0}$ and $\max\{\mathcal{H}, LF\} \leq \Sigma$, for fixed $C_0 \geq 1$. For every integer $m \geq 0$ and $\varepsilon > 0$,*

$$\mathcal{A}(W) \ll D^\varepsilon \Sigma \|W\|^2_{C^{4m+12}(I)} \Big(1 + \sup \frac{\mathcal{E}(\mathcal{H}', X', F'; \xi', U)}{\Sigma'}\Big) \tag{5.11}$$

where the supremum is over the family (4.9), with $\Sigma' = X'F'$, $\mathcal{H}', X', F' \geq 1$, and

$$\mathcal{H}' \leq \frac{\mathcal{H}L}{\Sigma F}, \qquad \frac{\mathcal{H}'}{\Sigma'} \leq \frac{\mathcal{H}}{\Sigma}, \qquad \Sigma' \leq L. \tag{5.12}$$

For $I = [u, v]$, the tests satisfy $U \in C_c^\infty(I')$, $I' = [u/16, 4v]$, and $\|U\|_{C^m(I')} \leq 1$; an empty supremum is zero. The estimate holds for every compact $I \subset (0, \infty)$ and $W \in C_c^\infty(I)$, with implied constant depending on $m, C_0, \varepsilon, I, \nu, S$ and the fixed ray class group.

The proof is given in Section 7, by combining Lemmas 7.1 and 7.3.

5.5 Proof of Proposition 5.1

Proof of Proposition 5.1. Fix $\kappa > 0$ and $C_0 \geq 1$. We prove by induction on $j \geq 0$ that the proposition holds under the additional restriction $\mathcal{H} \leq D^{j\kappa}$, with the derivative order and implied constant allowed to depend on j. More precisely, for every $\varepsilon > 0$ there is an integer $J = J(j, \kappa, C_0, \varepsilon) \geq 1$ such that

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I, \nu, S, j, \kappa, C_0, \varepsilon} D^\varepsilon \Sigma \|W\|^2_{C^J(I)}$$

for every compact interval $I = [u, v] \subset (0, \infty)$ and $W \in C_c^\infty(I)$, under the hypotheses of the proposition. The derivative order is independent of I; this allows us to apply the induction hypothesis on the enlarged interval in the transfer estimate. For the base case $j = 0$, we have $\mathcal{H} \leq 1$. In (4.9), the support of W restricts n to the $O_I(X)$ ideals with $\mathrm{N}_{K/\mathbb{Q}}(n) \in [uX, vX]$. The factors $a_\xi(n)$, $\chi_n(k)$, and $\chi_n(f)^4$ have absolute value at most one, so the triangle inequality bounds each inner sum by $O_I(X\|W\|_\infty)$. There are $O(F)$ choices of f and $O(\mathcal{H})$ choices of k. Consequently,

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_I \frac{F\mathcal{H}}{XF} \big(X\|W\|_\infty\big)^2 = \mathcal{H}X\|W\|^2_\infty.$$

Dividing by $\Sigma = XF$ and using $\mathcal{H} \leq 1$ and $F \geq 1$ proves the assertion for $j = 0$, with $J = 1$.

Suppose the assertion holds for j, and fix $\varepsilon > 0$. Let m be the derivative order supplied by the induction hypothesis with exponent $\varepsilon/3$. For $\mathcal{H} \leq D^{(j+1)\kappa}$, apply Lemma 5.3 with exponent $\varepsilon/3$. We must bound the mean squares in its supremum uniformly in b. If this supremum is nonempty, then $\mathcal{H}^2 > X$ and $H_c^3 = X^2/\mathcal{H}^2$. For each such b, recall that $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3 > 1$. Since $L_b \leq X$ and $\Sigma \geq \max\{\mathcal{H}, L_b F\}$, Proposition 5.4 applies at column scale L_b, with derivative order m, and exponent $\varepsilon/3$. Its estimate also bounds $\mathcal{E}(\mathcal{H}, L_b, F; \xi, W)$ because $\mathcal{E}(\mathcal{H}, L_b, F; \xi, W) \leq \mathcal{A}(W)$.

Let $\mathcal{H}', X', F'$ be any parameters in the supremum in (5.11), with $\Sigma' = X'F'$ as in that proposition. By (5.12) and $\mathrm{N}_{K/\mathbb{Q}}(b) > H_c$,

$$\mathcal{H}' \le \frac{\mathcal{H}L_b}{\Sigma F} = \frac{\mathcal{H}}{\mathrm{N}_{K/\mathbb{Q}}(b)^3F^2} < \mathcal{H}\Big(\frac{\mathcal{H}}{\Sigma}\Big)^2 \le D^{-2\kappa}\mathcal{H} \le D^{j\kappa},$$
$$\frac{\mathcal{H}'}{\Sigma'} \le \frac{\mathcal{H}}{\Sigma} \le D^{-\kappa}, \qquad \Sigma' \le L_b \le \Sigma \le D^{C_0}.$$

Thus each of these mean squares is covered by the induction hypothesis. Its weight U is supported in $[u/16, 4v]$ and has C^m norm at most one. Applying the induction hypothesis on this interval gives

$$\frac{\mathcal{E}(\mathcal{H}', X', F'; \xi', U)}{\Sigma'} \ll_{I,\nu,S,j,\kappa,C_0,\varepsilon} D^{\varepsilon/3}.$$

The enlarged interval is determined by I, so the implied constant has only the permitted dependence on the support. Substituting into (5.11) yields, uniformly in b,

$$\mathcal{E}(\mathcal{H}, L_b, F; \xi, W) \ll_{I,\nu,S,j,\kappa,C_0,\varepsilon} D^{2\varepsilon/3}\Sigma\|W\|^2_{C^{4m+12}(I)}.$$

Substitution into (5.7) contributes the remaining factor $D^{\varepsilon/3}$. Choose J at least $4m+12$ and at least the derivative order required by Lemma 5.3. We obtain

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I,\nu,S,j,\kappa,C_0,\varepsilon} D^{\varepsilon}\Sigma\|W\|^2_{C^J(I)}.$$

The choice of J depends only on $j, \kappa, C_0, \varepsilon$. If the supremum in (5.7) is empty, the same bound follows directly from that lemma. This completes the induction.

Finally, take $j = \lceil C_0/\kappa \rceil$. The hypothesis $\mathcal{H} \le \Sigma \le D^{C_0}$ ensures $\mathcal{H} \le D^{j\kappa}$, so the induction gives the proposition. □

5.6 The original mean square and the exponent 11/12

Proof of Proposition 3.1. Fix $0 < \vartheta \le 1/10$ and put $H = D^{1+\vartheta}$. The ranges in (4.11) give

$$\mathcal{H} \le \frac{CD^{1-\vartheta}}{B^2}, \qquad \Sigma = XF = \frac{D}{B}, \qquad \frac{\mathcal{H}}{\Sigma} \ll D^{-\vartheta}. \tag{5.13}$$

For large D, Proposition 5.1 applies with $\kappa = \vartheta/2$ and $C_0 = 2$. If $X < 1$, nonempty support forces $X \gg_I 1$, and counting gives $\mathcal{E} \ll_I \mathcal{H}\|W\|^2_\infty \ll \Sigma\|W\|^2_\infty$; if $\mathcal{H} < 1$, the sum is empty. Thus (4.12) holds, with derivative order depending only on ϑ, ε, and Proposition 4.5 proves (3.1). Bounded D is again covered by counting. □

6 Proof of the completed mean-square estimate

We prove Proposition 5.2. We first express the completed sum (5.3) using cubic theta coefficients and state the transformation formula. We then apply the quadratic large sieve and account for repeated prime factors in k.

6.1 Realization by the cubic theta function

With the row k and auxiliary index f fixed, we express $T(X; k, f)$ as a weighted sum of Fourier coefficients of the cubic theta function. Its automorphy then expresses this sum in terms of coefficients at other cusps, as in [7, §5 and Appendix A]. We use Ψ_k from (5.4), suppressing its dependence on the fixed f and ξ.

Write $(z, v) \in \mathbb{C} \times \mathbb{R}_{>0}$ for upper half-space coordinates, with horizontal coordinate z and height v. We use Kubota's cubic theta function in the normalization of [7, §5.1, (5.6)]:

$$\theta(z,v) = \frac{3^{5/2}}{2} v^{2/3} + \sum_{\substack{\ell \in \lambda^{-3}\mathcal{O} \\ \ell \neq 0}} \tau(\ell) v K_{1/3}(4\pi|\ell|v) \exp\bigl(2\pi\mathrm{i}(\ell z + \overline{\ell z})\bigr). \tag{6.1}$$

Here $K_{1/3}$ is the modified Bessel function of the second kind, and $\tau(\ell)$ is the coefficient sequence given explicitly in [7, (5.7)–(5.8)], following Patterson's calculation [38, Theorem 8.1].

We define $\Theta_k(z, v)$ by twisting the Fourier coefficients of $\overline{\theta}$. First define $\phi_k : \mathcal{O} \to \mathbb{C}$ by

$$\phi_k(n) = \begin{cases} \chi_n(\lambda)^2 \Psi_k(n), & n \equiv 1 \pmod 3, \quad (n, S) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then we define $\Theta_k(z, v)$ by multiplying the Fourier coefficient of $\overline{\theta}$ at ℓ, which is $\overline{\tau(-\ell)}$, by $\phi_k(\lambda^3 \ell)$:

$$\Theta_k(z,v) = \sum_{\substack{\ell \in \lambda^{-3}\mathcal{O} \\ \ell \neq 0}} \overline{\tau(-\ell)} \phi_k(\lambda^3 \ell) v K_{1/3}(4\pi|\ell|v) \exp\bigl(2\pi\mathrm{i}(\ell z + \overline{\ell z})\bigr). \tag{6.2}$$

Recall that $c_\theta(nb^3)$ is defined in (2.10). Choose a nonzero $q \in \mathcal{O}$ such that ϕ_k is periodic modulo (q), and put

$$\widehat{\phi_k}(h) = \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(q)} \sum_{n \bmod q} \phi_k(n) e(-hn/q).$$

Lemma 6.1. *With these definitions,*

$$T(X; k, f) = \frac{1}{3^{5/2}\sqrt{X}} \sum_{\substack{n, b \in \mathcal{O} \\ n, b \equiv 1 \ (3) \\ n \text{ squarefree} \\ (nb, S) = 1}} c_\theta(nb^3) \phi_k(nb^3) \overline{\alpha(nb^3)} W\Bigl(\frac{\mathrm{N}_{K/\mathbb{Q}}(nb^3)}{X}\Bigr). \tag{6.3}$$

Moreover,

$$\Theta_k(z,v) = \sum_{h \in \mathcal{O}/(q)} \widehat{\phi_k}(h) \overline{\theta(z + \lambda^2 h/q, v)}. \tag{6.4}$$

Proof. The coefficient formula (2.10) gives

$$c_\theta(nb^3)\phi_k(nb^3) = 3^{5/2}|b|\gamma_2(n)\Psi_k(n)\Psi_k(b)^3.$$

Substitution into (5.3), using $V_*(y) = \sqrt{y}\, W(y)$, proves (6.3). Finite Fourier inversion gives

$$\phi_k(n) = \sum_{h \in \mathcal{O}/(q)} \widehat{\phi_k}(h) e(nh/q), \qquad \sum_{h \in \mathcal{O}/(q)} \widehat{\phi_k}(h) = \phi_k(0) = 0.$$

Multiplication of the ℓth Fourier mode by $e(\lambda^3 \ell h/q)$ translates z to $z + \lambda^2 h/q$. The second identity cancels the constant term, proving (6.4). □

The resulting transformation formula is stated in the next subsection; its automorphy calculation is given in Appendix A.2.

6.2 The theta transformation

Recall that, for a primary prime $p \notin S$, $\chi_p(x) = (x/p)_6$ is the sextic residue character on $(\mathcal{O}/(p))^\times$. For every integer j, write χ_p^j for its jth power on this group, extended by zero on multiples of p; in particular, $\chi_p^0(x) = \mathbf{1}_{p\nmid x}$. Sextic reciprocity (4.1) expresses the factors of Ψ_k at primes outside S as such powers, with $0 \le j \le 5$.

For a primary prime $p \notin S$, $j \in \{0, \dots, 5\}$, and $x \in \mathcal{O}$, define the local factor

$$B_{p,j}(x) = \begin{cases} \chi_p(x)^{-j-2}, & j \neq 0, 4, \\ \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2}(-1 + \mathrm{N}_{K/\mathbb{Q}}(p)\mathbf{1}_{p|x}), & j = 4, \\ \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2}\chi_p(x)^{-2}, & j = 0. \end{cases} \tag{6.5}$$

The key case is the quadratic character

$$B_{p,1}(x) = \chi_p(x)^3.$$

For the smooth compactly supported weight V_* in (5.3), put $\widehat{V}_*(s) = \int_0^\infty V_*(x)x^s\,dx/x$. We write $\int_{(\sigma)}$ for integration upwards along the vertical line with real part σ. With Γ denoting Euler's gamma function, the accompanying transform of the weight is

$$V_*^\sharp(x) = \frac{1}{2\pi\mathrm{i}} \int_{(0)} \widehat{V}_*(-t) \frac{\Gamma(7/6+t)\Gamma(5/6+t)}{\Gamma(7/6-t)\Gamma(5/6-t)} \Big(\frac{(2\pi)^4 x}{27}\Big)^{-t} dt, \qquad x > 0. \tag{6.6}$$

Here θ is Kubota's cubic theta function, normalized in (6.1). The dual sums use the Fourier coefficients of $\overline{\theta}$ at three cusps. Take the representatives

$$\gamma_0 = I, \qquad \gamma_+ = \begin{pmatrix} 1 & 0 \\ \omega & 1 \end{pmatrix}, \qquad \gamma_- = \begin{pmatrix} 1 & 0 \\ \omega^2 & 1 \end{pmatrix}, \tag{6.7}$$

acting on upper half-space. For $\sigma \in \{0, +, -\}$, define $d_\sigma(\ell)$ by the Fourier expansion [7, (5.9), (5.15)]

$$\overline{\theta(\gamma_\sigma(z,v))} = \frac{3^{5/2}}{2}\mathbf{1}_{\sigma=0}v^{2/3} + \sum_{0\neq\ell\in\lambda^{-4}\mathcal{O}} d_\sigma(\ell) v K_{1/3}(4\pi|\ell|v) \exp\big(2\pi\mathrm{i}(\ell z + \overline{\ell z})\big). \tag{6.8}$$

Thus $d_0(\ell) = \overline{\tau(-\ell)}$, with τ extended by zero outside $\lambda^{-3}\mathcal{O}$. In particular, $c_\theta(nb^3) = d_0(\lambda^{-3}nb^3)$ for the indices in (2.10); the outline uses $d_\theta(m)$ for $d_\sigma(\lambda^{-4}m)$ with one of these three choices of σ. The arithmetic formulas for all three sequences are given by (A.6) and (A.7) in Appendix A.2.

We state the formula for a general product of local twists. Fix a ray class character whose conductor is supported on S, and let Ψ_0 be its extension by zero at every prime in S. Let $\mathcal{P}$ be a finite set of primes outside S, each represented by its primary generator in $\mathcal{O}$, and choose integers $0 \le j_p \le 5$ for each $p \in \mathcal{P}$. For primary $n \in \mathcal{O}$, set

$$\Psi(n) = \Psi_0(n) \prod_{p\in\mathcal{P}} \chi_p^{j_p}(n).$$

In the transformed sums, $\mathcal{A}$ will range over subsets satisfying

$$\{p \in \mathcal{P} : j_p \neq 0\} \subseteq \mathcal{A} \subseteq \mathcal{P}.$$

Thus only primes with $j_p = 0$ may be omitted from $\mathcal{A}$. We call the primes in $\mathcal{A}$ *active* and those in $\mathcal{P} \setminus \mathcal{A}$ *inactive*. For such a subset and $c_0 \in \mathcal{O} \setminus \{0\}$, write

$$c = c_0 \prod_{p\in\mathcal{A}} p.$$

Proposition 6.2. *The quantity $T(X;\Psi)$ defined in (5.3) is a sum of $O_{\Psi_0,S}(2^{|\mathcal{P}|})$ terms of the form*

$$C \sum_{0\neq \ell\in\lambda^{-4}\mathcal{O}} \frac{d(\ell)\alpha(\ell)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\ell)}}\,\psi(\lambda^4\ell)\prod_{p\in\mathcal{A}} B_{p,j_p}(\lambda^4\ell)V_*^{\sharp}\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(\ell)X}{\mathrm{N}_{K/\mathbb{Q}}(c)^2}\Big). \tag{6.9}$$

Here $\mathcal{A}$ and c are as above, and $|C| \ll_{\Psi_0,S} 1$. The triples (d,ψ,c_0) belong to a fixed finite family depending only on Ψ_0, S, where $d\in\{d_0,d_+,d_-\}$, $c_0\in\mathcal{O}\setminus\{0\}$, and ψ is a unit-modulus additive character on $\mathcal{O}$.

To average the transformed sums over k_0, we need to choose their cusp coefficients and additive characters consistently as k_0 varies. Fix a finite set $\mathcal{P}_{\mathrm{fix}}$ of primary primes outside S and exponents $j_p\in\{0,\dots,5\}$, and put

$$\Psi_{k_0}(n) = \Psi_0(n)\prod_{p\in\mathcal{P}_{\mathrm{fix}}}\chi_p^{j_p}(n)\prod_{p\mid k_0}\chi_p(n),$$

where k_0 is primary and squarefree, with $(k_0, S\prod_{p\in\mathcal{P}_{\mathrm{fix}}}p)=1$. Let $\mathcal{A}_{\mathrm{fix}}$ range over the subsets satisfying

$$\{p\in\mathcal{P}_{\mathrm{fix}} : j_p\neq 0\}\subseteq\mathcal{A}_{\mathrm{fix}}\subseteq\mathcal{P}_{\mathrm{fix}},\qquad \mathcal{A}=\mathcal{A}_{\mathrm{fix}}\cup\{p : p\mid k_0\}.$$

Every prime dividing k_0 belongs to $\mathcal{A}$, since its exponent is 1.

Lemma 6.3 (Uniformity in the twist)**.** *For the family Ψ_{k_0} above, the summands in Proposition 6.2 may be indexed by $(h,\mathcal{A}_{\mathrm{fix}})$, with h in a fixed finite set depending only on Ψ_0, S. Zero scalar coefficients are permitted. For each fixed index, the triple (d,ψ,c_0) depends on k_0 only through its ray class modulo a fixed ideal supported on S and depending only on Ψ_0, S.*

To estimate the dual sums, we need bounds for the cusp coefficients and the transformed weight.

Lemma 6.4. *For each $d\in\{d_0,d_+,d_-\}$, the coefficient $d(\ell)$ vanishes unless ℓ can be written as*

$$\ell = u\lambda^m n b^3, \tag{6.10}$$

where $u\in\mathcal{O}^\times$, $m\in\mathbb{Z}$ with $m\geq -4$, and $n,b\in\mathcal{O}$ are primary with n squarefree.

$$|d(\ell)| = |d(u\lambda^m n b^3)| \leq 27\cdot 3^{m/6}|b|. \tag{6.11}$$

For $A>0$, integers $j\geq 0$, and V_* supported in a fixed compact interval $I\subset(0,\infty)$, there is $J=J(A,j)$ such that

$$|(x\partial_x)^j V_*^{\sharp}(x)| \ll_{A,j,I} \|V_*\|_{C^J(I)}\min(x^{1/4},x^{-A}),\qquad x>0. \tag{6.12}$$

Proposition 6.2 and Lemmas 6.3 and 6.4 are proved in Appendix A.2.

6.3 The quadratic mean square on the dual side

We record the quadratic large-sieve estimate needed below.

Lemma 6.5. *Let $\mathcal{H}, U\geq 1$, and let $\beta(n)$ be complex coefficients on squarefree primary n with $\mathrm{N}_{K/\mathbb{Q}}(n)\asymp U$. For every $\varepsilon>0$,*

$$\sum_{\substack{k\equiv 1\ (3),\ (k,S)=1\\ \mathrm{N}_{K/\mathbb{Q}}(k)\leq\mathcal{H}}}^{*}\Big|\sum_{\substack{n\equiv 1\ (3),\ n\ \text{squarefree}\\ \mathrm{N}_{K/\mathbb{Q}}(n)\asymp U}}\beta(n)\chi_k(n)^3\Big|^2 \ll_{S,\varepsilon} (\mathcal{H}U)^{\varepsilon}(\mathcal{H}+U)\sum_n|\beta(n)|^2. \tag{6.13}$$

The star restricts k to squarefree primary elements.

Proof. This is Goldmakher and Louvel's quadratic large sieve [13, Theorem 1.1], after fixing the product of the prime factors of n lying in S, and finitely many ray classes. For completeness, let $e_k \in \{0,1\}$ according as $\mathrm{N}_{K/\mathbb{Q}}(k) \equiv 1,3 \pmod 4$, and let κ_λ be the nontrivial character modulo λ. The character $x \mapsto (x/k)_2 \kappa_\lambda(x)^{e_k}$ is trivial on units and has primitive conductor $k\lambda^{e_k}$. When $e_k = 0$, the factor $\kappa_\lambda(x)^{e_k}$ is omitted. Classes modulo $24\mathcal{O}$ fix the supplementary characters and reciprocity factors; for coprime k_1, k_2 in the same class, the product character has conductor $k_1 k_2$. These are the hypotheses in [13, Definition 1 and §2]. □

6.4 Squarefree rows and the completed bound

Write $N(a) = \mathrm{N}_{K/\mathbb{Q}}(a)$. In (5.4), allow any $g \in \mathcal{O}\setminus\{0\}$ in place of f, so $\Psi_k(n) = \xi(n)\chi_n(k)\chi_n(g)^4$. The zero extension at S is retained. Choose prime-ideal generators, primary away from 3, and extend multiplicatively to all ideals. We first bound squarefree rows.

Lemma 6.6. *For every $\varepsilon > 0$ and $C_0 \ge 1$ there is $J = J(\varepsilon, C_0) \ge 1$ such that*

$$\sum_{\substack{0<N(s)\le \mathcal{H} \\ s \text{ squarefree}}} |T(X; u_0 s, g)|^2 \ll_{I,\xi,S,\varepsilon,C_0} D^{\varepsilon} \|W\|_{C^J(I)}^2 \Big(\mathcal{H} + \frac{\mathcal{H}^2 N(g)}{X}\Big) \tag{6.14}$$

for $1 \le \mathcal{H}, X, N(g) \le D^{C_0}$, $u_0 \in \mathcal{O}^\times$, every compact interval $I \subset (0,\infty)$, and $W \in C_c^\infty(I)$. The sum uses the chosen generators of squarefree ideals, including those meeting S.

Proof. By homogeneity assume $\|W\|_{C^J(I)} \le 1$, with J chosen below. Write

$$s = tk_0, \qquad t = \prod_{\substack{p\mid s \\ p\mid g \text{ or } p\in S}} p, \qquad N(k_0) \le \mathcal{H}_0 := \frac{\mathcal{H}}{N(t)}.$$

Fix t; then k_0 is squarefree and primary, with $(k_0, g) = 1$ and $(k_0, S) = 1$. Discard empty ranges, so $\mathcal{H}_0 \ge 1$, and retain these restrictions below. For a coefficient function $A(n,b)$, a positive scale Y, and an integer $m \ge -4$, put

$$\mathscr{S}_m[A,Y](k_0) := \sum_{\substack{n,b\equiv 1 \ (3) \\ n \text{ squarefree}}} \frac{A(n,b)\chi_{k_0}(nb)^3}{3^{m/3}\sqrt{N(n)}\,N(b)} V_*^{\sharp}\Big(\frac{3^m N(n) N(b)^3 \mathcal{H}_0^2}{Y N(k_0)^2}\Big). \tag{6.15}$$

We shall obtain, after partitioning k_0 into fixed ray classes,

$$T(X; u_0 t k_0, g) = \sum_{\iota\in\mathcal{I}} \sum_{m\ge -4} c_{\iota,m}(k_0) \mathscr{S}_m[A_{\iota,m}, Y_\iota](k_0),$$

where $\mathcal{I}$ is independent of k_0, $|\mathcal{I}| \preccurlyeq 1$, $|c_{\iota,m}(k_0)| \ll 1$, and $A_{\iota,m}$ is independent of k_0. For some $a_\iota, Y_\iota > 0$, the coefficients and lengths satisfy

$$|A_{\iota,m}(n,b)| \ll a_\iota, \qquad a_\iota^2 \le 1, \qquad a_\iota^2 Y_\iota \ll \frac{\mathcal{H}_0^2}{X} N(t)^2 N(g). \tag{6.16}$$

All column restrictions are included by extending $A_{\iota,m}$ by zero. Put $k = u_0 t k_0$. For $\mathcal{P} = \{p \notin S : p \mid kg\}$, sextic reciprocity (4.1) gives

$$\Psi_k(n) = \Psi_0(n) \prod_{p\in\mathcal{P}} \chi_p^{j_p}(n), \qquad j_p \equiv v_p(k) + 4v_p(g) \pmod 6, \qquad 0 \le j_p \le 5. \tag{6.17}$$

Even when $j_p = 0$, the factor $\chi_p^0(n) = \mathbf{1}_{p\nmid n}$ retains the zero extension at $p \mid kg$. The fixed factor Ψ_0 contains ξ, the factors at S, the unit factors, and $n \mapsto \mathcal{R}(n, \prod_{p\notin S} p^{v_p(k)})$. The fourth power at g

contributes no reciprocity sign, and the factors at S depend only on exponents modulo six. Thus Ψ_0 ranges over a fixed finite family. In the notation of Lemma 6.3, take $\mathcal{P}_{\mathrm{fix}} = \{p \notin S : p \mid tg\}$, so that $\mathcal{P} = \mathcal{P}_{\mathrm{fix}} \sqcup \{p : p \mid k_0\}$; the exponents on $\mathcal{P}_{\mathrm{fix}}$ are fixed with t, g. Partitioning k_0 into fixed ray classes fixes Ψ_0 and, by Lemma 6.3, the data d, ψ, c_0 in each transformed term.

Fix one transformed term, with active primes $\mathcal{A}_{\mathrm{fix}}$ away from k_0, and put $c_* = c_0 \prod_{p \in \mathcal{A}_{\mathrm{fix}}} p$. Thus $c = c_* k_0$ in (6.9). By (6.10), write $\ell = u\lambda^m n b^3$, with u, m, n, b as there. Every prime dividing k_0 has $j_p = 1$, and hence

$$B_{p,1}(u\lambda^{m+4} n b^3) = \chi_p(u\lambda^{m+4})^3 \chi_p(nb)^3. \tag{6.18}$$

Indeed, the transformed exponent is $-1-2 \equiv 3 \pmod 6$ and $\chi_p(b)^9 = \chi_p(b)^3$, also when $p \mid b$ by zero extension. The coefficient bound and rapid decay in Lemma 6.4 give absolute convergence of the transformed series, so we may regroup its terms below. Set

$$A_{u,m}(n,b) = \frac{d(u\lambda^m n b^3)\alpha(u\lambda^m n b^3)\psi(u\lambda^{m+4} n b^3)}{3^{m/6}\sqrt{N(b)}}, \qquad |A_{u,m}(n,b)| \le 27.$$

The transformed term is therefore

$$C(k_0) \sum_{u \in \mathcal{O}^\times} \sum_{m \ge -4} \chi_{k_0}(u\lambda^{m+4})^3 \sum_{\substack{n,b \equiv 1 \ (3) \\ n \text{ squarefree}}} \frac{A_{u,m}(n,b)\chi_{k_0}(nb)^3}{3^{m/3}\sqrt{N(n)}\,N(b)}$$
$$\times \prod_{p \in \mathcal{A}_{\mathrm{fix}}} B_{p,j_p}(u\lambda^{m+4} n b^3) V_*^{\sharp}\Big(\frac{3^m N(n) N(b)^3 X}{N(c_*)^2 N(k_0)^2}\Big), \qquad |C(k_0)| \ll 1.$$

The representation is unique: away from λ, the prime exponents of ℓ are $3v_p(b)$ or $1 + 3v_p(b)$. Apart from the quadratic character and weight, the k_0-dependence is a bounded scalar.

Fix u, m and write $q = N(p)$ for an active prime $p \nmid k_0$. For $j_p = 4$, the local identity is

$$B_{p,4}(u\lambda^{m+4} n b^3) = -q^{-1/2} + q^{1/2}\mathbf{1}_{p \mid n} + q^{1/2}\mathbf{1}_{p \nmid n,\ p \mid b}.$$

To display the reindexing in (6.15), let $A^{(p)}(n,b)$ include all the other local factors and set

$$A^{(p,n)}(n,b) = \mathbf{1}_{p \nmid n} A^{(p)}(pn,b), \qquad A^{(p,b)}(n,b) = \mathbf{1}_{p \nmid n} A^{(p)}(n,pb).$$

Then the changes $n = pn'$ and $b = pb'$ give the exact identity

$$\begin{aligned}
&\mathscr{S}_m[A^{(p)}(n,b) B_{p,4}(u\lambda^{m+4} n b^3), q^2 Y](k_0) \\
&\quad = -q^{-1/2}\mathscr{S}_m[A^{(p)}, q^2 Y](k_0) + \chi_{k_0}(p)^3 \mathscr{S}_m[A^{(p,n)}, qY](k_0) \\
&\qquad + q^{-1/2}\chi_{k_0}(p)^3 \mathscr{S}_m[A^{(p,b)}, Y/q](k_0).
\end{aligned}$$

In the second term $p \nmid n'$ preserves squarefreeness; in the third, $p \nmid n$ is retained and b' is unrestricted at p. Both extracted phases have modulus one. The factors $q^{1/2}$ in the local identity cancel against $\sqrt{N(pn')}$ or leave $q^{-1/2}$ after division by $N(pb')$.

Let a track a bound $|A(n,b)| \le 27a$ for the coefficient in (6.15), and let Y be its scale. Before inserting the active primes, these are $a = 1$ and $Y = N(c_0)^2 \mathcal{H}_0^2 / X$. Each active prime contributes q^2 to the squared conductor norm. Including this factor, the updates are

$$(a^2, Y) \longmapsto \begin{cases} (a^2, q^2 Y), & j_p \notin \{0,4\}, \\ (a^2/q, q^2 Y), & j_p = 0, \\ (a^2/q, q^2 Y),\ (a^2, qY),\ (a^2/q, Y/q), & j_p = 4. \end{cases}$$

Thus a^2 never increases. If $p \mid t$ and $p \notin S$, then j_p is odd, so $a^2 Y$ costs at most $N(p)^2$. If $p \mid g$ and $p \nmid t$, the cost is $N(p)$ for $j_p = 0, 4$ and $N(p)^2$ for $j_p = 2$; the latter case requires $v_p(g) \geq 2$. Hence

$$a^2 Y \ll \frac{\mathcal{H}_0^2}{X} \prod_{\substack{p \mid t \\ p \notin S}} N(p)^2 \prod_{\substack{p \mid g \\ p \nmid t,\ p \notin S}} N(p)^{v_p(g)} \leq \frac{\mathcal{H}_0^2 N(t)^2 N(g)}{X}.$$

Reindexing at distinct primes preserves the form and zero extensions. By Lemma 6.3, the transformed terms, units, and at most three choices per prime $p \in \mathcal{A}_{\mathrm{fix}}$ with $j_p = 4$ form an index set of divisor-bounded size in tg. Here and below, divisor-bounded in a means bounded by $C\tau_{\mathrm{div}}(a)^A$ for fixed constants C, A; in particular, this is $\ll_\varepsilon \mathrm{N}_{K/\mathbb{Q}}(a)^\varepsilon$ for every $\varepsilon > 0$. For one index ι, abbreviate $a = a_\iota$, $Y = Y_\iota$, and $A_m = A_{\iota,m}$. The local updates give $Y \ll D^{C_1}$ for some fixed $C_1 = C_1(C_0)$. Split b into $B \leq N(b) < 2B$, and n by a smooth dyadic partition $V(N(n)/U)$, with $U, B \geq 1$ and uniformly bounded cutoffs. For this part of (6.15), write

$$\mathscr{S}_{m;U,B}(k_0) = 3^{-m/3} \sum_{B \leq N(b) < 2B} \frac{\chi_{k_0}(b)^3}{N(b)} F_b(k_0),$$

$$F_b(k_0) = \sum_{n \text{ squarefree}} \frac{A_m(n,b)\chi_{k_0}(n)^3}{\sqrt{N(n)}} V(N(n)/U) V_*^\sharp\Big(\frac{3^m N(n) N(b)^3 \mathcal{H}_0^2}{Y N(k_0)^2}\Big).$$

Here n, b remain primary. Put $z_* = 3^m U B^3 / Y$ and fix a decay exponent $A > 0$. With m, U, B, Y fixed, Lemma B.1, applied only in $N(n)/U$, gives the following representation in a real Mellin variable s:

$$\begin{aligned} F_b(k_0) &= \int_{\mathbb{R}} c_{b,k_0}(s) G_{b,s}(k_0)\, ds, \\ G_{b,s}(k_0) &= \sum_{\substack{n \text{ squarefree} \\ N(n) \asymp U}} \frac{A_m(n,b)\chi_{k_0}(n)^3}{\sqrt{N(n)}} (N(n)/U)^{\mathrm{i}s}, \\ |c_{b,k_0}(s)| &\ll_{I,A} (1+z_*)^{-A}(1+|s|)^{-2}. \end{aligned} \tag{6.19}$$

Indeed, the scale R in (B.2) is $3^m U N(b)^3 \mathcal{H}_0^2 / (Y N(k_0)^2) \geq z_*$. Lemma 6.4 supplies the required derivative bounds, independently of b, k_0, U, B, m. Since $\sum_{N(n) \asymp U} N(n)^{-1} \ll 1$, Lemma 6.5 gives, for any $\sigma > 0$,

$$\sideset{}{^*}\sum_{N(k_0) \leq \mathcal{H}_0} |G_{b,s}(k_0)|^2 \ll_{S,\sigma} a^2 (\mathcal{H}_0 U)^\sigma (\mathcal{H}_0 + U).$$

Apply weighted Cauchy–Schwarz to the Mellin integral, as recorded in (B.6), with the common majorant in (6.19), then weighted Cauchy–Schwarz in b, using $\sum_{B \leq N(b) < 2B} N(b)^{-1} \ll 1$:

$$\begin{aligned} \sideset{}{^*}\sum_{N(k_0) \leq \mathcal{H}_0} |\mathscr{S}_{m;U,B}(k_0)|^2 &\leq 3^{-2m/3} \Big(\sum_{B \leq N(b) < 2B} \frac{1}{N(b)} \Big) \sum_{B \leq N(b) < 2B} \frac{1}{N(b)} \sideset{}{^*}\sum_{N(k_0) \leq \mathcal{H}_0} |F_b(k_0)|^2 \\ &\ll_{I,S,A,\sigma} a^2 3^{-2m/3} (\mathcal{H}_0 U)^\sigma (\mathcal{H}_0 + U)(1+z_*)^{-2A}. \end{aligned} \tag{6.20}$$

Since $m \geq -4$, we have $U \leq 81 Y z_*$ and hence $\mathcal{H}_0 U \ll D^{C_0 + C_1}(1 + z_*)$. Choose $0 < \sigma \leq 1$ small in terms of $\varepsilon_1 > 0, C_0$ and take $A = 3$. Taking square roots in (6.20) gives

$$\Big(\sideset{}{^*}\sum_{N(k_0) \leq \mathcal{H}_0} |\mathscr{S}_{m;U,B}(k_0)|^2 \Big)^{1/2} \ll a D^{\varepsilon_1} 3^{-m/3} (\sqrt{\mathcal{H}_0} + \sqrt{U})(1 + 3^m U B^3 / Y)^{-2}.$$

The infinite dyadic sums converge: for dyadic $U, B \geq 1$ and every $Z > 0$,

$$\sum_{U,B \text{ dyadic}} (1 + UB^3/Z)^{-2} \ll \log^2(2+Z), \qquad \sum_{U,B \text{ dyadic}} \sqrt{U}(1 + UB^3/Z)^{-2} \ll \sqrt{Z}.$$

These estimates make the sum of $\ell^2(k_0)$ norms finite. The triangle inequality and the geometric sum over $m \ge -4$ therefore give

$$\Big(\sideset{}{^*}\sum_{N(k_0)\le \mathcal{H}_0} \Big| \sum_{m\ge -4} c_{\iota,m}(k_0)\mathscr{S}_m[A_m, Y](k_0)\Big|^2 \Big)^{1/2} \ll aD^{\varepsilon_1}\big(\sqrt{\mathcal{H}_0}\log^2(2+Y) + \sqrt{Y}\big).$$

Combine the divisor-bounded choices of ι and use (6.16) to obtain

$$\sideset{}{^*}\sum_{N(k_0)\le \mathcal{H}_0} |T(X; u_0 t k_0, g)|^2 \preccurlyeq \max_\iota a_\iota^2(\mathcal{H}_0 + Y_\iota) \ll \mathcal{H}_0 + \frac{\mathcal{H}_0^2 N(t)^2 N(g)}{X} \le \mathcal{H} + \frac{\mathcal{H}^2 N(g)}{X}.$$

Sum the fixed ray classes and the divisor-bounded choices $t \mid \mathrm{rad}(g\prod_{\mathfrak{p}\in S}\mathfrak{p})$, choosing ε_1 and the other small-power losses in terms of ε. The weight estimates above require a fixed number $J = J(\varepsilon, C_0)$ of derivatives. Homogeneity restores $\|W\|^2_{C^J(I)}$ and proves (6.14). □

Proof of Proposition 5.2. Write uniquely $k = u_0 s v^2$, with s squarefree and s, v among the chosen ideal generators. No condition $(s, v) = 1$ is imposed. Since $8 \equiv 2 \pmod 6$, including the zero extensions,

$$\chi_n(u_0 s v^2)\chi_n(f)^4 = \chi_n(u_0 s)\chi_n(f v^2)^4, \qquad T(X; u_0 s v^2, f) = T(X; u_0 s, f v^2).$$

Apply Lemma 6.6 with row bound $\mathcal{H}/N(v)^2$ and auxiliary twist $g = fv^2$. Here $N(g) \le \mathcal{H}N(f) \le D^{2C_0}$, so use that lemma with $2C_0$. Summing its bounds gives

$$\begin{aligned}\sum_{0<N(k)\le\mathcal{H}} |T(X;k,f)|^2 &= \sum_{u_0\in\mathcal{O}^\times} \sum_{N(v)^2\le\mathcal{H}} \sum_{\substack{N(s)\le \mathcal{H}/N(v)^2 \\ s \text{ squarefree}}} |T(X; u_0 s, f v^2)|^2 \\ &\ll D^{\varepsilon}\|W\|^2_{C^J(I)}\Big(\mathcal{H} + \frac{\mathcal{H}^2 N(f)}{X}\Big)\sum_v \frac{1}{N(v)^2}.\end{aligned}$$

The last sum is $\zeta_K(2) < \infty$, where $\zeta_K(s) = L_K(s, \mathbf{1})$ is the Dedekind zeta function of K. This proves (5.5). □

7 Proof of the transfer proposition

We prove Proposition 5.4 for $\mathcal{A}(W)$ defined in (5.10). Throughout this section, $\mathcal{H}, L, F, \Sigma, \xi$ satisfy the hypotheses of that proposition. We retain the fixed weight Φ and the support bound C_Φ chosen before (5.10). The enlargement by $\Sigma/(LF) \ge 1$ in the intermediate mean square makes the second Poisson summation return to (4.9) with row range at most $\mathcal{H}L/(\Sigma F)$. Lemma B.2 separates the weights, and Lemma 4.4 removes the remaining exclusion.

7.1 First application of Poisson summation

Write $N = \mathrm{N}_{K/\mathbb{Q}}$, $I = [u, v]$, and $I_* = [u/2, 2v]$. All ideal indices below are prime to S and represented by their primary generators; a star additionally requires squarefreeness. The variables k, h, y range over $\mathcal{O}$. For squarefree C, t with $(C, t) = 1$, and for $d \mid C$, put

$$\ell = \frac{L}{N(C)N(t)}, \qquad w_{C,d} = \frac{\mathcal{H}N(C)}{N(d)L^2F}, \qquad Y_{C,d} = c_I\frac{\Sigma LFN(d)}{\mathcal{H}N(C)^2},$$

where $c_I \ge \max(1, 4C_\Phi v^2)$ is fixed. Retain only $\ell \ge 1/(2v)$. For $U \in C_c^\infty(I_*)$ and a character ξ_1 of the fixed ray class group, define

$$\begin{aligned} p_y(n) &= \mu(n)\xi_1(n)\mathbf{1}_{(n,t)=1}\overline{\chi_n(y)}\chi_n(C)^4\chi_n(d), \\ P_{C,d,t}(y; U) &= \sideset{}{^*}\sum_n p_y(n) U(N(n)/\ell). \end{aligned} \tag{7.1}$$

The notation p_y suppresses its dependence on C, d, t, ξ_1. The nonnegative form required in the second Poisson calculation is

$$\mathcal{Q}_{\xi_1}(U) = \sum_{\substack{C,t \text{ squarefree} \\ (Ct,S)=1,(C,t)=1 \\ N(C)N(t)\le 2vL}} \sum_{d\mid C} w_{C,d} \sum_{y} \Phi(N(y)/Y_{C,d})|P_{C,d,t}(y;U)|^2. \tag{7.2}$$

The outer domain has no additional restriction coming from the support of $\widehat{\Phi}$. Here $Y_{C,d} > 0$ may be smaller than 1; the outer sums are finite, and the y-sum converges absolutely.

Lemma 7.1. *Under the hypotheses of Proposition 5.4, fix an integer $j \ge 0$ and $\varepsilon_0 > 0$. If $M \ge 0$ satisfies $\mathcal{Q}_{\xi_1}(U) \le M\|U\|^2_{C^j(I_*)}$ for every $U \in C_c^\infty(I_*)$ and every ξ_1, then*

$$\mathcal{A}(W) \ll D^{\varepsilon_0}(\Sigma + M)\|W\|^2_{C^{2j+4}(I)}.$$

Proof. Expand the square defining $\mathcal{A}(W)$, and write $n_i = Cu_i$, where $C = (n_1, n_2)$ and $(u_1, u_2) = (u_1u_2, C) = 1$. The row character is $\chi_{u_1}\overline{\chi_{u_2}}$, primitive modulo u_1u_2, with the extra restriction $(k, C) = 1$. Lemma 4.2 introduces $d \mid C$ and a frequency h. Let Z denote the zero-frequency contribution. It requires $u_1 = u_2 = 1$, so

$$|Z| \ll \frac{\mathcal{H}}{LF} \sideset{}{^*}\sum_{F\le N(f)<2F} \sideset{}{^*}\sum_{C} |W(N(C)/L)|^2 \ll_I \mathcal{H}\|W\|^2_\infty.$$

For the other frequencies, the Chinese remainder theorem, (4.4), and (4.7) give the paired identity

$$a_\xi(u_1)\overline{a_\xi(u_2)}\gamma(\chi_{u_1}\overline{\chi_{u_2}}) = \mu(u_1)\mu(u_2)(\xi G)(u_1u_2^{-1}). \tag{7.3}$$

Expand $(\xi G)(z) = \sum_{\xi_1} \ell_{\xi_1}\xi_1(z)$ on the fixed ray class group. Insert $\mathbf{1}_{(u_1,u_2)=1} = \sum_{t\mid u_1,\ t\mid u_2} \mu(t)$ and put $u_i = tx_i$. The inverse is $x_i = u_i/t$; squarefreeness imposes $(x_i, t) = 1$, but no condition $(x_1, x_2) = 1$ remains. The factors at C, t cancel against their conjugates, leaving $\mu(d)\mu(t)$ and the restrictions $(f, Ct) = (h, t) = 1$. Thus

$$\begin{aligned}\mathcal{A}(W) - Z = \sum_{\xi_1} \ell_{\xi_1} &\sum_{\substack{C,t \text{ squarefree} \\ (Ct,S)=1,(C,t)=1 \\ N(C)N(t)\le 2vL}} \sum_{d\mid C} \mu(d)\mu(t) w_{C,d} \\ \times \sideset{}{^*}\sum_{\substack{F\le N(f)<2F \\ (f,Ct)=1}} &\sum_{\substack{h\ne 0,(h,t)=1 \\ N(h)\le C_\Phi v^2L^2N(d)/(\mathcal{H}N(C)^2)}} \sideset{}{^*}\sum_{x_1,x_2} p_{hf^2}(x_1)\overline{p_{hf^2}(x_2)}\mathscr{K}_h(N(x_1)/\ell, N(x_2)/\ell),\end{aligned}$$

where, for each integer $j_0 \ge 0$,

$$\mathscr{K}_h(z_1, z_2) = (z_1z_2)^{-1/2}W(z_1)\overline{W(z_2)}\widehat{\Phi}\Big(\frac{\mathcal{H}N(h)N(C)^2}{L^2N(d)z_1z_2}\Big),$$
$$\sup_{C,d,h} \|\mathscr{K}_h\|_{C^{j_0}(I^2)} \ll_{I,\Phi,j_0} \|W\|^2_{C^{j_0}(I)}.$$

Here $\chi_{x_i}(C)^4$ supplies $(x_i, C) = 1$; the other original zeros are supplied by $p_{hf^2}(x_i)$. The displayed frequency range follows from the support of $\widehat{\Phi}$.

For fixed C, d, t, the map $(f, h) \mapsto y = hf^2$ has divisor-bounded multiplicity in y. Since $\Sigma/(LF) \ge 1$, its nonzero image satisfies $N(y) \le 4C_\Phi v^2L^2F^2N(d)/(\mathcal{H}N(C)^2) \le Y_{C,d}$. Since $\Phi \ge 1$ on $[0, 1]$ and is nonnegative,

$$\sum_{C,t}\sum_{d\mid C} w_{C,d} \sum_{f,h} |P_{C,d,t}(hf^2;U)|^2 \preccurlyeq \mathcal{Q}_{\xi_1}(U) \le M\|U\|^2_{C^j(I_*)},$$

with the preceding ranges on the left. Apply Lemma B.2 to the full displayed expression for $\mathcal{A}(W) - Z$. The kernel bound gives

$$|\mathcal{A}(W) - Z| \ll D^{\varepsilon_0/2} M \|W\|^2_{C^{2j+4}(I)}.$$

Together with $\mathcal{H} \le \Sigma$, this proves the result after allocating the divisor losses within ε_0. □

7.2 Second application of Poisson summation

Let C, t be coprime squarefree primary elements prime to S, and let $d \mid C$. Fix a character ξ_1 of the ray class group. For arbitrary scales $\ell, Y > 0$ and $U \in C_c^\infty(I_*)$, with $I_* = [a, b] \subset (0, \infty)$, put

$$\begin{aligned} P(y) &= \sideset{}{^*}\sum_{(n,S)=1} \mu(n)\xi_1(n)\mathbf{1}_{(n,t)=1}\overline{\chi_n(y)}\chi_n(C)^4\chi_n(d)U(N(n)/\ell), \\ \mathcal{M} &= \sum_{y\in\mathcal{O}} \Phi(N(y)/Y)|P(y)|^2. \end{aligned} \tag{7.4}$$

Here $P(y)$ abbreviates $P_{C,d,t}(y; U)$ from (7.1), with the scale ℓ now arbitrary.

The second Poisson summation turns the Möbius coefficients back into cubic Gauss-sum coefficients. To state the identity, put $U_0(x) = x^{-1/2}U(x)$ and expand

$$\xi_1(z)G(z^{-1}) = \sum_{\xi'} c_{\xi'}\xi'(z)$$

on the fixed ray class group.

On the transformed side, g is the common divisor of the original columns, $e \mid g$ comes from the row exclusion, and w removes the remaining coprimality condition. The index h is the nonzero Fourier frequency. Let g, w range over squarefree primary elements prime to S, e over divisors of g, and h over $\mathcal{O} \setminus \{0\}$, subject to

$$(g, Ct) = (w, gCth) = 1, \qquad N(gw) \le b\ell, \qquad N(h) \le \frac{C_\Phi b^2 \ell^2 N(e)}{YN(g)^2}.$$

These conditions ensure that tg/e and Cew are coprime and squarefree. For each such choice and each character ξ', the new column scale, coefficient, and coupled smooth weight are

$$\begin{aligned} X' &= \frac{\ell}{N(gw)}, \\ a^\sharp(n) &= a_{\xi'}(n)\mathbf{1}_{(n,tg/e)=1}\chi_n(deh)\chi_n(Cew)^4, \\ \mathscr{K}(x_1, x_2) &= U_0(x_1)\overline{U_0(x_2)}\widehat{\Phi}\Big(\frac{YN(h)N(g)^2}{N(e)\ell^2 x_1 x_2}\Big). \end{aligned} \tag{7.5}$$

Here $\mathscr{K}$ depends on U, Y, ℓ, g, e, h. A star on a sum over n_1, n_2 restricts each index separately to squarefree primary elements. Define also

$$Z = Y\widehat{\Phi}(0) \sideset{}{^*}\sum_{\substack{(g,Ct)=1\\(g,S)=1}} |U(N(g)/\ell)|^2 \prod_{p\mid g}\Big(1 - \frac{1}{N(p)}\Big).$$

Lemma 7.2 (Second Poisson identity)**.** *With the notation and index ranges above,*

$$\mathcal{M} = Z + \sum_{g,e,w,h,\xi'} \frac{Y\mu(e)\mu(w)c_{\xi'}N(g)}{N(e)\ell} \sideset{}{^*}\sum_{(n_1n_2,S)=1} a^\sharp(n_1)\overline{a^\sharp(n_2)}\mathscr{K}\Big(\frac{N(n_1)}{X'}, \frac{N(n_2)}{X'}\Big). \tag{7.6}$$

The zero-frequency term satisfies

$$|Z| \ll_{I_*,\Phi} Y\ell\|U\|^2_\infty.$$

Proof. Expand the square, put $g=(n_1,n_2)$ and $n_i=gz_i$. The nonzero terms satisfy

$$(z_1,z_2)=(z_1z_2,gCt)=(g,Ct)=1,\qquad \overline{\chi_{n_1}(y)}\chi_{n_2}(y)=\mathbf{1}_{(y,g)=1}\overline{\chi_{z_1}(y)}\chi_{z_2}(y).$$

Apply Lemma 4.2 to this last character, with exclusion g and divisor $e\mid g$. Its zero frequency occurs exactly when $z_1=z_2=1$ and gives the displayed Z; counting g gives its bound. For the nonzero frequencies, (4.5)–(4.7) and the Chinese remainder theorem give

$$\mu(z_1)\mu(z_2)\xi_1(z_1)\overline{\xi_1(z_2)}\gamma(\overline{\chi_{z_1}}\chi_{z_2})=\sum_{\xi'}c_{\xi'}a_{\xi'}(z_1)\overline{a_{\xi'}(z_2)}.$$

Use also

$$\chi_z(dh)\overline{\chi_z(e)}\chi_z(C)^4=\chi_z(deh)\chi_z(Ce)^4,$$
$$\frac{U(N(gz_1)/\ell)\overline{U(N(gz_2)/\ell)}}{\sqrt{N(z_1)N(z_2)}}=\frac{N(g)}{\ell}U_0(N(gz_1)/\ell)\overline{U_0(N(gz_2)/\ell)}.$$

Thus the full nonzero contribution is

$$\begin{aligned}\mathcal{M}-Z=\sum_{\xi'}c_{\xi'}\sideset{}{^*}\sum_{\substack{g\\(g,Ct)=1}}\sum_{e\mid g}\frac{Y\mu(e)N(g)}{N(e)\ell}\sum_{h\neq0}&\\ \times\sideset{}{^*}\sum_{\substack{z_1,z_2\\(z_1,z_2)=1\\(z_1z_2,gCt)=1}}&a_{\xi'}(z_1)\overline{a_{\xi'}(z_2)}\chi_{z_1}(deh)\overline{\chi_{z_2}(deh)}\chi_{z_1}(Ce)^4\overline{\chi_{z_2}(Ce)^4}\\ \times U_0\Big(\frac{N(gz_1)}{\ell}\Big)\overline{U_0\Big(\frac{N(gz_2)}{\ell}\Big)}&\widehat{\Phi}\Big(\frac{YN(h)}{N(e)N(z_1)N(z_2)}\Big).\end{aligned}$$

All starred ideal indices are prime to S. Insert

$$\mathbf{1}_{(z_1,z_2)=1}=\sum_{w\mid z_1,\ w\mid z_2}\mu(w),\qquad z_i=wn_i.$$

Here $(w,gCt)=1$. Equation (4.6) and the zero-extended characters give

$$a_{\xi'}(wn)=a_{\xi'}(w)a_{\xi'}(n)\chi_n(w)^4\quad((w,n)=1),$$
$$|a_{\xi'}(w)\chi_w(deh)\chi_w(Ce)^4|^2=\mathbf{1}_{(w,h)=1},$$
$$\mathbf{1}_{(n,gCtw)=1}\chi_n(deh)\chi_n(Cew)^4=\mathbf{1}_{(n,tg/e)=1}\chi_n(deh)\chi_n(Cew)^4.$$

Consequently the preceding full sum becomes

$$\begin{aligned}\mathcal{M}-Z=\sum_{\xi'}c_{\xi'}\sideset{}{^*}\sum_{\substack{g,w\\(g,Ct)=(w,gCt)=1}}\sum_{e\mid g}\frac{Y\mu(e)\mu(w)N(g)}{N(e)\ell}\sum_{\substack{h\neq0\\(h,w)=1}}&\\ \times\sideset{}{^*}\sum_{(n_1n_2,S)=1}a^\sharp(n_1)\overline{a^\sharp(n_2)}U_0\Big(\frac{N(gwn_1)}{\ell}\Big)\overline{U_0\Big(\frac{N(gwn_2)}{\ell}\Big)}&\widehat{\Phi}\Big(\frac{YN(h)}{N(e)N(w)^2N(n_1)N(n_2)}\Big).\end{aligned}$$

There is no remaining condition $(n_1,n_2)=1$. The column support gives $N(gw)\le b\ell$ and $N(z_1z_2)\le b^2\ell^2/N(g)^2$; the support of $\widehat{\Phi}$ therefore imposes the stated bound on $N(h)$. Substituting (7.5) gives (7.6), with only finitely many terms. □

Lemma 7.3. *Return to $I=[u,v]$ and $I_*=[u/2,2v]$. For an integer $m\ge0$, let S_m be the supremum in (5.11). Then, for every $\varepsilon_0>0$,*

$$\mathcal{Q}_{\xi_1}(U)\ll D^{\varepsilon_0}\Sigma(1+S_m)\|U\|^2_{C^{2m+4}(I_*)}\qquad(U\in C_c^\infty(I_*)).$$

The test functions in S_m are supported in $I'=[u/16,4v]$ and have $C^m(I')$ norm at most one; an empty supremum is zero.

Proof. Write $N = \mathrm{N}_{K/\mathbb{Q}}$. Apply Lemma 7.2 with input length ℓ, row scale $Y_{C,d}$, and the same C, d, t, using the character ξ_1. Here h is the new Fourier variable for the sum over y. Its nonzero output has squarefree g, w, $e \mid g$, and

$$(g, Ct) = (w, gCth) = 1, \qquad r = tg/e, \quad f' = Cew, \quad k' = deh.$$

Thus r, f' are coprime and squarefree, and the coefficient $a^\sharp$ from (7.5) becomes

$$a^\sharp(n) = a_{\xi'}(n)\mathbf{1}_{(n,r)=1}\chi_n(k')\chi_n(f')^4.$$

Write X_0' for the individual column scale X' in (7.5); below, X' will denote a common dyadic scale. The length, coefficient, and Fourier parameter simplify to

$$\begin{aligned} X_0' &= \frac{\ell}{N(g)N(w)} = \frac{L}{N(r)N(f')}, \\ w_{C,d}\frac{Y_{C,d}N(g)}{N(e)\ell} &= \frac{c_I\Sigma N(r)}{L^2}, \\ \frac{Y_{C,d}N(h)N(g)^2}{N(e)\ell^2} &= \frac{c_I\Sigma F N(k')N(r)^2}{\mathcal{H}L}. \end{aligned}$$

Put $U_0(x) = x^{-1/2}U(x)$. The coupled kernel is therefore

$$\mathscr{K}_{r,f',k'}(x_1, x_2) = U_0(x_1)\overline{U_0(x_2)}\widehat{\Phi}\Big(\frac{c_I\Sigma F N(k')N(r)^2}{\mathcal{H}Lx_1x_2}\Big),$$

with both columns evaluated at $x_j = N(n_j)/X_0'$. In particular its nonzero support requires

$$N(r)N(f') \le 2vL, \qquad 0 < N(k') \le \frac{4C_\Phi v^2}{c_I}\frac{\mathcal{H}L}{\Sigma F N(r)^2} \le \frac{\mathcal{H}L}{\Sigma F N(r)^2},$$

since $C_\Phi(2v)^2/c_I \le 1$.

For fixed r, f', k', all preimages are obtained by choosing

$$t \mid r, \qquad f' = Cew, \qquad e \mid k', \qquad (w, k') = 1, \qquad d \mid (C, k'),$$

and setting $g = e(r/t)$, $h = k'/(de)$. The factorization $f' = Cew$ is disjoint and squarefree. All these preimages have the same kernel: $Ctgw = rf'$, and the Fourier parameter displayed above depends only on r, k'. Thus the support restrictions discard only zero kernels. The choices $t \mid r$ contribute $\tau_{\mathrm{div}}(r)$. At a prime $p \mid f'$, the choices $p \mid C$, $p \mid e$, and $p \mid w$, respectively, contribute

$$\big(1 + \mathbf{1}_{p\mid k'}\big) - \mathbf{1}_{p\mid k'} - \mathbf{1}_{p\nmid k'} = \mathbf{1}_{p\mid k'}$$

to the sum of $\mu(e)\mu(w)$ over the preimages; the two choices inside the first term are $p \nmid d$ and $p \mid d$. Let $\mathcal{Z}$ be the total zero-frequency contribution: the sum of the terms Z from Lemma 7.2, with the outer weights $w_{C,d}$ and ranges in (7.2). Consequently, regrouping before taking absolute values gives

$$\begin{aligned} \mathcal{Q}_{\xi_1}(U) - \mathcal{Z} = \frac{c_I\Sigma}{L^2}\sum_{\xi'} c_{\xi'} \sum_{\substack{r,f' \text{ squarefree} \\ (r,f')=1}} N(r)\tau_{\mathrm{div}}(r) \sum_{\substack{k'\neq 0 \\ f'\mid k'}} \\ \times \sideset{}{^*}\sum_{n_1,n_2} a^\sharp(n_1)\overline{a^\sharp(n_2)}\mathscr{K}_{r,f',k'}\Big(\frac{N(n_1)}{X_0'}, \frac{N(n_2)}{X_0'}\Big). \end{aligned}$$

Fix $R \le N(r) < 2R$, $F' \le N(f') < 2F'$ dyadically, and put

$$X' = \frac{L}{RF'}, \qquad \Sigma' = X'F' = \frac{L}{R}, \qquad \mathcal{H}' = \frac{\mathcal{H}L}{\Sigma F R^2}.$$

Retain only those r for which a nonzero term occurs in these ranges. Since $f' \mid k'$, each such r satisfies

$$F' \le N(f') \le N(k') \le \frac{\mathcal{H}L}{\Sigma F N(r)^2}.$$

In particular $\mathcal{H}' \ge 1$. For every $j \mid r$, using $\mathcal{H} \le \Sigma$, we also have

$$\frac{X'}{N(j)} \ge \frac{\Sigma F N(r)^2}{\mathcal{H} R N(j)} \ge \frac{F\Sigma}{\mathcal{H}} \ge 1. \tag{7.7}$$

Moreover $1 \le X'/X'_0 < 4$. The rescaled kernels

$$\mathcal{K}_{r,f',k'}(x_1,x_2) = \mathscr{K}_{r,f',k'}\Big(\frac{X'}{X'_0}x_1, \frac{X'}{X'_0}x_2\Big)$$

have support in $[u/8, 2v]^2 \subset \mathrm{int}(I'^2)$ and satisfy

$$\sup_{r,f',k'} \|\mathcal{K}_{r,f',k'}\|_{C^{2m+4}(I'^2)} \ll_{I_*,m,\Phi} \|U\|^2_{C^{2m+4}(I_*)},$$

where the supremum is over the retained indices in the dyadic block. This follows from the Schwartz bounds for $\widehat{\Phi}$. The common parameters satisfy exactly

$$\frac{\mathcal{H}'}{\Sigma'} = \frac{\mathcal{H}}{\Sigma F R} \le \frac{\mathcal{H}}{\Sigma}, \qquad \Sigma' \le L, \qquad \mathcal{H}' \le \frac{\mathcal{H}L}{\Sigma F}.$$

Fix $\delta > 0$, to be chosen in terms of ε_0 at the end. For each retained r, enlarge the f', k' ranges by positivity, keeping this set of r fixed. Lemma 4.4 then gives, for $V \in C_c^\infty(I')$,

$$\sum_{F' \le N(f') < 2F'}^{*} \sum_{0 < N(k') \le \mathcal{H}'} \Big|\sum_n{}^{*} a^\sharp(n) V(N(n)/X')\Big|^2 = \Sigma' \mathcal{E}_{(r)}(\mathcal{H}', X', F'; \xi', V)$$
$$\ll_\delta D^\delta (\Sigma')^2 S_m \|V\|^2_{C^m(I')}.$$

Indeed exclusion removal replaces (X', F') by $(X'/N(j), F'N(j))$, $j \mid r$, preserving $\mathcal{H}'$ and Σ'. Equation (7.7) gives $X'/N(j) \ge 1$, and $F'N(j) \ge 1$ as well. Thus every resulting mean square lies in the defining supremum.

There are $O(R)$ ideals r in the dyadic range. Apply Lemma B.2 with row index (r, f', k'), the preceding mean-square bound, and the uniform kernel bound. The absolute contribution of this dyadic block is at most

$$D^{2\delta}\frac{\Sigma R}{L^2} R(\Sigma')^2 S_m \|U\|^2_{C^{2m+4}(I_*)} = D^{2\delta}\Sigma S_m \|U\|^2_{C^{2m+4}(I_*)},$$

because $(\Sigma R/L^2)R(L/R)^2 = \Sigma$.

Finally, the zero-frequency bound in Lemma 7.2 gives

$$|\mathcal{Z}| \ll_{I_*,\Phi} \|U\|^2_\infty \sum_{N(Ct) \le 2vL} \sum_{d \mid C} w_{C,d} Y_{C,d} \ell$$
$$\ll_{I_*} \Sigma \|U\|^2_\infty \sum_C \frac{\tau_{\mathrm{div}}(C)}{N(C)^2} \sum_{N(t) \le 2vL} \frac{1}{N(t)} \ll_{I_*,\delta} D^\delta \Sigma \|U\|^2_\infty.$$

This uses only $Y_{C,d} > 0$, not $Y_{C,d} \ge 1$. Sum the $O_{I_*,C_0}((\log(2D))^2)$ dyadic blocks and the fixed finite character set, and take $\delta = \varepsilon_0/4$. The stated bound follows, with implied constant depending only on $I_*, m, C_0, \varepsilon_0$, the fixed cutoffs and arithmetic data. □

Proof of Proposition 5.4. Use S_m as in Lemma 7.3, with m as in the proposition. Lemma 7.3, with loss $D^{\varepsilon/4}$, supplies the hypothesis of Lemma 7.1 with $j = 2m+4$ and $M \ll D^{\varepsilon/4}\Sigma(1+S_m)$. Applying that lemma with the same loss gives

$$\mathcal{A}(W) \ll D^{\varepsilon/2}\Sigma(1+S_m)\|W\|^2_{C^{4m+12}(I)},$$

which proves (5.11). □

A Arithmetic identities and theta calculations

A.1 Reciprocity and Gauss sums

Proof of Lemma 4.1. Proof of (4.2). For squarefree n, set $G(n) = \overline{\chi_n(4)}\gamma_3(n)$. Its dependence on a fixed ray class will be verified below. Fix a primary prime $p \equiv 1 \pmod 3$ outside S. For multiplicative characters A, B of $(\mathcal{O}/(p))^\times$, extended by zero at 0, write

$$J(A,B) = \sum_{x \bmod p} A(x)B(1-x).$$

For each $y \in \mathcal{O}/(p)$, the substitution $u = 2x-1$ gives

$$\#\{x \bmod p : 4x(1-x) = y\} = \#\{u \bmod p : u^2 = 1-y\} = 1 + \chi_p^3(1-y),$$

since $p \nmid 2$ and χ_p^3 is the quadratic character, extended by zero at 0. Hence

$$\chi_p(4)J(\chi_p,\chi_p) = \sum_{x \bmod p} \chi_p\big(4x(1-x)\big) = \sum_{y \bmod p} \chi_p(y)\big(1+\chi_p^3(1-y)\big) = J(\chi_p,\chi_p^3).$$

The Gauss–Jacobi identity (cf. [22, (3.18)]) gives

$$\frac{\gamma_1(p)\gamma_3(p)}{\gamma_4(p)} = \frac{J(\chi_p,\chi_p^3)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = \chi_p(4)\frac{J(\chi_p,\chi_p)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = \chi_p(4)\frac{\gamma_1(p)^2}{\gamma_2(p)}.$$

For $j \not\equiv 0 \pmod 6$, the normalized Gauss sums satisfy

$$\gamma_j(p)\gamma_{-j}(p) = \chi_p(-1)^j.$$

Taking $j = 2$ gives $\gamma_2(p)\gamma_4(p) = \chi_p^2(-1) = 1$, since χ_p^2 is cubic. Combining this with the Gauss–Jacobi relation above gives, after cancelling $\gamma_1(p)$ and rearranging,

$$\gamma_1(p)\gamma_2(p) = \overline{\chi_p(4)}\gamma_3(p)\gamma_2(p)^3.$$

To finish the prime case of (4.2), it remains to evaluate $\gamma_2(p)^3$. We do so by computing $J(\chi_p^2,\chi_p^2)$. Put $q = \mathrm{N}_{K/\mathbb{Q}}(p)$. We have $J(\chi_p^2,\chi_p^2) \in \mathcal{O}$ and $\mathrm{N}_{K/\mathbb{Q}}\big(J(\chi_p^2,\chi_p^2)\big) = q$. For $0 \le \ell \le (q-1)/3$, the exponent $(q-1)/3+\ell$ lies strictly between 0 and $q-1$, so $\sum_{x \bmod p} x^{(q-1)/3+\ell} = 0$ in $\mathcal{O}/(p)$. Reducing modulo p and expanding the binomial therefore gives

$$\begin{aligned} J(\chi_p^2,\chi_p^2) &\equiv \sum_{x \bmod p} x^{(q-1)/3}(1-x)^{(q-1)/3} \\ &= \sum_{\ell=0}^{(q-1)/3} (-1)^\ell \binom{(q-1)/3}{\ell} \sum_{x \bmod p} x^{(q-1)/3+\ell} = 0 \pmod p. \end{aligned}$$

Thus $J(\chi_p^2,\chi_p^2)/p \in \mathcal{O}$ has norm one, so $J(\chi_p^2,\chi_p^2)$ is a unit multiple of p. To determine the unit, write $\chi_p^2(x(1-x)) = \omega^{j_x}$ with $j_x \in \{0,1,2\}$ for $x \in \mathcal{O}/(p) \setminus \{0,1\}$. Then

$$\prod_{x \ne 0,1} x(1-x) = 1 \quad\Longrightarrow\quad \sum_{x \ne 0,1} j_x \equiv 0 \pmod 3.$$

Since $(\omega-1)^2$ generates (3), expansion modulo this ideal gives

$$J(\chi_p^2,\chi_p^2) = \sum_{x \ne 0,1} \omega^{j_x} \equiv (q-2) + (\omega-1)\sum_{x \ne 0,1} j_x \equiv -1 \pmod 3.$$

Together with $p \equiv 1 \pmod 3$, this fixes the unit:

$$J(\chi_p^2, \chi_p^2) = -p.$$

Consequently

$$\gamma_2(p)^3 = \frac{\gamma_2(p)^2}{\gamma_4(p)} = \frac{J(\chi_p^2, \chi_p^2)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = -\alpha(p).$$

The Chinese remainder theorem extends these prime-modulus identities to (4.2) for squarefree n, with one minus sign for each prime factor accounting for $\mu(n)$.

Proof of (4.6). For coprime squarefree primary a, b prime to S, the Chinese remainder theorem gives

$$\gamma_2(ab) = \gamma_2(a)\gamma_2(b)\chi_a(b)^2\chi_b(a)^2,$$

and cubic reciprocity gives $\chi_a(b)^2 = \chi_b(a)^2$. Multiplying by $\overline{\alpha(ab)}\xi(ab)$ therefore proves (4.6).

Proof of (4.1). We now prove (4.1) and the formula for $G(n)$ in (4.3), including their dependence on a fixed ray class. We begin by evaluating the normalized quadratic Gauss sum. For $c \in \mathcal{O}$ coprime to 2, we will show that

$$\Gamma_{\mathrm{quad}}(c) := |c|^{-1} \sum_{x \bmod c} e(x^2/c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4). \tag{A.1}$$

To justify (A.1), apply Poisson summation over $z \in \mathcal{O}$ to $e(z^2/c)e^{-\pi\eta|z|^2}$, with $\eta > 0$. The Gaussian makes the sum convergent. The Fourier transform of the product is

$$\frac{|c|}{2\sqrt{r_\eta}} e^{-\pi\eta \mathrm{N}_{K/\mathbb{Q}}(c)|y|^2/(4r_\eta)} e(-cy^2/(4r_\eta)), \qquad r_\eta = 1 + 3\mathrm{N}_{K/\mathbb{Q}}(c)\eta^2/16.$$

Divide both sides by the Gaussian mass

$$\sum_{z \in \mathcal{O}} e^{-\pi\eta|z|^2} \asymp \eta^{-1}.$$

On the Fourier side, replacing r_η by 1 has total error

$$O_c\Big(\eta^2 \sum_{y \in \mathcal{O}} |y|^2 e^{-C_c\eta|y|^2}\Big) = O_c(1),$$

where $C_c > 0$ is fixed. The normalized error is therefore $O_c(\eta) \to 0$. Grouping the original sum modulo c and the Fourier sum modulo $2\mathcal{O}$, then letting $\eta \downarrow 0$, gives (A.1).

For $c = a + b\omega$, formula (A.1) becomes

$$\Gamma_{\mathrm{quad}}(a + b\omega) = \frac{1 + \mathrm{i}^{-b} + \mathrm{i}^{a} + \mathrm{i}^{b-a}}{2}.$$

Thus $\Gamma_{\mathrm{quad}}(c)$ depends only on $c \bmod 4\mathcal{O}$ and is invariant under multiplication by a square in $(\mathcal{O}/4\mathcal{O})^\times$. Representatives for the four square classes and their values are

c	1	-1	λ	$-\lambda$
$\Gamma_{\mathrm{quad}}(c)$	1	1	i	$-\mathrm{i}$

Consequently the function

$$\mathcal{R}(c_1, c_2) := \frac{\Gamma_{\mathrm{quad}}(c_1c_2)}{\Gamma_{\mathrm{quad}}(c_1)\Gamma_{\mathrm{quad}}(c_2)}$$

on these square classes satisfies

$$\mathcal{R}((-1)^e\lambda^f, (-1)^g\lambda^h) = (-1)^{eh+fg+fh}, \qquad e, f, g, h \in \{0, 1\},$$

and is a symmetric bicharacter. For a prime p, each $y \in \mathcal{O}/(p)$ has $1+\chi_p^3(y)$ square roots. Grouping by $y = x^2$ gives

$$\Gamma_{\mathrm{quad}}(p) = \frac{1}{|p|} \sum_{y \bmod p} \big(1+\chi_p^3(y)\big) e(y/p) = \gamma_3(p),$$

since $\sum_{y \bmod p} e(y/p) = 0$. The Chinese remainder theorem extends this equality to squarefree n. Together with cubic reciprocity, it identifies the above $\mathcal{R}$ with the quotient $\chi_b(a)/\chi_a(b)$ for coprime primary a, b. Furthermore $\chi_c(4) = (-2/c)_3 = (c/(-2))_3$ is a character modulo 2. Consequently $G(c) = \overline{\chi_c(4)}\Gamma_{\mathrm{quad}}(c)$ factors through a fixed ray class group and satisfies (4.1), with $\mathcal{R}(c,c) = \chi_c(-1)$.

Proof of (4.7). In the fixed ray class group, the multiplicative relation for G gives

$$G(a^{-1}) = \chi_a(-1)\overline{G(a)}.$$

Using the symmetry and multiplicativity of $\mathcal{R}$, we obtain

$$G(ba^{-1}) = G(b)G(a^{-1})\mathcal{R}(b,a^{-1}) = \chi_a(-1)\overline{G(a)}G(b)\mathcal{R}(a,b),$$

which is (4.7).

Proof of (4.4)–(4.5). The second identity in (4.3) follows from the inverse-character Gauss identity: $\gamma_1(n)\gamma_{-1}(n) = \chi_n(-1)$. It remains to prove (4.4)–(4.5). Multiplying the second identity in (4.2) by $\overline{\alpha(n)}$ gives (4.4), and combining it with this inverse-character identity gives (4.5). □

A.2 The cubic theta transformation with fixed ray class twists

We prove the transformation formula of Proposition 6.2 for the completed sum $T(X;\Psi)$ in (5.3), together with the uniformity assertion of Lemma 6.3 and the coefficient and weight bounds of Lemma 6.4. The proof adapts the theta-transformation method of Dunn and Radziwiłł [7, §5 and Appendix A], which extends Patterson [38] and Yoshimoto [51]. The treatment of the fixed ray class twists and the uniformity assertions are supplied below. We use the setup preceding the proposition and the dependence of constants specified in these three statements: Ψ_0, S are fixed, while $\mathcal{P}$ is the varying finite set of primes outside S and j_p are the exponents of their character factors in Ψ. Character powers follow the zero-extension convention preceding (6.5); in particular, $\chi_p^0(x) = \mathbf{1}_{p\nmid x}$. This convention also applies when we write $\chi_p(x)^j$.

Write $w = (z,v) \in \mathbb{C}\times\mathbb{R}_{>0}$, and use θ from (6.1). The three sequences d_σ, $\sigma \in \{0,+,-\}$, are defined by the Fourier expansions of $\overline{\theta(\gamma_\sigma w)}$ in (6.8), with the representatives (6.7). For the additive characters, write $\check{e}(z) = \exp(2\pi\mathrm{i}(z+\bar{z}))$; thus $e(z) = \check{e}(z/\lambda)$, with e as in (2.6).

Proof of Proposition 6.2 and Lemmas 6.3 and 6.4. Finite Fourier expansion. Define $\phi : \mathcal{O} \to \mathbb{C}$ by

$$\phi(n) = \begin{cases} \chi_n(\lambda)^2\Psi_0(n), & n \equiv 1 \pmod 3,\quad (n,S)=1,\\ 0, & \text{otherwise.}\end{cases}$$

For $\Psi(n) = \Psi_0(n)\prod_{p\in\mathcal{P}}\chi_p^{j_p}(n)$, define

$$\Theta_\Psi(z,v) = \sum_{\substack{\ell\in\lambda^{-3}\mathcal{O}\\ \ell\neq 0}} \overline{\tau(-\ell)}\phi(\lambda^3\ell)\Big(\prod_{p\in\mathcal{P}}\chi_p^{j_p}(\lambda^3\ell)\Big) v K_{1/3}(4\pi|\ell|v)\check{e}(\ell z). \tag{A.2}$$

When $\Psi = \Psi_k$, the multiplier of $\overline{\tau(-\ell)}$ is $\phi_k(\lambda^3\ell)$, so this definition recovers Θ_k in (6.2). We first write Θ_Ψ as a finite sum of translates of $\overline{\theta}$ and determine their reduced denominators. Choose once and for all a nonzero $L \in \mathcal{O}$, with prime divisors in S, divisible by the conductor of Ψ_0, every prime in S, and sufficiently high powers of the primes above 2 and 3. The supplementary law of

cubic reciprocity for λ [7, (1.5)], which evaluates $\chi_n(\lambda)^2 = (\lambda/n)_3$, makes ϕ periodic modulo L. As before (6.4), define

$$\widehat{\phi}(h_0) = \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(L)} \sum_{x \bmod L} \phi(x) e(-h_0 x/L), \qquad h_0 \in \mathcal{O}/(L).$$

Since $|\phi| \le 1$, we have $|\widehat{\phi}(h_0)| \le 1$.

For $p \in \mathcal{P}$ and $0 \le j \le 5$, we expand χ_p^j on $\mathcal{O}/(p)$ in additive characters. Its Fourier coefficients are defined, for $h_p \in \mathcal{O}/(p)$, by

$$C_{p,j}(h_p) := \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(p)} \sum_{y \bmod p} \chi_p^j(y) e(-h_p y/p).$$

Finite Fourier inversion then gives, for $x \in \mathcal{O}$,

$$\chi_p^j(x) = \sum_{h_p \in \mathcal{O}/(p)} C_{p,j}(h_p) e(h_p x/p). \tag{A.3}$$

Multiplying (A.3) over the primes in $\mathcal{P}$ gives, for $x \in \mathcal{O}$,

$$\prod_{p \in \mathcal{P}} \chi_p^{j_p}(x) = \sum_{\substack{h_p \in \mathcal{O}/(p) \\ p \in \mathcal{P}}} \Big(\prod_{p \in \mathcal{P}} C_{p,j_p}(h_p) \Big) e\Big(x \sum_{p \in \mathcal{P}} \frac{h_p}{p} \Big).$$

Each summand is indexed by a tuple $(h_p)_{p \in \mathcal{P}}$. At $x = \lambda^3 \ell$, its additive character gives the shift $\lambda^2 \sum_{p \in \mathcal{P}} h_p/p$ of $\overline{\theta}$. Writing $\boldsymbol{h} = (h_0, (h_p)_{p \in \mathcal{P}})$ with $h_0 \in \mathcal{O}/(L)$, we claim that

$$\Theta_\Psi(z, v) = \sum_{\boldsymbol{h}} c_{\mathrm{F}}(\boldsymbol{h}) \overline{\theta(z + z_{\boldsymbol{h}}, v)},$$

where

$$z_{\boldsymbol{h}} = \lambda^2 \Big(h_0/L + \sum_{p \in \mathcal{P}} h_p/p \Big), \qquad c_{\mathrm{F}}(\boldsymbol{h}) = \widehat{\phi}(h_0) \prod_{p \in \mathcal{P}} C_{p,j_p}(h_p). \tag{A.4}$$

To verify this identity, expand the fixed factor ϕ as well. The finite Fourier expansions identify all nonzero Fourier coefficients, and the constant terms of the translates cancel because

$$\sum_{\boldsymbol{h}} c_{\mathrm{F}}(\boldsymbol{h}) = \phi(0) \prod_{p \in \mathcal{P}} \chi_p^{j_p}(0) = 0.$$

For $j \not\equiv 0 \pmod 6$, we use $\gamma_j(p)$ from (2.6). Changing y to $-y$ gives

$$\frac{1}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} \sum_{y \bmod p} \chi_p(y)^j e(-y/p) = \chi_p(-1)^j \gamma_j(p), \qquad |\gamma_j(p)| = 1.$$

For $h_p \ne 0$, substitute $y = h_p^{-1} u$ in the definition of $C_{p,j}(h_p)$. For $h_p = 0$ and $j \ne 0$, use character orthogonality; for $j = 0$, sum the additive character over nonzero residues directly. These calculations give

$$C_{p,j}(h_p) = \begin{cases} \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2} \chi_p(-1)^j \gamma_j(p) \chi_p(h_p)^{-j}, & j \ne 0,\ h_p \ne 0, \\ 0, & j \ne 0,\ h_p = 0, \\ -\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}, & j = 0,\ h_p \ne 0, \\ 1 - \mathrm{N}_{K/\mathbb{Q}}(p)^{-1}, & j = 0,\ h_p = 0. \end{cases} \tag{A.5}$$

For the tuple $\boldsymbol{h}$, define its set of *active primes* by $\mathcal{A}(\boldsymbol{h}) := \{p \in \mathcal{P} : h_p \neq 0\}$. By (A.5), $c_{\mathrm{F}}(\boldsymbol{h}) \neq 0$ implies $\{p \in \mathcal{P} : j_p \neq 0\} \subseteq \mathcal{A}(\boldsymbol{h})$. Thus only primes with $j_p = 0$ can be inactive, as asserted in the proposition. Put $r = \prod_{p\in\mathcal{A}(\boldsymbol{h})} p$, and let c_0 be the reduced denominator of $\lambda^2 h_0/L$. At each active prime p,

$$v_p(\lambda^2 h_p/p) = -1, \qquad v_p(z_{\boldsymbol{h}} - \lambda^2 h_p/p) \geq 0,$$

so the factor p cannot cancel from the denominator. At primes dividing L, all terms $\lambda^2 h_p/p$ are integral. Thus the reduced denominator is $c = c_0 r$ up to a unit, including when $(h_0, L) \neq 1$. The finite Fourier expansion has therefore expressed Θ_Ψ as $O_L(2^{|\mathcal{P}|})$ groups of translates $\overline{\theta(z + z_{\boldsymbol{h}}, v)}$, with shifts $z_{\boldsymbol{h}}$ and coefficients $c_{\mathrm{F}}(\boldsymbol{h})$ given by (A.4). The groups are indexed by $h_0 \in \mathcal{O}/(L)$ and the active set $\mathcal{A}$. Each group has a common reduced denominator $c = c_0 \prod_{p\in\mathcal{A}} p$ up to a unit. For each translate $\overline{\theta(z + z_{\boldsymbol{h}}, v)}$, we next identify which of the three Fourier expansions in (6.8) will be used after changing coordinates at $z_{\boldsymbol{h}}$.

Reduced denominators and cusp coefficients. We next express each translate $\overline{\theta(z + z_{\boldsymbol{h}}, v)}$ using one of the three cusp expansions identified above. Our representatives $\gamma_0, \gamma_+, \gamma_-$ correspond to $\gamma_1, \gamma_{10}, \gamma_{19}$, respectively, in the numbering of [7, §5.1], which follows [38, Table II]. We allow $(h_0, L) \neq 1$ and keep the reduced denominator c_0 of $\lambda^2 h_0/L$ in the calculation; this is the extension beyond [7, Corollary 5.1].

Set $M = \lambda^{12} L^4$. Since $(p, M) = 1$ for $p \in \mathcal{A}(\boldsymbol{h})$, the Chinese remainder theorem lets us choose representatives $h_p \in \mathcal{O}$ with $h_p \equiv 0 \pmod{M^2}$. Changing representatives modulo p changes $z_{\boldsymbol{h}}$ by an element of $\lambda^2\mathcal{O} = 3\mathcal{O}$, under which θ is periodic. To track dependence on the primes in r, restrict r to a residue class modulo M^2. For fixed h_0, active set, and this residue class, write $z_{\boldsymbol{h}} = a/c$ in lowest terms, normalizing $a \equiv 1 \pmod 3$ if $\lambda \mid c$, and $c \equiv 1 \pmod 3$ otherwise. Since

$$a = \lambda^2 c\Big(h_0/L + \sum_{p\mid r} h_p/p\Big),$$

these choices fix the residue of a modulo Mc_0. Since $(a, c) = 1$ and $(r, M) = 1$, the Chinese remainder theorem gives $\delta' \in \mathcal{O}$ satisfying the following congruences, where q ranges over prime divisors of the indicated elements:

$$\begin{aligned} a\delta' &\equiv 1 \pmod{q^{v_q(Mc_0)}} && (q \mid c_0),\\ \delta' &\equiv 0 \pmod{q^{v_q(M)}} && (q \mid M,\ q \nmid c_0),\\ a\delta' &\equiv 1 \pmod{r}. \end{aligned}$$

Put

$$b_g = \frac{a\delta' - 1}{c}, \qquad g = \begin{pmatrix} a & b_g \\ c & \delta' \end{pmatrix} \in \mathrm{SL}_2(\mathcal{O}).$$

The congruences for δ' give

$$b_g \equiv \begin{cases} 0, & q \mid c_0, \\ -c^{-1}, & q \mid M,\ q \nmid c_0, \end{cases} \pmod{q^{v_q(M)}}.$$

To choose the cusp expansion at a/c, put

$$H = \begin{cases} I, & 3 \mid c, \\ \left(\begin{smallmatrix} 1 & 0 \\ u_0 & 1 \end{smallmatrix}\right), & v_\lambda(c) = 1, \quad u_0 \in \{\lambda, -\lambda\}, \quad u_0 \equiv c \pmod 3, \\ \left(\begin{smallmatrix} u_0 & -1 \\ 1 & 0 \end{smallmatrix}\right), & (c, \lambda) = 1, \quad u_0 \equiv a \pmod 3. \end{cases}$$

In the last case choose $u_0 \in \mathcal{O}$ from a fixed set of representatives modulo 3. In each case, put $g_1 = gH^{-1}$. Then $g = g_1 H$ and $g_1 \equiv I \pmod 3$, as in [7, §5.1]. The invariances

$$\theta(\gamma w) = \theta(w) \quad (\gamma \in \mathrm{SL}_2(\mathbb{Z})), \qquad \theta(z + t, v) = \theta(z, v) \quad (t \in \mathbb{Z} + 3\mathcal{O})$$

reduce $\overline{\theta}(Hw)$ to one of the three functions $\overline{\theta(\gamma_\sigma w)}$, $\sigma \in \{0,-,+\}$, with the matrices γ_σ from (6.7). These representatives give the infinity expansion and the two additional cusp expansions needed here: translating by ω or $-\omega$, then applying inversion, gives the representatives labeled $-$ and $+$, respectively [7, (5.9)–(5.15), Appendix A]. Write $t_\sigma(\ell)$ for the coefficient of $vK_{1/3}(4\pi|\ell|v)\breve{e}(\ell z)$ in $\theta(\gamma_\sigma w)$. Let $\tau_1, \tau_2 : \lambda^{-4}\mathcal{O} \setminus \{0\} \to \mathbb{C}$ be the coefficient sequences defined in [7, (5.13), (5.14)]. Patterson's cusp calculation [38, Theorem 8.1 and Table III] then gives

$$t_0(\ell) = \tau(\ell), \qquad t_-(\ell) = \omega^2 \tau_1(\omega^2\ell)\breve{e}(\ell), \qquad t_+(\ell) = \omega\tau_2(\omega\ell)\breve{e}(\ell). \tag{A.6}$$

Complex conjugation changes the Fourier frequency from ℓ to $-\ell$. Thus the coefficients in the normalization (6.8) are

$$d_\sigma(\ell) = \overline{t_\sigma(-\ell)}, \qquad \sigma \in \{0,+,-\}. \tag{A.7}$$

For indices $\ell = u\lambda^m n b^3$ as in (6.10), the coefficient magnitudes satisfy

$$|t_0(\ell)| \le \begin{cases} 3^{k/2+2}|b|, & m = 3k-4,\ k \ge 1, \\ 3^{k/2+5/2}|b|, & m = 3k-3,\ k \ge 0. \end{cases}$$

The $m = -4$ coefficients of t_- and t_+ have magnitude at most $9|b|$. For the chosen H, let $\sigma \in \{0,+,-\}$ index the corresponding expansion in (6.8). For each translate $\overline{\theta(z + a/c, v)}$, we have identified the Fourier expansion in z of $\overline{\theta(H(z,v))}$ as one of the three expansions in (6.8):

$$\overline{\theta(H(z,v))} = \frac{3^{5/2}}{2}\mathbf{1}_{\sigma=0}v^{2/3} + \sum_{0\neq\ell\in\lambda^{-4}\mathcal{O}} d_\sigma(\ell) vK_{1/3}(4\pi|\ell|v)\breve{e}(\ell z).$$

These coefficients satisfy the support restriction (6.10) and bound (6.11), proving the coefficient assertions of Lemma 6.4. The next step uses $g = g_1 H$ to express the original translate through this Fourier series evaluated at $g^{-1}(z + a/c, v)$, and computes the accompanying multiplier and additive phase. The matrix g maps ∞ to a/c, which is why this is called an expansion at the cusp a/c.

The local character transformation. The matrix factorization $g = g_1 H$ lets us apply the automorphy law

$$\theta(g_1 w) = \kappa(g_1)\theta(w), \qquad \kappa(g_1) = (c_1/a_1)_3, \qquad g_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \equiv I \pmod 3,$$

where κ is Kubota's cubic character [7, (5.4), (5.6)]. We will combine its conjugate with the finite Fourier coefficients $C_{p,j}(h_p)$ in (A.5) to obtain the factors $B_{p,j}$ of (6.5). For the translated theta function, the automorphy law gives

$$\overline{\theta(z + a/c, v)} = \overline{\kappa(g_1)}\,\overline{\theta\big(Hg^{-1}(z + a/c, v)\big)}, \tag{A.8}$$

where

$$g^{-1}(z + a/c, v) = \Big(-\frac{\delta'}{c} - \frac{\bar{z}}{c^2(v^2 + |z|^2)}, \frac{v}{\mathrm{N}_{K/\mathbb{Q}}(c)(v^2 + |z|^2)}\Big). \tag{A.9}$$

At $z = 0$, the term with Fourier index ℓ in the expansion of $\overline{\theta}(Hw)$ therefore acquires the phase $\breve{e}(-\delta'\ell/c)$. We claim that the multiplier is given by

$$\kappa(g_1) = \begin{cases} (c_0/a)_3(a/r)_3, & 3 \mid c, \\ (-u_0/(a - u_0 b_g))_3\,((c_0/u_0)/a)_3(a/r)_3, & v_\lambda(c) = 1, \\ (a/c_0)_3(a/r)_3, & (c,\lambda) = 1. \end{cases} \tag{A.10}$$

To verify (A.10), use the determinant equation and cubic reciprocity. For the case $v_\lambda(c) = 1$, the required congruences are

$$a(c - u_0\delta') \equiv -u_0 \pmod{a - u_0 b_g}, \qquad b_g c \equiv -1 \pmod a.$$

For $(c,\lambda)=1$, the congruence

$$-b_g c = 1 - a\delta' \equiv 1 \pmod 9$$

removes the supplementary factors; reciprocity at the remaining primes then gives $(\delta'/(-b_g c))_3 = 1$. The factors other than $(a/r)_3$ depend only on the fixed residues at primes in S. We may therefore write

$$\kappa(g_1) = \kappa_0 (a/r)_3 = \kappa_0 \prod_{p\mid r} \chi_p(a)^2, \qquad |\kappa_0| = 1,$$

where κ_0 is fixed once h_0, the active set, and $r \bmod M^2$ are fixed.

Fix h_0 and the active set, so that the denominator $c = c_0 r$ is fixed while the nonzero residues h_p vary. Put $D_0 = \lambda^3 c_0$. For each active prime p, define in $\mathcal{O}/(p)$

$$\sigma_p = \lambda^2 c/p, \qquad \epsilon_p = -\big((\lambda^3 c/p)\sigma_p\big)^{-1}.$$

The inverse in ϵ_p is taken in the field $\mathcal{O}/(p)$; it exists because r is squarefree and $p \nmid \lambda c_0$. The expression for a gives $a \equiv \sigma_p h_p \pmod p$. For a dual Fourier index $\ell \in \lambda^{-4}\mathcal{O}$, the additive form of the Chinese remainder theorem yields

$$\breve{e}(-\delta'\ell/c) = \psi(\lambda^4\ell) \prod_{p\mid r} e(\epsilon_p h_p^{-1} \lambda^4 \ell/p), \tag{A.11}$$

where

$$\psi(\lambda^4\ell) := e(-\delta_0' r^{-1} \lambda^4 \ell / D_0), \qquad \delta_0' = \delta' \bmod D_0.$$

The residues δ_0' and $r^{-1} \bmod D_0$ are fixed by the choices above, so ψ ranges over a fixed finite family of additive characters of $\mathcal{O}$. We claim the local transformation identity

$$\sum_{h_p \neq 0} C_{p,j}(h_p) \chi_p(a)^{-2} e(\epsilon_p h_p^{-1} \lambda^4 \ell/p) = \chi_p(\sigma_p)^{-2} \omega_{p,j} B_{p,j}(\lambda^4\ell), \tag{A.12}$$

where $B_{p,j}$ is defined in (6.5), and

$$\omega_{p,j} = \begin{cases} \chi_p(-1)^j \gamma_j(p)\gamma_{j+2}(p)\chi_p(\epsilon_p)^{-j-2}, & j \neq 0, 4, \\ \gamma_4(p), & j = 4, \\ -\gamma_2(p)\chi_p(\epsilon_p)^{-2}, & j = 0 \text{ active}. \end{cases}$$

The indices of $\gamma_j(p)$ are read modulo six; in the second case, $\chi_p(-1)^4 = 1$. In each case $|\omega_{p,j}| = 1$. To prove (A.12), combine (A.5), the conjugate of (A.10), and (A.11). Setting $y = h_p^{-1}$ changes the character exponent as follows:

$$\chi_p(h_p)^{-j}\chi_p(a)^{-2} = \chi_p(\sigma_p)^{-2}\chi_p(h_p)^{-j-2} = \chi_p(\sigma_p)^{-2}\chi_p(y)^{j+2}, \qquad y = h_p^{-1}.$$

For $j \neq 0, 4$, the character χ_p^{j+2} is nontrivial, and its Gauss sum gives the first case of $B_{p,j}$. For $j = 4$, it is trivial on nonzero residues; after rescaling by $\epsilon_p \neq 0$, the sum is

$$\sum_{y \neq 0} e(\lambda^4 \ell y/p) = -1 + \mathrm{N}_{K/\mathbb{Q}}(p) \mathbf{1}_{p \mid \lambda^4 \ell}.$$

For active $j = 0$, the coefficient $-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}$ combines with a cubic Gauss sum to give

$$B_{p,0}(\lambda^4\ell) = \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2} \chi_p(\lambda^4\ell)^{-2},$$

with the unit factors included in $\omega_{p,0}$. This proves (A.12) in every case. An inactive prime has $j = 0$ and contributes only the scalar $1 - \mathrm{N}_{K/\mathbb{Q}}(p)^{-1}$. Consequently, for each fixed h_0 and active set $\mathcal{A}$, combining the Fourier coefficients $d_\sigma(\ell)$ of $\overline{\theta(H(z,v))}$ with the sums over $h_p \neq 0$ gives

$$d_\sigma(\ell)\psi(\lambda^4\ell)\prod_{p\in\mathcal{A}} B_{p,j_p}(\lambda^4\ell),$$

up to a scalar independent of ℓ. These are the arithmetic factors in the dual sum (6.9). In particular, $B_{p,1}(\lambda^4\ell) = \chi_p(\lambda^4\ell)^3$ is the quadratic factor used in (6.18).

The archimedean transform. We now derive the transformed weight $V_*^\sharp$ in (6.6) and the scalar multiplying the dual sum. For $\operatorname{Re} s > 1$, introduce the Dirichlet series associated with the completed sum (5.3):

$$\mathcal{T}(s,\Psi) = \Big(\sideset{}{^*}\sum_{\substack{n\in\mathcal{O}\\ n\equiv 1\ (3)\\ (n,S)=1}} \overline{\alpha(n)}\gamma_2(n)\Psi(n)\mathrm{N}_{K/\mathbb{Q}}(n)^{-s}\Big)\Big(\sum_{\substack{b\in\mathcal{O}\\ b\equiv 1\ (3)\\ (b,S)=1}} \overline{\alpha(b)}^{\,3}\Psi(b)^3\mathrm{N}_{K/\mathbb{Q}}(b)^{-3s+1/2}\Big).$$

With V_* from (5.3) and its Mellin transform defined before (6.6), Mellin inversion gives

$$T(X;\Psi) = \frac{1}{2\pi\mathrm{i}}\int_{(\sigma)} \widehat{V}_*(s-\tfrac12)\mathcal{T}(s,\Psi)X^{s-1/2}\,ds, \qquad \sigma > 1. \tag{A.13}$$

The identity (6.3), with Ψ in place of Ψ_k, identifies these coefficients with those of Θ_Ψ after inserting the factor $\overline{\alpha(nb^3)}$. We now compute the normalization relating $\mathcal{T}(s,\Psi)$ to the Mellin transform of the derivative of Θ_Ψ.

For $z = x + \mathrm{i}y$, use $\partial_{\bar z} = (\partial_x + \mathrm{i}\partial_y)/2$ and $\partial_z = (\partial_x - \mathrm{i}\partial_y)/2$. Differentiation in $\bar z$ supplies a factor $\bar\ell$ in each Fourier mode. Combined with the factor $|\ell|^{-1}$ from the Bessel integral below, this produces the required angular factor $\overline{\alpha(\ell)}$. We therefore define, initially for $\operatorname{Re} s > 1$, the Mellin transform

$$\mathcal{J}(s) = \int_0^\infty \partial_{\bar z}\Theta_\Psi(z,v)\Big|_{z=0} v^{2s-1}\,dv.$$

For each nonzero Fourier mode, differentiation and the Mellin integral for $K_{1/3}$ (see [35, (10.43.19)]) give the following formulas:

$$\partial_{\bar z}\breve e(\ell z) = 2\pi\mathrm{i}\bar\ell\,\breve e(\ell z), \tag{A.14}$$

$$\int_0^\infty v^{2s}K_{1/3}(4\pi|\ell|v)\,dv = \frac{2^{2s-1}\Gamma(s+1/3)\Gamma(s+2/3)}{(4\pi|\ell|)^{2s+1}} \qquad \operatorname{Re} s > 0. \tag{A.15}$$

For $\operatorname{Re} s > 1$, the coefficient formula (2.10) and (A.15) show that the sum of the integrals of the absolute values is finite. We may therefore integrate the differentiated Fourier series term by term.

For $\ell = \lambda^{-3}nb^3$, the phase and scale simplify to

$$\mathrm{i}\overline{\alpha(\ell)}|\ell|^{-2s} = 27^s\overline{\alpha(n)}\,\overline{\alpha(b)}^{\,3}\mathrm{N}_{K/\mathbb{Q}}(n)^{-s}\mathrm{N}_{K/\mathbb{Q}}(b)^{-3s}.$$

Combining this with (2.10) gives

$$\mathcal{J}(s) = \frac{3^{5/2}}{4}\Big(\frac{27}{(2\pi)^2}\Big)^s\Gamma(s+1/3)\Gamma(s+2/3)\mathcal{T}(s,\Psi). \tag{A.16}$$

We next show that $\mathcal{J}(s)$ is entire and use (A.16) to continue $\mathcal{T}(s,\Psi)$.

Write $(z',v') = g^{-1}(z + a/c, v)$. Differentiating (A.9) at $z = 0$ gives

$$\frac{\partial z'}{\partial\bar z}\Big|_{z=0} = -\frac{1}{c^2v^2}, \qquad \frac{\partial\overline{z'}}{\partial\bar z}\Big|_{z=0} = 0, \qquad \frac{\partial v'}{\partial\bar z}\Big|_{z=0} = 0.$$

Thus the derivative in cusp coordinates is $-(cv)^{-2}\partial_{z'}$, with no height-derivative term. The defining Fourier series (A.2) for Θ_Ψ controls $v \to \infty$. For $v \to 0$, use (A.8) and the cusp expansions (6.8). In each expansion the horizontal derivative removes the constant mode. The remaining modes decay exponentially as $v \to \infty$; at $v \to 0$, the transformed height $1/(\mathrm{N}_{K/\mathbb{Q}}(c)v)$ tends to infinity, giving exponential decay in $1/v$. Thus the integral defining $\mathcal{J}(s)$ converges for every s and defines an entire function. Equation (A.16), after division by its gamma factors, also continues $\mathcal{T}(s,\Psi)$ to an entire function.

For the term indexed by $\boldsymbol{h}$ in (A.4), write $c_{\boldsymbol{h}}, \delta'_{\boldsymbol{h}}, g_{1,\boldsymbol{h}}, H_{\boldsymbol{h}}$ for the corresponding choices above. Let $\sigma_{\boldsymbol{h}}$ be the cusp index determined by $H_{\boldsymbol{h}}$. Define its cusp Mellin transform by

$$\mathcal{J}_{\boldsymbol{h}}^{\vee}(s) = \int_0^\infty \partial_z \big\{\overline{\theta(H_{\boldsymbol{h}}(z,v))}\big\}\Big|_{z=-\delta'_{\boldsymbol{h}}/c_{\boldsymbol{h}}} v^{2s-1}\,dv. \tag{A.17}$$

Substituting $v \mapsto (\mathrm{N}_{K/\mathbb{Q}}(c_{\boldsymbol{h}})v)^{-1}$ in each translated term of $\mathcal{J}(s)$, using (A.8), gives

$$\mathcal{J}(s) = -\sum_{\boldsymbol{h}} c_{\mathrm{F}}(\boldsymbol{h})\overline{\kappa(g_{1,\boldsymbol{h}})}\,\overline{\alpha(c_{\boldsymbol{h}})}^{\,2}\mathrm{N}_{K/\mathbb{Q}}(c_{\boldsymbol{h}})^{1-2s}\mathcal{J}_{\boldsymbol{h}}^{\vee}(1-s). \tag{A.18}$$

Here $c_{\mathrm{F}}(\boldsymbol{h})$ is the coefficient in (A.4). The cusp coefficients $d_{\sigma_{\boldsymbol{h}}}(\ell)$ and (A.15) give, for $\operatorname{Re} s > 1$,

$$\mathcal{J}_{\boldsymbol{h}}^{\vee}(s) = \frac{\mathrm{i}\,\Gamma(s+1/3)\Gamma(s+2/3)}{4(2\pi)^{2s}} \sum_{0\neq\ell\in\lambda^{-4}\mathcal{O}} \frac{d_{\sigma_{\boldsymbol{h}}}(\ell)\alpha(\ell)}{\mathrm{N}_{K/\mathbb{Q}}(\ell)^s}\check{e}(-\delta'_{\boldsymbol{h}}\ell/c_{\boldsymbol{h}}). \tag{A.19}$$

The Dirichlet series $\mathcal{T}(s,\Psi)$ converges absolutely for $\operatorname{Re} s > 1$, while the series in (A.19) at $1-s$ converges absolutely for $\operatorname{Re} s < 0$. The functional equation (A.18) and Stirling's formula [22, §5.A.4] therefore give polynomial bounds in $|\operatorname{Im} s|$ for $\mathcal{T}(s,\Psi)$ on both sides of the strip $0 \le \operatorname{Re} s \le 1$. Splitting the integral defining $\mathcal{J}(s)$ at $v = 1$ gives finite order; Phragmén–Lindelöf [22, Theorem 5.53] then gives the same type of bound inside the strip (compare the arguments of Dunn and Radziwiłł in [7, Propositions 5.1–5.2]). Together with the rapid decay of $\widehat{V}_*$ on vertical lines, these bounds justify moving the s-contour in (A.13) to $\operatorname{Re} s < 0$. No poles are crossed, since $\mathcal{T}(s,\Psi)$ is entire. Setting $t = \frac{1}{2} - s$ then gives a line $\operatorname{Re} t > 1/2$, where the dual coefficient series converges absolutely.

For fixed h_0 and active set, insert (A.19) into (A.18) and use (A.12) to sum over the nonzero h_p. After the normalization in (A.16), the scalar C in (6.9) for this group of translates is

$$C = -\frac{\mathrm{i}}{81}\overline{\alpha(c)}^{\,2}\widehat{\phi}(h_0)\overline{\kappa_0} \prod_{p \text{ inactive}} (1-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}) \prod_{p \text{ active}} \chi_p(\sigma_p)^{-2}\omega_{p,j_p}, \qquad |C| \le 1/81.$$

Dividing (A.16) by its gamma factors and setting $t = \frac{1}{2} - s$ produces the gamma quotient in (6.6). Its numerator gamma factors have their first pole at $t = -5/6$, and its reciprocal denominator gamma factors are entire. Thus no poles lie between the current contour $\operatorname{Re} t > 1/2$ and $\operatorname{Re} t = 0$. The rapid decay of $\widehat{V}_*$, together with Stirling's formula, allows us to shift each kernel contour to $\operatorname{Re} t = 0$. This gives the weight $V_*^\sharp$ defined in (6.6), evaluated at $\mathrm{N}_{K/\mathbb{Q}}(\ell)X/\mathrm{N}_{K/\mathbb{Q}}(c)^2$. Together with (A.12), this gives the dual sum (6.9).

For k_0 as in Lemma 6.3, fix h_0, the factors of Ψ other than χ_{k_0}, and the active/inactive choices at their primes. Every prime dividing k_0 has exponent $j_p = 1$ and is therefore active, so

$$r = k_0 \prod_{p\in\mathcal{A}_{\mathrm{fix}}} p.$$

Thus r/k_0 is fixed, and fixing $k_0 \bmod M^2$ fixes $r \bmod M^2$. The choices of H, c_0 and the residues in (A.11) are therefore fixed in each such class. Thus $d = d_\sigma$, ψ, and c_0 have precisely the asserted dependence on k_0.

It remains to bound the transformed weight. The first numerator pole of the gamma quotient in (6.6) is at $t=-5/6$. We may therefore shift the kernel contour to $\operatorname{Re} t=-1/4$ for $0<x\le 1$, obtaining the factor $x^{1/4}$, and to $\operatorname{Re} t=A$ for $x\ge 1$, obtaining x^{-A}. Each application of $x\partial_x$ introduces a factor $-t$. Stirling's formula on $-1/4\le \operatorname{Re} t\le A$ then gives, for every $A>0$ and $j\ge 0$,

$$|(x\partial_x)^j V_*^\sharp(x)| \ll_{A,j} \min\{x^{1/4},(1+x)^{-A}\} \times \sup_{-A\le\eta\le 1/4}\int_{\mathbb{R}}(1+|u|)^{\lceil 4A\rceil+j+2}|\widehat{V}_*(\eta+\mathrm{i}u)|\,du. \tag{A.20}$$

When V_* is supported in a fixed compact interval $I\subset(0,\infty)$, repeated integration by parts in its Mellin transform gives rapid decay in $|u|$, uniformly for $-A\le\eta\le 1/4$. Thus, for a sufficiently large $J=J(A,j)$, the supremum of integrals in (A.20) is $\ll_{A,j,I}\|V_*\|_{C^J(I)}$. This proves (6.12). Thus (6.9) expresses $T(X;\Psi)$ as $O_{\Psi_0,S}(2^{|\mathcal{P}|})$ dual sums with $|C|\le 1/81$, the asserted ray class dependence of (d,ψ,c_0), and the weight decay just established. Together with the support and coefficient bounds proved above, this completes the proof of Proposition 6.2 and Lemmas 6.3 and 6.4. □

B Separating variables in smooth weights

This appendix justifies the separation of smooth weights in the completed mean-square estimate of Section 6, and the treatment of kernels depending on the row in the Poisson reductions of Sections 4 and 7.

B.1 Separating the variables

We use Mellin inversion to separate the variables of a smooth weight, with coefficient bounds controlled by finitely many derivatives.

Lemma B.1. *Fix a box $\mathcal{I}=I_1\times\cdots\times I_d$, where each I_j is a compact interval in $(0,\infty)$. Every $\mathcal{K}\in C_c^\infty((0,\infty)^d)$ supported in $\mathcal{I}$ has a representation*

$$\mathcal{K}(\mathbf{x})=\int_{\mathbb{R}^d} b(\mathbf{t})\prod_{j=1}^d x_j^{\mathrm{i}t_j}\,d\mathbf{t},\qquad x_j>0. \tag{B.1}$$

For $J\ge 0$ and every even integer $q>J+d$, the coefficient satisfies

$$\int_{\mathbb{R}^d}|b(\mathbf{t})|(1+|\mathbf{t}|)^J\,d\mathbf{t}\ll_{\mathcal{I},J,q,d}\|\mathcal{K}\|_{C^q(\mathcal{I})}.$$

Proof. Take the Fourier transform in logarithmic coordinates:

$$b(\mathbf{t})=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\mathcal{K}(e^{y_1},\dots,e^{y_d})e^{-\mathrm{i}\mathbf{t}\cdot\mathbf{y}}\,d\mathbf{y}.$$

Applying Fourier inversion to this gives (B.1). Since the intervals I_j are fixed and bounded away from zero, the C^q norm in logarithmic coordinates is bounded by a constant times $\|\mathcal{K}\|_{C^q(\mathcal{I})}$. Applying $(1-\Delta_{\mathbf{y}})^{q/2}$ under the integral therefore gives

$$|b(\mathbf{t})|\ll_{\mathcal{I},q,d}(1+|\mathbf{t}|)^{-q}\|\mathcal{K}\|_{C^q(\mathcal{I})}\qquad(q\ge 0\text{ even}).$$

Multiplication by $(1+|\mathbf{t}|)^J$ and integration proves the bound when $q>J+d$. See also the multivariable Mellin formulation in [39, §10.1, (130)–(131)]. □

In our applications, the weight has the more specific form

$$\mathcal{K}_R(\mathbf{x}) = \prod_{j=1}^{d} W_j(x_j) F\Big(R \prod_{j=1}^{d} x_j^{a_j}\Big), \qquad R > 0,$$

where the exponents a_j are fixed real numbers and $W_j \in C_c^\infty((0,\infty))$ is supported in I_j. Assume that $F \in C^\infty((0,\infty))$ satisfies

$$\|F\|_{A,q} := \max_{0\le m\le q} \sup_{u>0} (1+u)^A \big|(u\partial_u)^m F(u)\big| < \infty \qquad (A \ge 0,\ q \in \mathbb{Z}_{\ge 0}).$$

On the fixed box $\mathcal{I}$, the argument of F is comparable to R. The chain rule and the pointwise bound in the proof give coefficients b_R satisfying, for $A \ge 0$ and even $q \ge 0$,

$$|b_R(\mathbf{t})| \ll (1+R)^{-A}(1+|\mathbf{t}|)^{-q} \|F\|_{A,q} \prod_{j=1}^{d} \|W_j\|_{C^q(I_j)}. \tag{B.2}$$

Consequently, for $J \ge 0$ and even $q > J + d$,

$$\int_{\mathbb{R}^d} |b_R(\mathbf{t})|(1+|\mathbf{t}|)^J \, d\mathbf{t} \ll (1+R)^{-A} \|F\|_{A,q} \prod_{j=1}^{d} \|W_j\|_{C^q(I_j)}. \tag{B.3}$$

The constants depend only on A, J, q, d, the intervals I_j, and the exponents a_j. Thus separation preserves the decay in R, and its cost is controlled by finitely many derivatives of the original weights.

B.2 Weights depending on the row

The next lemma extends mean-square bounds for a common test function to smooth kernels depending on the row, with losses controlled by uniform bounds on their derivatives.

Fix compact intervals $I \subset \operatorname{int} I_*$ in $(0,\infty)$ and an integer $m \ge 0$.

Lemma B.2. *Consider two families of finite sums*

$$S_{j,r}(U) = \sum_n a_{j,r}(n) U(x_{j,r,n}), \qquad j = 1, 2, \quad x_{j,r,n} > 0,$$

with a finite set of rows r and nonnegative weights w_r. Suppose that, for every $U \in C_c^\infty(I_*)$,

$$\sum_r w_r |S_{j,r}(U)|^2 \le M_j \|U\|^2_{C^m(I_*)}, \qquad j = 1, 2.$$

Then for any $|c_r| \le w_r$ and smooth kernels $\mathcal{K}_r$ supported in I^2,

$$\Big|\sum_r c_r \sum_{n_1,n_2} a_{1,r}(n_1)\overline{a_{2,r}(n_2)} \mathcal{K}_r(x_{1,r,n_1}, x_{2,r,n_2})\Big| \ll_{I,I_*,m} \sqrt{M_1 M_2} \sup_r \|\mathcal{K}_r\|_{C^{2m+4}(I^2)}. \tag{B.4}$$

Proof. Choose a real $V \in C_c^\infty(I_*)$ equal to one on I, and set $U_t(x) = V(x)x^{it}$. Apply (B.1) of Lemma B.1 to each $\mathcal{K}_r$ with $d = 2$ and $\mathcal{I} = I^2$. Replacing the second Mellin variable by its negative and multiplying by $V(x)V(y)$, which equals one on the support of $\mathcal{K}_r$, gives

$$\mathcal{K}_r(x,y) = \int_{\mathbb{R}^2} b_r(s,t) U_s(x) \overline{U_t(y)} \, ds\, dt,$$

with the following uniform bound, obtained from the pointwise coefficient estimate in the proof of Lemma B.1 with $d = 2$ and $\mathcal{I} = I^2$:

$$\sup_r |b_r(s,t)| \ll_{I,q} \frac{\sup_r \|\mathcal{K}_r\|_{C^q(I^2)}}{(1+|s|+|t|)^q} \qquad (q \ge 0 \text{ even}).$$

The hypothesis applies to each U_t, and $\|U_t\|_{C^m(I_*)} \ll_{I,I_*,m} (1+|t|)^m$. Taking the common bound for b_r before integrating and applying Cauchy–Schwarz in r therefore bounds the absolute value in (B.4) by a constant depending on I, I_*, m, q times

$$\sqrt{M_1 M_2} \sup_r \|\mathcal{K}_r\|_{C^q(I^2)} \int_{\mathbb{R}^2} \frac{(1+|s|)^m (1+|t|)^m}{(1+|s|+|t|)^q}\, ds\, dt. \tag{B.5}$$

Taking $q = 2m+4$ makes the integral converge and proves the claim. □

B.3 Recombining the separated sums

We use weighted Cauchy–Schwarz to pass from mean-square bounds for the separated sums to a bound for their weighted integral.

Lemma B.3. *Let r, ι range over finite sets, let $d \geq 1$, and let $w_r \geq 0$. Suppose the measurable coefficients $c_{\iota,r}$ satisfy*

$$|c_{\iota,r}(\mathbf{t})| \leq b_\iota(\mathbf{t}), \qquad M := \sum_\iota \int_{\mathbb{R}^d} b_\iota(\mathbf{t})\, d\mathbf{t} < \infty,$$

where $b_\iota : \mathbb{R}^d \to [0,\infty)$. For measurable $F_{\iota,\mathbf{t}}(r)$, whenever the right-hand side is finite, one has

$$\sum_r w_r \Big| \sum_\iota \int c_{\iota,r}(\mathbf{t}) F_{\iota,\mathbf{t}}(r)\, d\mathbf{t} \Big|^2 \leq M \sum_\iota \int b_\iota(\mathbf{t}) \sum_r w_r |F_{\iota,\mathbf{t}}(r)|^2\, d\mathbf{t}. \tag{B.6}$$

The same assertion holds for finite sums with the integrals omitted.

Proof. For each fixed r, weighted Cauchy–Schwarz gives

$$\Big| \sum_\iota \int c_{\iota,r}(\mathbf{t}) F_{\iota,\mathbf{t}}(r)\, d\mathbf{t} \Big|^2 \leq M \sum_\iota \int b_\iota(\mathbf{t}) |F_{\iota,\mathbf{t}}(r)|^2\, d\mathbf{t}.$$

Multiply by w_r and sum over r to obtain (B.6). The same argument with sums in place of integrals proves the finite version. □

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Appendix B. Transfer parameters for the eleven-twelfths half-plane

The purpose of this appendix is to check the numerical consequences of changing the zero-free boundary. The arithmetic here does not prove the zero-free theorem. It checks that the subsequent deductions use only the weaker included statement. No claim of novelty is made for these elementary parameter choices.

B.1. *The contour and unsmoothing margins.* Put

$$\theta = \frac{11}{12}, \qquad \delta = \frac{1}{2000}, \qquad a = \theta - 2\delta, \qquad \sigma_0 = a + 8\delta = \frac{2759}{3000}.$$

We have $1/2 < a < 1$, $0 < \delta < 1/1000$, and $a + 2\delta = \theta$. Thus the hypothesis of Proposition 3.7 follows from the included nonvanishing theorem. That proposition gives $|S(\chi; x)| \le K x^{\sigma_0} \log(q+2)$.

In Corollary 3.8 choose $h = x^{47/50}$. The two exponents arising from unsmoothing satisfy

$$\frac{49}{50} - \left(1 + \sigma_0 - \frac{47}{50}\right) = \frac{1}{3000} > 0, \qquad \frac{47}{50} + \frac{1}{100} = \frac{19}{20} < \frac{49}{50}.$$

The first inequality controls the difference of two Riesz means divided by h; the second absorbs the logarithm in the short-interval error. The prime-power correction is smaller still, since $1/2+1/100 < 49/50$. Consequently the proof gives a twisted prime-sum bound with exponent $49/50$, with constants depending only on the fixed choice of δ.

B.2. *Major arcs.* Partial summation gives, for $M \ge 1$,

$$\sum_{m \le M} m^{-49/50} \le 1 + \int_1^M u^{-49/50}\, du = 50M^{1/50} - 49 \le 50M^{1/50}.$$

The proof of Lemma 5.3 therefore gives its stated bound with $Y^{1/50}$ in the denominator and $C_2 = 100C_3$.

Let $L = \log q \ge 200$ and $M = \lceil L \rceil$, as in Proposition 5.4. Set $R = M^{16}$ and $Y = M^{1800}$. The elementary inequalities $L \le M \le 2L$ and $e^{L/16} \ge (L/16)^4/24 > L + 1$ show $2 \le R < q$. For the last strict inequality it suffices that $L^3 > 48 \cdot 16^4$, which holds for $L \ge 200$.

In the major-arc contribution, $\log(qR+2) \le 18M$, $2 + 2\pi R \le 8M^{16}$ and $1 + L \le 2M$. Hence

$$k(1+L) \le \frac{288M^{34}}{1800M^{36}\log M} \le 1.$$

Here $1800/50 = 36$, leaving two powers of M beyond the required numerator exponent 34. The minor-arc estimate is unchanged: $\rho(1+L) \le 2/M \le 1$. Finally $\log Y = 1800 \log M \le 3600 \log L$, so the two smooth-number terms

contribute at most $7200e^{11}\log L$. These are exactly the constants used in Proposition 5.4.

The old choice $Y = M^{700}$ would give $Y^{1/50} = M^{14}$, which does not establish the displayed major-arc bound. Changing the boundary without changing this parameter would leave an actual gap.

B.3. *The cyclotomic tail and progression sieve.* For the tail in Theorem 1.3(ii), choose $Y = q^{25}$. The exact margin is

$$25(1-\sigma_0) - 2 = \frac{1}{120} > 0.$$

Thus $q\log(q+2)Y^{\sigma_0-1} \le \log(q+2)/q \le 1$ for $q \ge 3$. The corresponding old margin, with 17 in place of 25, is negative; the old tail estimate cannot simply be retained.

Now $n = 1 + mq \le q^{25}$ implies $m < q^{24}$. Put $J = \lfloor 24\log_2 q \rfloor + 1$. Since $\log 2 > 56/81$ and $\log q > 1$, we have

$$J \le \left(\frac{24\cdot 81}{56} + 1\right)\log q < 36\log q.$$

The bound for $\log 2$ follows from the first two terms of

$$\log 2 = 2\sum_{j\ge 0}\frac{(1/3)^{2j+1}}{2j+1}.$$

For $j \ge 48$ use $u = j/48$ and $z = 2^u$ in Lemma 6.2. Then $1 < z < 2^{1/48}q^{1/2} < q$, and

$$\frac{2\cdot 2^j}{\log z} = \frac{96\cdot 2^j}{j\log 2} \le \frac{150\cdot 2^j}{j+1}.$$

Indeed, $\log 2 > 2/3$ and $(j+1)/j \le 49/48$ make the ratio at most $96(3/2)(49/48) = 147 < 150$. For the remaining term,

$$(2+u)^2(48u+1) \le 9u^2\cdot 49u = 441u^3 \le 441\,2^{3u} \quad (u \ge 1),$$

and therefore

$$2^u(2+u)^2 \le \frac{441\cdot 2^{4u}}{48u+1} \le \frac{441\cdot 2^j}{j+1}.$$

The elementary inequality $u \le 2^u$ for $u \ge 1$ follows, for example, by minimizing $u\log 2 - \log u$. We obtain $N_j \le 591\cdot 2^j/(j+1) \le 600\cdot 2^j/(j+1)$. For $j < 48$ the bound with 600 is immediate from $N_j \le 2^j$.

Summing the blocks as in Lemma 6.3 gives

$$\sum_{\substack{1<n\le q^{25}\\ n\equiv 1 \pmod q}}\frac{\Lambda(n)}{n} \le \frac{600}{q}\big(56\log q + \log q\log\log q\big).$$

Here $2(\log 2)J \le 72(\log 2)\log q < 51\log q$ and $1 + \log J \le 5 + \log\log q$. Since $\log\log 3 > 1/11$, the constant $C_0 = 600\cdot 617$ suffices. The lower-bound integral

now contributes $4\log(q^{25}) = 100\log q$ in place of $68\log q$. This completes the parameter verification used in the article.

B.4. *What was formally checked.* The three rational margin statements, including the two failures of the old cutoffs, were proved using `norm_num` in the accompanying Lean file `ParameterAudit.lean`. Fresh axiom output lists only `propext`, `Classical.choice` and `Quot.sound`. The file does not claim to formalise all analytic estimates in this appendix or to prove the included zero-free theorem. The exact rational Python program is a separate consistency check, with the same limits.

Appendix C. Independent replacement attempts and their remaining obligations

This appendix records two attempted substitutions for the zero-free argument. It proves limitations of those specific substitutions. It does not claim that all possible independent approaches are ruled out, and it does not prove a new zero-free half-plane.

C.1. *The Burgess-shaped bound at polylogarithmic lengths.* Consider an upper-bound expression of the classical Burgess shape

$$B_r(N,q) = N^{1-1/r}q^{(r+1)/(4r^2)}, \qquad r \geq 1.$$

This is only an examination of that expression; no hypothesis about when a particular version of the Burgess theorem applies is needed. Let $A > 0$, $L = \log q$, and $N = L^A$. If $L > 4A\log L$, then for every $r \geq 1$,

$$\log\frac{B_r(N,q)}{N} = \frac{L/4 - A\log L}{r} + \frac{L}{4r^2} > 0.$$

Thus the expression alone is larger than the trivial bound N for every r, even before introducing its implicit constants. Using it at these lengths does not supply the uniform cancellation needed for the major-arc argument. This calculation does not refute cancellation in the actual character sums, or prove that another method cannot give it.

C.2. *Coherent sixth-power rows.* An attempted coefficient-uniform replacement for the specific Möbius mean square in the included proof also faces a concrete obstruction. Write $\mathcal{O} = \mathbb{Z}[\omega]$ and let $\chi_{\mathfrak{n}}$ be the sextic residue characters in the primary convention used there. For a primary generator v coprime to $\mathfrak{n}$,

$$\chi_{\mathfrak{n}}(v^6) = 1.$$

Suppose $H = D^{1+\vartheta}$ with $0 < \vartheta < 1/5$. Select prime ideal columns $\mathfrak{p}$ of norm in $[D, 2D]$, outside a fixed finite excluded set, and put $a_{\mathfrak{p}} = 1$, with all other coefficients zero. The prime ideal theorem in the fixed field gives

$\asymp D/\log D$ such columns. The elementary lattice count for primary generators gives $\asymp H^{1/6}$ distinct rows v^6 of norm at most H; choosing one generator per unit class makes these rows distinct. Since $N(v) \le H^{1/6} < D$ for sufficiently large D, these generators are coprime to every selected prime ideal. Every selected column therefore contributes 1 in each of these rows.

Consequently the unrestricted-coefficient row energy is at least

$$c\,H^{1/6}\left(\frac{D}{\log D}\right)^2.$$

It cannot in general be bounded by

$$C_\varepsilon(H+D)D^\varepsilon \sum_{\mathfrak{n}} |a_{\mathfrak{n}}|^2$$

when $\varepsilon < (1-5\vartheta)/6$: dividing the lower bound by the proposed upper-bound expression leaves a constant times $D^{(1-5\vartheta)/6-\varepsilon}/\log D$, which tends to infinity. The counting facts used here are the classical fixed-field prime ideal theorem and the lattice count recorded in the included argument; this observation is not claimed as a new large-sieve result.

The actual proposition in the included proof has the special coefficients $\mu(\mathfrak{n})\nu(\mathfrak{n})W(N\mathfrak{n}/D)$, with a fixed smooth weight and controlled seminorms. It does not assert the false coefficient-uniform estimate. The obstruction above is therefore not a counterexample to that proposition. An independent replacement must use the special coefficients, or supply some other genuinely stronger mechanism; that step has not been constructed here.

Data, provenance and verification

The original source archive [2] is the coauthored repository https://github.com/creelie/Breakthrough-problems-3, pinned for this review at commit `ce39df5c8b06423beb7ea665fe320158f6efd475`. The author's requested preprint is https://hexagonmath.org/2610.00256v2. The revised text, four vector figures, credited proof supplement, source and license notices are supplied together. No new Hexagon version identifier is asserted.

The three rational parameter statements in Appendix B were freshly checked in Lean 4.34.1 with only `propext`, `Classical.choice` and `Quot.sound`; an independent exact rational program checked the same margins. These checks do not themselves prove nonvanishing or the character-sum estimates. The full source-level audit of the previously bundled zero-free development is separately logged; this review does not assert that it has completed successfully. The original Lean statements use the 7/8 route and do not yet formalise all changes in this 11/12 presentation. Consequently this document is an author-review revision, not a certificate that all proposed research tasks are complete.

Use of AI tools

The original coauthored manuscript and Lean development used Claude under the authors' direction, as recorded in the source archive. OpenAI Codex prepared the 11/12 parameter conversion in Section 3, Lemma 5.3, Proposition 5.4, and the progression and tail estimates in Section 6; Appendix B identifies their exact margins. It also reconstructed the elementary obstruction arguments in Appendix C and prepared this layout. The zero-free proof reproduced in Appendix A is OpenAI's published repository text, unchanged and explicitly credited; it is not claimed as a new discovery by these authors. The authors remain responsible for reviewing the mathematics and references.

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